The area of a circle

$S = \pi r^2$ — where the formula comes from, and why the radius is squared.

LESSON 97–99 | 3 × 40 MIN | MATEMATIKA 6, §21 | S7 15

LESSON CLOCK

25' | 30' | 35' | 25' | 5

Where the formula comes from	25 min
Using it	30 min
Backwards	35 min
Doubling the radius	25 min
Homework	5 min

LEARNING OBJECTIVES

- Use $S = \pi r^2$ to find the area of a disc.
- Find the radius from the area.
- Explain why doubling the radius quadruples the area.
- Solve practical problems about circular regions.

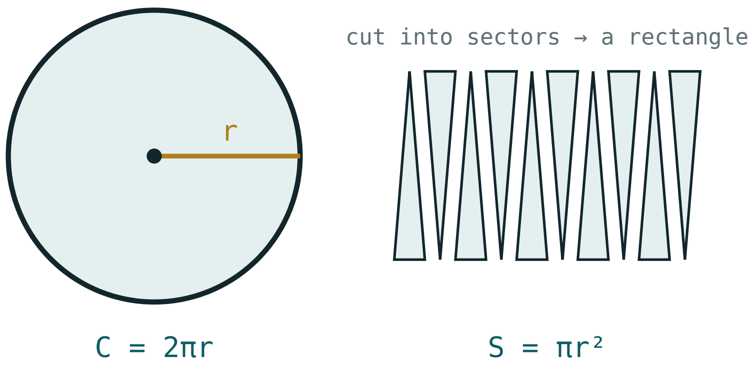
TEXTBOOK

National	pp. 281–290
Cambridge	Stage 7, pp. 148–154
Workbook	Exercise 15.2

01 Explanation

Where the formula comes from

Cut a disc into many thin sectors and lay them alternately point-up and point-down: they form a shape close to a rectangle.



A disc cut into sectors and rearranged

SIDE OF THE RECTANGLE	LENGTH
the long side	half the circumference, πr
the short side	the radius, r
so the area	$\pi r \cdot r = \pi r^2$

$$S = \pi r^2$$

THE THINNER THE SECTORS, THE BETTER THE RECTANGLE

With eight sectors the shape is lumpy; with sixty it is almost exactly a rectangle. The formula is what that process arrives at.

Using it

R	R^2	AREA ($\Pi = \frac{22}{7}$)	AREA ($\Pi = 3.14$)
7	49	154	153.86
14	196	616	615.44
10	100	314.29	314
21	441	1386	1384.74

SQUARE THE RADIUS, NOT ΠR

πr^2 means $\pi \cdot r \cdot r$, not $(\pi r)^2$. With $r = 7$ the area is 154, not 484.

Backwards

GIVEN AREA	$r^2 = S \div \pi$	r
154	49	7
616	196	14
314	100	10
1386	441	21

DIVIDE BY π, THEN TAKE THE SQUARE ROOT

Two steps, in that order. Stopping after the division gives r^2, which is a common half-done answer.

Doubling the radius

r	CIRCUMFERENCE	AREA
7	44	154
14	88	616
doubled	doubled	four times

LENGTHS SCALE BY k, AREAS BY k^2

A pizza of twice the diameter is four times as much pizza. It is the same rule met with rectangles, and it applies to every shape.

02 Worked examples

EXAMPLE 1 **Find the area of a disc of radius *7 cm*, taking $\pi = \frac{22}{7}$.**

1. $S = \pi r^2$
2. $= \frac{22}{7} \cdot 49$
3. $= 154\ cm^2$

   Square units.

ANSWER

$154\ cm^2$

EXAMPLE 2 **A disc has area *616 cm²*. Find its radius.**

1. $r^2 = S \div \pi = 616 \div \frac{22}{7}$
2. $= 196$
3. $r = \sqrt{196} = 14$ cm.

ANSWER

$14\ cm$

EXAMPLE 3 **A circular pond of radius *14 m* is to be covered with netting. Find the area needed.**

1. $S = \frac{22}{7} \cdot 196$

2. $= 616$ m².

3. Netting is sold by the square metre, so *616* m² is the order.

ANSWER

$616\ m^2$

03 Interactive model

Cut a paper disc into twelve sectors and rearrange them on the board; the near-rectangle makes πr^2 obvious in a way no derivation can.

Area, forwards and backwards

The area of a disc is:

- $2\pi r$
- πr^2
- πd
- πr

With $r = 7$ and $\pi = \frac{22}{7}$:

- 44
- 154
- 49
- 484

With $r = 10$ and $\pi = 3.14$:

- 31.4
- 62.8
- 314
- 100

An area of 154 gives r^2 equal to:

- 7
- 49
- 154
- 22

So the radius is:

- 7
- 49
- 14
- 22

Doubling the radius multiplies the area by:

- 2
- 3
- 4
- 8

And the circumference by:

- 2
- 4
- 8
- 1

An answer in cm for an area is:

- fine
- wrong units
- a radius
- a diameter

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Area	Yuza	Площадь
Disc	Doira	Круг
Radius	Radius	Радиус
Square units	Kvadrat birlik	Квадратные единицы
To square	Kvadratga ko‘tarish	Возвести в квадрат
Square root	Kvadrat ildiz	Квадратный корень
Scale factor	O‘xshashlik koeffitsiyenti	Коэффициент подобия
Circular region	Doiraviy soha	Круговая область

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$S = \pi r^2$ means:

- $(\pi r)^2$
- $\pi \cdot r \cdot r$
- $2\pi r$
- $\pi + r^2$

A disc of radius 14 has area:

- 88
- 616
- 196
- 308

To find r from S you:

- divide by π
- divide by π then take the root
- take the root
- multiply

A disc of area *314* has radius:

- *10*
- *100*
- *50*
- *31.4*

Trebling the radius multiplies the area by:

- *3*
- *6*
- *9*
- *27*

Area is measured in:

- cm
- cm^2
- cm^3
- degrees

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $r = 7$: the area ($\pi = \frac{22}{7}$)

   ANSWER $154\ cm^2$

2. $r = 14$: the area

   ANSWER $616\ cm^2$

3. $r = 10$: the area ($\pi = 3.14$)

   ANSWER $314\ cm^2$

4. $r = 21$: the area ($\pi = \frac{22}{7}$)

   ANSWER $1386\ cm^2$

5. $r = 5$: the area ($\pi = 3.14$)

   ANSWER $78.5\ cm^2$

6. The formula for the area of a disc

   ANSWER πr^2

7. The units of an area

   ANSWER cm^2

Medium · the standard question

7 PROBLEMS

1. Area *154*: the radius

   ANSWER *7*

2. Area *616*: the radius

   ANSWER *14*

3. Area *314*: the radius

   ANSWER *10*

4. A pond of radius *14* m: the netting needed

   ANSWER *616* m^2

5. $d = 14$: the area

   ANSWER $154\ cm^2$

6. $d = 28$: the area

   ANSWER $616\ cm^2$

7. Doubling r multiplies the area by

   ANSWER *4*

Hard · stretch and reason

7 PROBLEMS

1. A ring between radii 7 and 14

   ANSWER 462 cm²

2. A ring between radii 10 and 6 ($\pi = 3.14$)

   ANSWER 200.96 cm²

3. A disc of circumference 44: its area

   ANSWER 154 cm²

4. A disc of area 154: its circumference

   ANSWER 44 cm

5. Trebling the radius multiplies the area by

   ANSWER 9

6. A pizza of 30 cm across against two of 15 cm: which is more?

   ANSWER The one of 30 cm, by twice

7. Why is the radius squared?

   ANSWER Area has two directions, and both scale with r

07 Homework

Homework — 5 tasks

Square the radius before multiplying by π, and write cm^2.

1. Find the area of discs of radius $3.5\ cm$ and $21\ cm$.
2. Find the area of a disc of diameter $28\ cm$.
3. A disc has area $1386\ cm^2$. Find its radius.
4. Find the area of a ring between circles of radii 14 and 7 cm.
5. Explain in one sentence why doubling the radius quadruples the area.

The area of a semicircle and of a quarter circle

A fraction of the disc — and the fraction is the central angle over *360*.

LESSON 100–102 | 3 × 40 MIN | MATEMATIKA 6, §22 | S7 15

LESSON CLOCK

25' | 30' | 35' | 25' | 5

Halves and quarters	25 min
Any sector	30 min
Area against perimeter	35 min
Problems	25 min
Homework	5 min

LEARNING OBJECTIVES

- Find the area of a semicircle and of a quadrant.
- Find the area of any sector from its central angle.
- Distinguish area from perimeter in the same figure.
- Solve problems about circular parts of real objects.

TEXTBOOK

National	pp. 291–300
Cambridge	Stage 7, pp. 148–154
Workbook	Exercise 15.3

01 Explanation

Halves and quarters

semicircle $S = \dfrac{1}{2}\pi r^2$ quadrant $S = \dfrac{1}{4}\pi r^2$

R	WHOLE DISC	SEMICIRCLE	QUADRANT
7	*154*	*77*	*38.5*
14	*616*	*308*	*154*
10	*314*	*157*	*78.5*

FIND THE WHOLE DISC FIRST, THEN TAKE THE FRACTION

One calculation and one division is safer than trying to remember a separate formula for each fraction.

Any sector

$$\text{sector } S = \frac{\alpha}{360} \times \pi r^2$$

ANGLE	FRACTION	AREA WITH $R = 7$
180°	$\frac{1}{2}$	*77*
120°	$\frac{1}{3}$	*51.33*
90°	$\frac{1}{4}$	*38.5*
60°	$\frac{1}{6}$	*25.67*
45°	$\frac{1}{8}$	*19.25*

THE SAME FRACTION AS FOR THE ARC

A *120°* sector is a third of the disc's area and a third of its circumference. One fraction does both jobs.

Area against perimeter

FIGURE, $R = 7$	PERIMETER	AREA
whole disc	*44 cm*	*154 cm²*
semicircle	*36 cm*	*77 cm²*
quadrant	*25 cm*	*38.5 cm²*

HALVING THE DISC HALVES THE AREA BUT NOT THE PERIMETER

The semicircle keeps half the curve and gains a diameter, so its perimeter is *36*, not *22*. Area is halved cleanly; boundary is not.

Problems

PROBLEM	WORKING	ANSWER
a semicircular window of radius 70 cm: the glass	$\frac{1}{2} \cdot \frac{22}{7} \cdot 4900$	$7700\ cm^2$
a quadrant flower bed of radius 14 m	$\frac{1}{4} \cdot 616$	$154\ m^2$
a $60°$ slice of a pizza of radius 21 cm	$\frac{1}{6} \cdot 1386$	$231\ cm^2$
a fan sweeping $120°$ with a blade of 35 cm	$\frac{1}{3} \cdot 3850$	$1283.33\ cm^2$

READ THE QUESTION: GLASS OR FRAME?

Glass is an area, framing is a perimeter. The same window gives two different answers depending on which is asked for.

02 Worked examples

EXAMPLE 1 **Find the area of a semicircle of radius *14 cm*.**

1. The whole disc: $\frac{22}{7} \cdot 196 = 616$ cm².
2. Half of it.
3. $308\ cm^2$

ANSWER

$308\ cm^2$

EXAMPLE 2 **Find the area of a $60°$ sector of a circle of radius 21 cm.**

1. Whole disc: $\frac{22}{7} \cdot 441 = 1386$ cm².

2. The fraction is $\frac{60}{360} = \frac{1}{6}$.

3. $1386 \div 6 = 231$ cm².

ANSWER

231 cm²

EXAMPLE 3 **A semicircular window has radius 70 cm. Find the area of glass and the length of framing.**

1. Glass: $\frac{1}{2} \cdot \frac{22}{7} \cdot 4900 = 7700$ cm².

   An area.

2. Framing: $\pi r + 2r = 220 + 140$.

   A perimeter.

3. 360 cm of framing.

ANSWER

7700 cm² and 360 cm

03 Interactive model

Ask for the glass and the framing of the same window in one question; the class learns to read for area or perimeter rather than for numbers.

Which fraction, and area or perimeter?

A semicircle of radius 7 has area:

- 154
- 77
- 38.5
- 22

A quadrant of radius 7 has area:

- 154
- 77
- 38.5
- 11

A $120°$ sector is what fraction?

- $\frac{1}{2}$
- $\frac{1}{3}$
- $\frac{1}{4}$
- $\frac{1}{6}$

A $60°$ sector of radius 21 has area:

- 231
- 462
- 1386
- 33

A semicircle of radius 7 has perimeter:

- 22
- 36
- 44
- 77

Halving a disc halves:

- the perimeter
- the area
- both
- neither

A $90°$ sector of radius 14 has area:

- 77
- 154
- 308
- 616

Glass in a window is measured as:

- a perimeter
- an area
- a length
- an angle

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Semicircle	Yarim doira	Полукруг
Quadrant	Chorak doira	Четверть круга
Sector	Sektor	Сектор
Central angle	Markaziy burchak	Центральный угол
Fraction of the disc	Doiraning qismi	Часть круга
Area	Yuza	Площадь
Perimeter	Perimetr	Периметр
Square units	Kvadrat birlik	Квадратные единицы

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The area of a semicircle is:

- πr^2
- $\frac{1}{2}\pi r^2$
- πr
- $2\pi r$

The area of a quadrant is:

- $\frac{1}{2}\pi r^2$
- $\frac{1}{4}\pi r^2$
- $\frac{1}{4}\pi r$
- πr^2

A sector of angle α has area:

- $\frac{\alpha}{360}\pi r^2$
- $\frac{\alpha}{180}\pi r^2$
- $\alpha\pi r^2$
- $\frac{360}{\alpha}\pi r^2$

A semicircle of radius 14 has area:

- 154
- 308
- 616
- 88

A $45°$ sector of radius 7 has area:

- 19.25
- 38.5
- 77
- 154

Perimeter and area of the same figure are:

- the same
- different measurements
- proportional
- both lengths

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. A semicircle of radius 7: the area

   ANSWER $77\ cm^2$

2. A quadrant of radius 7

   ANSWER $38.5\ cm^2$

3. A semicircle of radius 14

   ANSWER $308\ cm^2$

4. A quadrant of radius 14

   ANSWER $154\ cm^2$

5. A semicircle of radius 10 ($\pi = 3.14$)

   ANSWER $157\ cm^2$

6. A $90°$ sector as a fraction

   ANSWER $\frac{1}{4}$

7. A $60°$ sector as a fraction

   ANSWER $\frac{1}{6}$

Medium · the standard question

7 PROBLEMS

1. A $60°$ sector of radius 21

ANSWER *231 cm²*

2. A $120°$ sector of radius 7

ANSWER *51.33 cm²*

3. A $45°$ sector of radius 7

ANSWER *19.25 cm²*

4. A semicircular window of radius 70 cm: the glass

ANSWER *7700 cm²*

5. And the framing

ANSWER *360 cm*

6. A quadrant flower bed of radius 14 m

ANSWER *154 m²*

7. A $120°$ fan sweep with a 35 cm blade

ANSWER *1283.33 cm²*

Hard · stretch and reason

7 PROBLEMS

1. A $30°$ sector of radius 12 ($\pi = 3.14$)

   ANSWER $37.68\ cm^2$

2. A sector of area 77 in a circle of radius 7: its angle

   ANSWER $180°$

3. A sector of area 38.5 in the same circle

   ANSWER $90°$

4. Half a ring between radii 7 and 14

   ANSWER $231\ cm^2$

5. A quadrant of radius 21: area and perimeter

   ANSWER $346.5\ cm^2$ and $75\ cm$

6. Why does a semicircle keep half the area but more than half the boundary?

   ANSWER The diameter is added to the boundary

7. A $270°$ sector of radius 14

   ANSWER $462\ cm^2$

07 Homework

Homework — 5 tasks

Find the whole disc first, then take the fraction.

1. Find the area of a semicircle of radius $21\ cm$.
2. Find the area of a quadrant of radius $28\ cm$.
3. Find the area of a $120°$ sector of a circle of radius $14\ cm$.
4. A semicircular table top has radius $60\ cm$. Find its area and the length of edging.
5. Find the area of a $30°$ sector of a circle of radius $6\ cm$, taking $\pi = 3.14$.

The area and perimeter of compound figures

Cut the shape into pieces you know, add or subtract, and trace the boundary separately.

LESSON 103–105 | 3 × 40 MIN | MATEMATIKA 6, §23 | S7 15

LESSON CLOCK

25' | 30' | 35' | 25' | 5

Cutting into pieces	25 min
Subtracting	30 min
Perimeter by tracing	35 min
Mixed problems	25 min
Homework	5 min

LEARNING OBJECTIVES

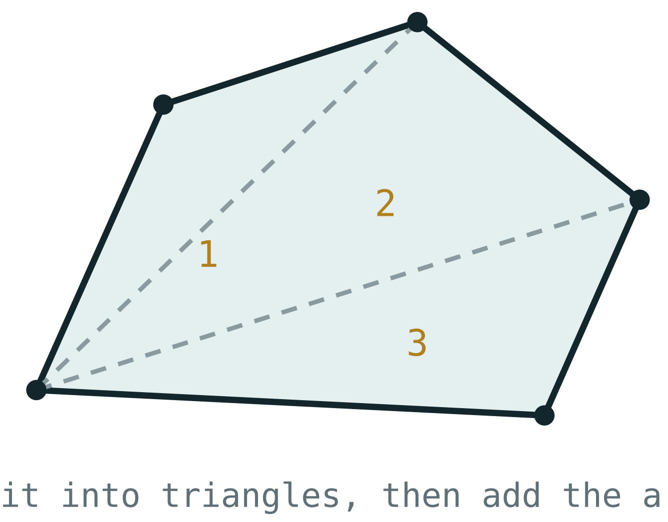
- Split a compound figure into rectangles, triangles and parts of circles.
- Find an area by adding or by subtracting.
- Find the perimeter by tracing the boundary.
- Keep area and perimeter apart in the same question.

TEXTBOOK

National	pp. 301–310
Cambridge	Stage 7, pp. 148–156
Workbook	Exercise 15.4

01 Explanation

Cutting into pieces



A compound shape cut into known pieces

SHAPE	PIECES	AREA
an L-shape 10×8 with a 4×3 corner removed	two rectangles	68
a house outline: 6×4 rectangle under a triangle of height 3	rectangle + triangle	33
a rectangle 14×10 with a semicircle of radius 7 on top	rectangle + semicircle	217
a running-track shape: 80×70 rectangle with two semicircular ends of radius 35	rectangle + circle	9450

WRITE THE AREA OF EACH PIECE ON THE SKETCH

Almost every lost mark in compound-area questions is a piece counted twice or left out — not an arithmetic error.

Subtracting

SHAPE	WORKING	AREA
a 20×15 plot with a 4×3 shed	$300 - 12$	288
a square of side 14 with a disc of radius 7 cut out	$196 - 154$	42
a 1 m path round a 5×4 m plot	$42 - 20$	22
a ring between radii 7 and 14	$616 - 154$	462

SUBTRACTING IS OFTEN QUICKER THAN CUTTING

A path round a plot could be split into four rectangles, but subtracting the inner area from the outer is one line of working instead of five.

Perimeter by tracing

Put a finger at one point of the boundary and travel round, writing down every edge you cross. Internal cuts are not part of the perimeter.

SHAPE	BOUNDARY PIECES	PERIMETER
rectangle 14×10 with a semicircle of radius 7 on top	$10 + 14 + 10 + 22$	56
a square of side 14 with a disc cut out	four sides + the circle	$56 + 44 = 100$
an L-shape 10×8 less a 4×3 corner	six sides	36

THE CUT EDGE IS NOT ON THE BOUNDARY

Where the semicircle meets the rectangle, that 14 cm line is inside the shape. It counts for neither perimeter piece — but the bottom 14 cm does.

Mixed problems

PROBLEM	AREA	PERIMETER
a semicircle of radius 7 on a 14×10 rectangle	217 cm²	56 cm
a 20×12 plot with a quadrant lawn of radius 7	$240 - 38.5 = 201.5$ m²	—
a track: 80×70 rectangle plus two semicircular ends of radius 35	9450 m²	380 m

TWO ANSWERS, TWO UNITS

An area is in m^2 and a perimeter in m. Writing the units is the last check that the right question has been answered.

02 Worked examples

EXAMPLE 1 **A semicircle of radius *7 cm* sits on top of a rectangle *14* cm by *10* cm. Find the area.**

1. Rectangle: $14 \cdot 10 = 140$ cm².
2. Semicircle: $\frac{1}{2} \cdot \frac{22}{7} \cdot 49 = 77$ cm².
3. $140 + 77 = 217$ cm².

ANSWER

217 cm²

EXAMPLE 2 **Find the perimeter of the same shape.**

1. Trace the boundary: two sides of 10, the bottom of 14, then the curve.
2. The curve is $\pi r = 22$ cm.
3. $10 + 14 + 10 + 22 = 56$ cm.

   The top 14 is internal.

ANSWER

56 cm

EXAMPLE 3 **A square of side *14 cm* has a disc of radius *7 cm* cut out of it. Find the area left.**

1. Square: 196 cm².
2. Disc: 154 cm².
3. $196 - 154 = 42$ cm².

   The four corner pieces.

ANSWER

$42\ cm^2$

03 Interactive model

Ask for area and perimeter of the same compound shape in every exercise; separating the two is the whole difficulty of the topic.

Add, subtract, or trace?

An L-shape 10×8 less a 4×3 corner has area:

- 68
- 80
- 92
- 12

A 14×10 rectangle with a semicircle of radius 7 on top has area:

- 140
- 154
- 217
- 294

Its perimeter is:

- 48
- 56
- 70
- 62

A square of side 14 less a disc of radius 7 has area:

- 42
- 154
- 196
- 350

Its perimeter is:

- 56
- 44
- 100
- 12

A 1 m path round a 5×4 plot has area:

- 20
- 22
- 42
- 9

An internal cut edge counts in:

- the area
- the perimeter
- both
- neither

Areas are written in:

- m
- m^2
- m^3
- no units

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Compound figure	Murakkab shakl	Составная фигура
To decompose	Bo‘laklarga ajratish	Разбить на части
To subtract	Ayirmoq	Вычесть
Boundary	Chegara	Граница
Internal edge	Ichki chegara	Внутренняя линия
Rectangle	To‘g‘ri to‘rtburchak	Прямоугольник
Semicircle	Yarim doira	Полукруг
Sum of the parts	Qismlar yig‘indisi	Сумма частей

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A compound area is found by:

- measuring
- cutting into known pieces
- guessing
- tracing

A compound perimeter is found by:

- adding all the pieces’ perimeters
- tracing the boundary
- subtracting
- halving

A 20×15 plot less a 4×3 shed:

- 288
- 300
- 312
- 12

A ring between radii *7* and *14*:

- *154*
- *462*
- *616*
- *770*

Where a semicircle meets a rectangle, the shared edge:

- counts once
- counts twice
- does not count
- is halved

The best first step is:

- calculate
- sketch and label the pieces
- estimate
- measure

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. An L-shape 10×8 less 4×3

   ANSWER 68

2. A house: 6×4 plus a triangle of height 3

   ANSWER 33

3. A 20×15 plot less a 4×3 shed

   ANSWER 288

4. A 1 m path round a 5×4 plot

   ANSWER $22\ m^2$

5. A square of side 14 less a disc of radius 7

   ANSWER $42\ cm^2$

6. A ring between radii 7 and 14

   ANSWER $462\ cm^2$

7. Units for an area

   ANSWER cm^2

Medium · the standard question

7 PROBLEMS

1. A 14×10 rectangle plus a semicircle of radius 7

   ANSWER $217\ cm^2$

2. Its perimeter

   ANSWER $56\ cm$

3. The square-less-disc shape: its perimeter

   ANSWER $100\ cm$

4. The L-shape: its perimeter

   ANSWER 36

5. A track: 80×70 plus ends of radius 35: the area

   ANSWER $9450\ m^2$

6. And its perimeter

   ANSWER $380\ m$

7. A 20×12 plot less a quadrant of radius 7

   ANSWER $201.5\ m^2$

Hard · stretch and reason

7 PROBLEMS

1. A 2 m path round a 10×6 m garden

   ANSWER $80\ m^2$

2. A 28 cm square with quadrants of radius 14 cut from two corners

   ANSWER $476\ cm^2$

3. A shape of two semicircles of radius 7 on opposite sides of a 14×20 rectangle: the area

   ANSWER $434\ cm^2$

4. Its perimeter

   ANSWER $84\ cm$

5. A disc of radius 14 with a 14 cm square hole

   ANSWER $420\ cm^2$

6. Why is subtracting often better than cutting?

   ANSWER One line of working instead of several

7. A $60°$ sector of radius 21 cut from a 21×21 square

   ANSWER $210\ cm^2$

07 Homework

Homework — 5 tasks

Sketch, label every piece, then answer area and perimeter separately.

1. Find the area of an L-shape 12×9 with a 5×4 corner removed, and its perimeter.
2. A semicircle of radius 10 cm sits on a 20×15 cm rectangle. Find the area ($\pi = 3.14$).
3. Find the perimeter of the same shape.
4. A 1.5 m path runs round a 12 m by 8 m lawn. Find the area of the path.
5. A square of side 28 cm has a disc of radius 14 cm cut out. Find the area remaining.

Home / Grade 6 / Geometry / Quarter 3

Think — compound figures in the world

Sports pitches, windows, tiles and gardens — real shapes, cut into known pieces.

LESSON 106 | 1 × 40 MIN | MATEMATIKA 6, O‘YLAB KO‘R | S7 15 IN CONTEXT

LESSON CLOCK

10' | 12' | 12' | 6'

Modelling a real shape	10 min
Area or length?	12 min
Costing a job	12 min
How accurate?	6 min

LEARNING OBJECTIVES

- Model a real object as a compound of simple shapes.
- Decide whether the question wants an area or a length.
- Estimate a real quantity from a measured plan.
- Judge the accuracy of the answer.

TEXTBOOK

National	pp. 311–313
Cambridge	Stage 7, pp. 148–156
Workbook	Exercise 15.5

01 Explanation

Modelling a real shape

OBJECT	MODELLED AS
a running track	a rectangle with two semicircular ends
an arched window	a rectangle with a semicircle on top
a football pitch	a rectangle, with a circle in the middle
a paved corner	a square less a quadrant
a garden with a round pond	a rectangle less a disc

EVERY MODEL LEAVES SOMETHING OUT

A real track has a kerb, a real window has a frame of some thickness. Saying what has been ignored is part of a good answer, not an apology for it.

Area or length?

WHAT IS WANTED	MEASUREMENT	UNITS
grass seed for a lawn	area	m^2
fencing round it	perimeter	m
glass for a window	area	cm^2
beading round the glass	perimeter	cm
paint for a wall	area	m^2
skirting board	length	m

THE MATERIAL TELLS YOU WHICH

Anything sold by the roll or the metre is a length; anything sold by the tin or the square metre is an area. Read the material, not the shape.

Costing a job

PROBLEM	WORKING	ANSWER
turf at $35\ 000$ sum a m^2 for a 12×8 m lawn	$96 \cdot 35\ 000$	$3\ 360\ 000$
fencing at $25\ 000$ a metre round the same lawn	$40 \cdot 25\ 000$	$1\ 000\ 000$
the same lawn with a 2 m round pond removed	$96 - 12.57$	$83.43\ m^2$
tiles 20 cm square for a 3×4 m floor	$12 \div 0.04$	300 *tiles*

CONVERT UNITS BEFORE DIVIDING

A 20 cm tile is 0.04 m², not 20 m². Mixing centimetres and metres in one division is where a factor of ten thousand goes missing.

How accurate?

ASSUMPTION	EFFECT
the corners are exactly square	small
$\pi = 3.14$ rather than more places	tiny
no waste when cutting tiles	can be 10%
the ground is flat	can be large on a slope

ADD FOR WASTE, THEN ROUND UP

Nobody orders 83.43 m² of turf. The professional answer is “about 90 m², allowing for waste” — and saying so is a mathematical judgement, not a vague one.

02 Worked examples

EXAMPLE 1 **A lawn is 12 m by 8 m with a circular pond of radius 2 m in it. Find the grassed area.**

1. Rectangle: 96 m².
2. Pond: $3.14 \cdot 4 = 12.57$ m².
3. $96 - 12.57 = 83.43$ m².
4. Order about 90 m² of turf, allowing for waste.

ANSWER

$83.43\ m^2$

EXAMPLE 2 **How many 20 cm square tiles cover a floor 3 m by 4 m?**

1. Floor: 12 m².

2. One tile: $0.2 \cdot 0.2 = 0.04$ m².
   Convert first.

3. $12 \div 0.04 = 300$ tiles.

ANSWER

300 tiles

EXAMPLE 3 **An arched window is a 80 cm by 120 cm rectangle with a semicircle of radius 40 cm on top. Find the glass area.**

1. Rectangle: $9\,600$ cm².

2. Semicircle: $\frac{1}{2} \cdot 3.14 \cdot 1\,600 = 2\,512$ cm².

3. $9\,600 + 2\,512 = 12\,112$ cm².
   About 1.2 m².

ANSWER

$12\,112\ cm^2$

03 Interactive model

Measure a real room or courtyard and cost a job in it; the arithmetic is the same as the textbook and the answer can be checked against a real price.

Area, length, or number of pieces?

Grass seed for a lawn is bought by:

- the metre
- the square metre
- the piece
- the litre

Fencing is bought by:

- the metre
- the square metre
- the tin
- the tile

A 12×8 m lawn has area:

- 40
- 96
- 20
- 192

Its perimeter is:

- 20
- 40
- 96
- 48

A pond of radius 2 m has area about:

- 6.28
- 12.57
- 4
- 25.1

A 20 cm square tile is:

- 20 m²
- 0.4 m²
- 0.04 m²
- 400 m²

Tiles for a 12 m² floor:

- 60
- 300
- 600
- 30

Ordering exactly the calculated area is:

- sensible
- risky — allow for waste
- required
- impossible

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Model	Model	Модель
Plan	Reja	План
Estimate	Baholash	Оценка
Material	Material	Материал
Cost	Narx	Стоимость
Per square metre	Bir kvadrat metrga	За квадратный метр
Accuracy	Aniqlik	Точность
Assumption	Faraz	Допущение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A running track is modelled as:

- a circle
- a rectangle with semicircular ends
- a square
- an oval with no formula

Paint is bought by:

- the metre
- the area to cover
- the perimeter
- the tile

Skirting board is bought by:

- the area
- the length
- the piece
- the tin

A 20 cm tile covers:

- 0.04 m²
- 0.2 m²
- 4 m²
- 20 m²

A model of a real shape:

- is exact
- ignores some details
- is useless
- needs no assumptions

A sensible order for 83.43 m² of turf is:

- 83.43
- about 90
- 80
- 83

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. A 12×8 m lawn: the area

ANSWER $96\ m^2$

2. Its perimeter

ANSWER $40\ m$

3. Turf at $35\ 000$ a m^2

ANSWER $3\ 360\ 000$ sum

4. Fencing at $25\ 000$ a metre

ANSWER $1\ 000\ 000$ sum

5. A 20 cm tile in m^2

ANSWER 0.04

6. Tiles for 12 m²

ANSWER 300

7. Grass seed is measured by

ANSWER area

Medium · the standard question

7 PROBLEMS

1. A 12×8 lawn with a pond of radius 2 m

ANSWER $83.43\ m^2$

2. An arched window 80×120 with a semicircle of radius 40

ANSWER $12\,112\ cm^2$

3. A 3×4 m floor in 25 cm tiles

ANSWER 192 tiles

4. Paint covering 10 m² a litre for 45 m²

ANSWER 4.5 litres

5. A 5×4 m room: skirting board

ANSWER $18\ m$

6. A track 80×70 with ends of radius 35: the area

ANSWER $9450\ m^2$

7. Which needs the perimeter: turf or fencing?

ANSWER Fencing

Hard · stretch and reason

7 PROBLEMS

1. A lawn 20×15 m with two round beds of radius 3 m

   ANSWER $243.5\ m^2$

2. Turf for that lawn at $40\ 000$ a m^2, allowing 10% waste

   ANSWER About $10\ 700\ 000$ sum

3. A room 6×4 m with a door 0.9 m wide: the skirting

   ANSWER 19.1 m

4. A wall 8 m by 2.5 m with a 2×1.5 m window: the paint area

   ANSWER $17\ m^2$

5. Why allow for waste?

   ANSWER Cut pieces at the edges cannot always be reused

6. A 30 cm tile for the same 12 m² floor

   ANSWER 134 tiles, rounded up

7. State two assumptions in the lawn problem

   ANSWER The ground is flat and the corners are square

07 Homework

Homework — 5 tasks

Say what you are assuming, and round the final order sensibly.

1. Measure a room at home and find its floor area and perimeter.
2. Work out how many 30 cm square tiles would cover that floor.
3. A lawn 15×10 m has a circular pond of radius 2.5 m. Find the grassed area.
4. Cost turf for that lawn at $38\ 000$ sum a square metre, allowing 10% for waste.
5. Write two sentences saying what your model of the lawn has ignored.

Revision — the circle and its measures

Circumference, area, arcs, sectors and compound shapes on one page.

LESSON 107–108 | 2 × 40 MIN | MATEMATIKA 6, TAKRORLASH | S7 8 AND 15 CONSOLIDATION

LESSON CLOCK

20' | 25' | 25' | 8' | 2

The two formulae	20 min
Fractions of a circle	25 min
Compound shapes	25 min
The errors that remain	8 min
Homework	2 min

LEARNING OBJECTIVES

- Choose between $C = 2\pi r$ and $S = \pi r^2$ from the wording.
- Work forwards and backwards with both formulae.
- Find arc lengths and sector areas as fractions.
- Handle a compound shape confidently.

TEXTBOOK

National	pp. 314–318
Cambridge	Stage 7, pp. 80–84, 148–156
Workbook	Revision 15

01 Explanation

The two formulae

WANTED	FORMULA	UNITS	BACKWARDS
circumference	$C = 2\pi r = \pi d$	cm	$r = C \div 2\pi$
area	$S = \pi r^2$	cm^2	$r = \sqrt{(S \div \pi)}$

R	C	S
7	44	154
14	88	616
21	132	1386

THE UNITS DECIDE THE FORMULA

Fencing, edging and string are lengths; turf, glass and paint are areas. Reading the material tells you which formula before any number is written.

Fractions of a circle

FIGURE	ARC OR CURVE	AREA	PERIMETER
semicircle, $r = 7$	22	77	36
quadrant, $r = 7$	11	38.5	25
$120°$ sector, $r = 21$	44	462	86

AREA IS A CLEAN FRACTION; PERIMETER IS NOT

A semicircle has half the area of the disc but not half the perimeter — the straight edge is added. This is the single most repeated error of the chapter.

Compound shapes

SHAPE	AREA	PERIMETER
a semicircle of radius 7 on a 14×10 rectangle	217	56
a square of side 14 less a disc of radius 7	42	100
a ring between radii 7 and 14	462	132

SKETCH, LABEL, THEN CALCULATE

Writing the area of each piece on the sketch, and tracing the boundary with a finger, catches almost every error before it happens.

The errors that remain

ERROR	LOOKS LIKE	CORRECT
diameter used as radius	$\pi \cdot 14^2$	$\pi \cdot 7^2$
$(\pi r)^2$	484	$\pi r^2 = 154$
semicircle perimeter without the diameter	22	36
square root forgotten	$r = 49$	$r = 7$
units missing or wrong	154 *cm*	154 *cm²*

02 Worked examples

EXAMPLE 1 **A circle has radius *21 cm*. Find its circumference and area.**

1. $C = 2 \cdot \frac{22}{7} \cdot 21 = 132$ cm.
2. $S = \frac{22}{7} \cdot 441 = 1386$ cm².
3. One in *cm*, one in *cm²*.

ANSWER

132 cm and *1386 cm²*

EXAMPLE 2 **A $120°$ sector has radius 21 cm. Find its arc, area and perimeter.**

1. Fraction: $\frac{1}{3}$.
2. Arc: $132 \div 3 = 44$ cm.
3. Area: $1386 \div 3 = 462$ cm².
4. Perimeter: $44 + 42 = 86$ cm.

   Two radii added.

ANSWER

44 cm, 462 cm², 86 cm

EXAMPLE 3 **A disc has area 1386 cm². Find its circumference.**

1. $r^2 = 1386 \div \frac{22}{7} = 441$
2. $r = 21$ cm.

   The square root — easily forgotten.
3. $C = 2 \cdot \frac{22}{7} \cdot 21 = 132$ cm.

ANSWER

132 cm

03 Interactive model

Give six questions and ask only whether each needs C or S; sorting them takes two minutes and removes most of the errors in the control work.

Length or area?

Fencing round a circular pond needs:

- C
- S
- both
- neither

Turf for the same pond area needs:

- C
- S
- both
- neither

With $r = 21$, C is:

- 66
- 132
- 1386
- 441

With $r = 21$, S is:

- 132
- 441
- 1386
- 66

A semicircle of radius 7 has perimeter:

- 22
- 36
- 44
- 77

Its area is:

- 22
- 36
- 77
- 154

An area of 1386 gives r equal to:

- 21
- 441
- 132
- 66

154 cm as an answer for an area is:

- fine
- the wrong units
- a radius
- a perimeter

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Circumference	Aylana uzunligi	Длина окружности
Area	Yuza	Площадь
Arc	Yoy	Дуга
Sector	Sektor	Сектор
Semicircle	Yarim doira	Полукруг
Quadrant	Chorak doira	Четверть круга
Compound shape	Murakkab shakl	Составная фигура
Units	O‘lchov birligi	Единицы измерения

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

C is measured in:

- cm
- cm^2
- cm^3
- degrees

S is measured in:

- cm
- cm^2
- cm^3
- degrees

A disc of radius 14 has area:

- 88
- 308
- 616
- 196

A semicircle of radius 14 has perimeter:

- 44
- 72
- 88
- 28

A $120°$ sector is what fraction?

- $\frac{1}{2}$
- $\frac{1}{3}$
- $\frac{1}{4}$
- $\frac{1}{6}$

To find r from the area you:

- divide by π
- divide by π then take the root
- divide by 2π
- take the root

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $r = 7$: C

   ANSWER 44 cm

2. $r = 7$: S

   ANSWER 154 cm^2

3. $r = 14$: C

   ANSWER 88 cm

4. $r = 14$: S

   ANSWER 616 cm^2

5. $r = 21$: C

   ANSWER 132 cm

6. $r = 21$: S

   ANSWER 1386 cm^2

7. Units for an area

   ANSWER cm^2

Medium · the standard question

7 PROBLEMS

1. A semicircle of radius 7: area and perimeter

   ANSWER 77 cm², 36 cm

2. A quadrant of radius 7

   ANSWER 38.5 cm², 25 cm

3. A $120°$ sector of radius 21

   ANSWER 462 cm², 86 cm

4. $C = 88$: the radius

   ANSWER 14

5. $S = 616$: the radius

   ANSWER 14

6. A ring between radii 7 and 14: the area

   ANSWER 462 cm²

7. A square of side 14 less a disc of radius 7

   ANSWER 42 cm²

Hard · stretch and reason

7 PROBLEMS

1. A disc of area 1386: its circumference

   ANSWER *132 cm*

2. A disc of circumference 132: its area

   ANSWER $1386\ cm^2$

3. A semicircle of radius 7 on a 14×10 rectangle: area and perimeter

   ANSWER $217\ cm^2$, $56\ cm$

4. A $60°$ sector of radius 14: the area

   ANSWER $102.67\ cm^2$

5. A ring between radii 7 and 14: its outer and inner boundary total

   ANSWER *132 cm*

6. Doubling the radius: what happens to C and S?

   ANSWER C doubles, S quadruples

7. Why is *154 cm* wrong for an area?

   ANSWER Areas need square units

07 Homework

Homework — 5 tasks

Write the units first, and let them tell you which formula to use.

1. Find C and S for circles of radius *3.5 cm* and *28 cm*.
2. A circle has circumference *66 cm*. Find its radius and area.
3. Find the area and perimeter of a semicircle of radius *14 cm*.
4. Find the area of a $45°$ sector of a circle of radius *28 cm*.

5. A rectangle 20×14 cm has a semicircle of radius 7 cm cut from one short side. Find the area left.

Control work 6 — the circle, and work on the mistakes

Circumference, area, sectors and a compound shape, under time.

LESSON 109–110 | 2 × 40 MIN | MATEMATIKA 6, NAZORAT ISHI 6 | S7 8 AND 15 REVIEW

LESSON CLOCK

3 | 35' | 12' | 25' | 5'

Instructions	3 min
The paper	35 min
Answers	12 min
Diagnosis and rewrite	25 min
The map	5 min

LEARNING OBJECTIVES

- Use the two circle formulae accurately and with units.
- Work backwards from a circumference or an area.
- Find a sector area and a semicircle perimeter.
- Classify each lost mark and rewrite the whole solution.

TEXTBOOK

National	pp. 250–318
Cambridge	Stage 7, pp. 80–84, 148–156
Workbook	Control paper 6

01 Explanation

The paper — 25 marks, 35 minutes

Q	TASK	MARKS	FROM
1	Name the centre, a radius, a diameter and a chord on a given circle	*3*	L89–91
2	Find C and S for a circle of radius 14 cm	*4*	L92–99
3	A circle has $C = 66$ cm: find its radius	*3*	L92–94
4	A disc has area 154 cm²: find its radius	*3*	L97–99
5	Find the area and perimeter of a semicircle of radius 7 cm	*4*	L95–102
6	Find the area of a $120°$ sector of radius 21 cm	*4*	L100–102
7	A semicircle of radius 7 sits on a 14×10 rectangle: find the area	*4*	L103–105

THE ANSWERS

88 cm and *616* cm²; *10.5* cm; *7* cm; *77* cm² and *36* cm; *462* cm²; *217* cm².

Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
diameter used for radius	$\pi \cdot 28^2$	$\pi \cdot 14^2$
formulae swapped	$C = \pi r^2$	$C = 2\pi r$
square root forgotten	$r = 49$	$r = 7$
semicircle perimeter without the diameter	22	36
sector fraction wrong	$\frac{120}{180}$	$\frac{120}{360}$
compound shape: pieces not added	140	$140 + 77$
units missing	616	$616\ cm^2$

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

The chapter as one map

IDEA	FORMULA OR RULE
circumference	$C = 2\pi r = \pi d$
area of a disc	$S = \pi r^2$
arc	$\frac{\alpha}{360} \cdot C$
sector area	$\frac{\alpha}{360} \cdot \pi r^2$
semicircle perimeter	$\pi r + 2r$
quadrant perimeter	$\frac{1}{2}\pi r + 2r$
compound shapes	add or subtract; trace the boundary

LOOKING FORWARD

Next come two Cambridge lessons on constructions and symmetry, and then the largest block of the quarter: speed, distance and time.

02 Worked examples

EXAMPLE 1 **Model answer, Q3: $C = 66$ cm.**

1. $C = 2\pi r$

2. $66 = 2 \cdot \frac{22}{7} \cdot r$

3. $66 = \frac{44}{7} r$

4. $r = 10.5$ cm.

   Check: $2 \cdot \frac{22}{7} \cdot 10.5 = 66$ ✓

ANSWER

$10.5\ cm$

EXAMPLE 2 **Model answer, Q5: a semicircle of radius 7 cm.**

1. Area: $\frac{1}{2} \cdot 154 = 77$ cm².
2. Curve: $\pi r = 22$ cm.
3. Perimeter: $22 + 14 = 36$ cm.

   The diameter is part of the boundary.

ANSWER

77 cm^2 and 36 cm

EXAMPLE 3 **Model answer, Q7: the compound shape.**

1. Rectangle: 140 cm².
2. Semicircle: 77 cm².
3. $140 + 77 = 217$ cm².

ANSWER

217 cm^2

03 Interactive model

Mark Q5 in two halves — area and perimeter — and report them separately; the class sees which of the two it actually finds hard.

The chapter in eight questions

C for $r = 14$:

- 44
- 88
- 616
- 196

S for $r = 14$:

- 88
- 196
- 308
- 616

$C = 66$ gives r:

- 10.5
- 21
- 7
- 33

$S = 154$ gives r:

- 7
- 49
- 14
- 11

A semicircle of radius 7 has area:

- 154
- 77
- 44
- 36

And perimeter:

- 22
- 36
- 44
- 77

A $120°$ sector of radius 21:

- 462
- 1386
- 44
- 132

The compound shape of Q7 has area:

- *140*
- *77*
- *217*
- *294*

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Circumference	Aylana uzunligi	Длина окружности
Area	Yuza	Площадь
Sector	Sektor	Сектор
Semicircle	Yarim doira	Полукруг
Compound shape	Murakkab shakl	Составная фигура
Units	O‘lchov birligi	Единицы
Diagnosis	Tashxis	Диагностика

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Q2 needs two formulae because it asks for:

- two circles
- a length and an area
- two radii
- a check

Q3 divides by:

- π
- 2π
- r
- 2

Q4 ends with:

- a division
- a square root
- a multiplication
- a subtraction

Q5 perimeter includes:

- the curve only
- the curve and the diameter
- two radii
- the whole circle

Q6 fraction is:

- $\frac{120}{180}$
- $\frac{120}{360}$
- $\frac{360}{120}$
- $\frac{1}{2}$

Q7 is answered by:

- subtracting
- adding two pieces
- tracing
- measuring

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $r = 14$: C

   ANSWER $88\ cm$

2. $r = 14$: S

   ANSWER $616\ cm^2$

3. $C = 66$: the radius

   ANSWER $10.5\ cm$

4. $S = 154$: the radius

   ANSWER $7\ cm$

5. A semicircle of radius 7: the area

   ANSWER $77\ cm^2$

6. And the perimeter

   ANSWER $36\ cm$

7. A $120°$ sector of radius 21

   ANSWER $462\ cm^2$

Medium · the standard question

7 PROBLEMS

1. A semicircle of radius 7 on a 14×10 rectangle

   ANSWER $217\ cm^2$

2. Its perimeter

   ANSWER $56\ cm$

3. $r = 3.5$: C and S

   ANSWER $22\ cm,\ 38.5\ cm^2$

4. A quadrant of radius 14: the area

   ANSWER $154\ cm^2$

5. And its perimeter

   ANSWER $50\ cm$

6. $C = 44$: the area of the disc

   ANSWER $154\ cm^2$

7. $S = 616$: the circumference

   ANSWER $88\ cm$

Hard · stretch and reason

7 PROBLEMS

1. A ring between radii 14 and 21

   ANSWER $770\ cm^2$

2. A $45°$ sector of radius 28

   ANSWER $308\ cm^2$

3. A square of side 28 less a disc of radius 14

   ANSWER $168\ cm^2$

4. A $270°$ sector of radius 14: area and perimeter

   ANSWER $462\ cm^2$, $94\ cm$

5. Doubling the radius: C and S

   ANSWER Doubled and quadrupled

6. A track 80×70 with ends of radius 35: one lap

   ANSWER $380\ m$

7. Why does an area answer need cm^2?

   ANSWER It measures two directions

07 Homework

Homework — 5 tasks

Rewrite every question you lost a mark on, with its units.

1. Rewrite in full every question on which you lost a mark.
2. Find C and S for a circle of radius $10.5\ cm$.
3. A disc has area $616\ cm^2$. Find its radius and circumference.
4. Find the area and perimeter of a quadrant of radius $21\ cm$.
5. Find the area of a $150°$ sector of a circle of radius $14\ cm$.

Angles and constructions — intersecting lines and quadrilaterals

A Cambridge insert: drawing accurately with compasses, ruler and protractor.

LESSON 111–113 | 3 × 40 MIN | MATEMATIKA 6, QO'SHIMCHA MAVZU | S7 5.2–5.3

LESSON CLOCK

25' | 30' | 35' | 25' | 5

Angles where lines cross	25 min
Constructing triangles	30 min
Constructing quadrilaterals	35 min
Checking	25 min
Homework	5 min

LEARNING OBJECTIVES

- Use the angle facts at intersecting lines.
- Construct a triangle from given sides and angles.
- Construct a rectangle, a parallelogram and a rhombus.
- Check a construction by measuring what was not given.

TEXTBOOK

National	pp. 319–328
Cambridge	Stage 7, pp. 58–65
Workbook	Exercise 5.3

01 Explanation

Angles where lines cross

FACT	STATEMENT
vertically opposite	the two opposite angles are equal
on a line	neighbours add to $180°$
at the crossing	all four add to $360°$
perpendicular lines	all four are $90°$

ONE ANGLE	THE OTHER THREE
$70°$	$110°, 70°, 110°$
$125°$	$55°, 125°, 55°$
$90°$	$90°, 90°, 90°$

TWO CROSSING LINES GIVE ONLY TWO DIFFERENT SIZES

An angle and its supplement, each appearing twice. Knowing one of the four gives all of them.

Constructing triangles

GIVEN	TOOLS	STEPS
three sides	ruler and compasses	draw one side, two arcs, join
two sides and the angle between	ruler and protractor	draw the angle, mark both lengths
a side and two angles	ruler and protractor	draw the side, both angles at its ends

DRAW THE GIVEN ANGLE BEFORE MEASURING THE SIDES

Marking two lengths first and hoping the angle comes out right does not work. The given data fix the order of the construction.

Constructing quadrilaterals

SHAPE	METHOD
rectangle 6×4	two perpendiculars at the ends of a 6 cm side
square of side 5	the same, with equal sides
parallelogram, sides 6 and 4, angle $60°$	draw the angle, mark both sides, complete with parallels
rhombus of side 5, angle $70°$	as a parallelogram with all sides equal

OPPOSITE SIDES ARE MADE EQUAL, NOT MEASURED TWICE

In a parallelogram, set the compasses to the side length and step it off; measuring each side separately introduces small errors that leave the figure open.

Checking

CONSTRUCTION	CHECK BY MEASURING	SHOULD BE
triangle from three sides	the three angles	they add to $180°$
rectangle	the two diagonals	equal
parallelogram	opposite sides	equal
rhombus	the diagonals	they cross at $90°$

CHECK SOMETHING YOU WERE NOT GIVEN

Measuring what you drew tells you nothing; measuring a diagonal, or an angle you did not set, is a real test of the construction.

02 Worked examples

EXAMPLE 1 **Two lines cross and one angle is $125°$. Find the other three.**

1. Vertically opposite: $125°$.
2. On the line: $180 - 125 = 55°$.
3. Its opposite: $55°$.
4. So $125°, 55°, 125°, 55°$.

ANSWER

$55°, 125°, 55°$

EXAMPLE 2 **Construct a triangle with sides *6*, *5* and *4* cm, and check it.**

1. Draw the 6 cm side.
2. Arc of 5 from one end, arc of 4 from the other.
3. Join the crossing to both ends.
4. Measure the three angles: they should total $180°$.
   The check.

ANSWER

Constructed and checked

EXAMPLE 3 **Construct a parallelogram with sides *6* and *4* cm and an angle of *60°*.**

1. Draw a 6 cm side.
2. At one end, set off $60°$ and mark 4 cm.
3. From the two free ends, step off 6 and 4 cm with compasses.
4. Check: opposite sides equal, and the other angle is $120°$.

ANSWER

Constructed

03 Interactive model

Ask every pupil to measure a diagonal of their rectangle and compare with a neighbour; the two answers agreeing is a better verdict than any teacher’s tick.

Crossing lines and constructions

Two lines cross; one angle is $70°$. Its opposite is:

- $70°$
- $110°$
- $90°$
- $20°$

Its neighbour is:

- $70°$
- $110°$
- $20°$
- $290°$

How many different sizes appear?

- 1
- 2
- 3
- 4

To construct a triangle from three sides you need:

- a protractor
- compasses
- a set square
- nothing

To construct one from two sides and the angle between:

- compasses only
- a protractor and a ruler
- a set square
- a calculator

A rectangle is checked by measuring:

- the sides
- the diagonals
- the angles only
- nothing

A rhombus is checked by:

- equal sides
- diagonals crossing at $90°$
- equal diagonals
- right angles

A parallelogram with one angle $60°$ has another of:

- $60°$
- $120°$
- $30°$
- $90°$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Construction	Yasash	Построение
Intersecting lines	Kesishuvchi chiziqlar	Пересекающиеся прямые
Vertically opposite	Vertikal burchaklar	Вертикальные углы
Parallelogram	Parallelogramm	Параллелограмм
Rhombus	Romb	Ромб
Perpendicular	Perpendikulyar	Перпендикуляр
To check by measuring	O‘lchab tekshirish	Проверить измерением
Accuracy	Aniqlik	Точность

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Vertically opposite angles are:

- supplementary
- equal
- complementary
- right

The four angles at a crossing add to:

- $90°$
- $180°$
- $270°$
- $360°$

One angle of $125°$ gives the others:

- $55, 125, 55$
- $125, 125, 125$
- $35, 125, 35$
- $55, 55, 55$

Three given sides need:

- a protractor
- compasses
- a set square
- a calculator

A construction should be checked by measuring:

- what was given
- something not given
- nothing
- the paper

In a rhombus the diagonals:

- are equal
- cross at $90°$
- are parallel
- bisect the sides

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. One angle at a crossing is *70°*: its opposite

   ANSWER *70°*

2. And its neighbour

   ANSWER *110°*

3. One angle is *125°*: the neighbour

   ANSWER *55°*

4. Perpendicular lines: all four angles

   ANSWER *90°*

5. Angles at a crossing add to

   ANSWER *360°*

6. Sizes appearing at a crossing

   ANSWER *2*

7. A parallelogram with one angle *60°*: the next

   ANSWER *120°*

Medium · the standard question

7 PROBLEMS

1. Which tool for three given sides?

   ANSWER Compasses

2. Which for two sides and the included angle?

   ANSWER Protractor and ruler

3. Which for a side and two angles?

   ANSWER Protractor and ruler

4. A rectangle is checked by

   ANSWER measuring the diagonals

5. A rhombus is checked by

   ANSWER the diagonals crossing at $90°$

6. A parallelogram is checked by

   ANSWER opposite sides being equal

7. A triangle from three sides is checked by

   ANSWER the angles adding to $180°$

Hard · stretch and reason

7 PROBLEMS

1. Can a triangle be constructed with sides 3, 4, 8?

   ANSWER No — the arcs do not meet

2. A parallelogram of sides 6 and 4 with angle $60°$: all four angles

   ANSWER $60°$, $120°$, $60°$, $120°$

3. Its perimeter

   ANSWER 20 cm

4. A rhombus of side 5 with angle $70°$: the other angles

   ANSWER $110°$, $70°$, $110°$

5. Its perimeter

   ANSWER 20 cm

6. Why measure a diagonal to check a rectangle?

   ANSWER It was not given, so it is an independent test

7. A square of side 5: the diagonal, to the nearest mm

   ANSWER 7.1 cm

07 Homework

Homework — 5 tasks

Leave every construction line and arc visible on the page.

1. Two lines cross and one angle is $142°$. Find the other three.
2. Construct a triangle with sides 7, 6 and 5 cm and measure its angles.
3. Construct a rectangle 7 cm by 4 cm and measure both diagonals.
4. Construct a parallelogram with sides 7 and 5 cm and an angle of $50°$.
5. Construct a rhombus of side 6 cm with one angle $60°$ and measure its diagonals.

The symmetry of 2D shapes; congruent shapes

A Cambridge insert: lines of symmetry, order of rotation, and when two shapes are the same.

LESSON 114–115 | 2 × 40 MIN | MATEMATIKA 6, QO'SHIMCHA MAVZU | S7 8.1, 8.3

LESSON CLOCK

20' | 25' | 25' | 8' | 2

Lines of symmetry	20 min
Rotational symmetry	25 min
Congruent shapes	25 min
In the world	8 min
Homework	2 min

LEARNING OBJECTIVES

- Find the lines of symmetry of a shape.
- Find the order of rotational symmetry.
- Decide whether two shapes are congruent.
- Recognise symmetry in ornament and in nature.

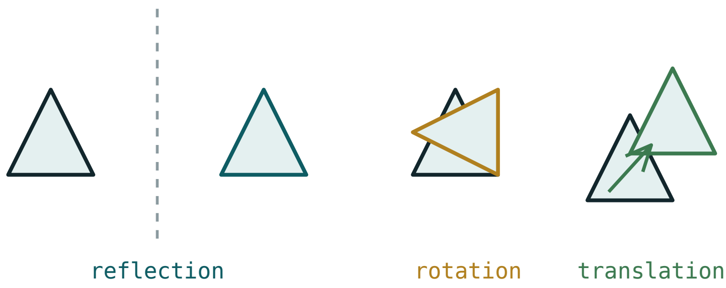
TEXTBOOK

National	pp. 329–334
Cambridge	Stage 7, pp. 78–86
Workbook	Exercise 8.1

01 Explanation

Lines of symmetry

A **line of symmetry** folds the shape onto itself exactly.



Reflection in a line of symmetry

SHAPE	LINES OF SYMMETRY
square	*4*
rectangle (not square)	*2*
equilateral triangle	*3*
isosceles triangle	*1*
scalene triangle	*0*
rhombus	*2*
regular hexagon	*6*
circle	infinitely many

A RECTANGLE HAS TWO LINES, NOT FOUR

Its diagonals do not fold it onto itself — the folded corner does not land on a corner. Testing with a folded piece of paper settles it.

Rotational symmetry

The **order** is the number of positions, in one full turn, where the shape looks the same.

SHAPE	ORDER	ANGLE OF ROTATION
square	*4*	*90°*
rectangle	*2*	*180°*
equilateral triangle	*3*	*120°*
regular hexagon	*6*	*60°*
parallelogram	*2*	*180°*
isosceles triangle	*1*	none but the full turn

ORDER *1* MEANS NO ROTATIONAL SYMMETRY

Every shape returns to itself after $360°$, so the order is never zero. A regular n-sided polygon has order n and n lines of symmetry.

Congruent shapes

Two shapes are **congruent** when one can be moved exactly onto the other by a translation, a rotation, a reflection, or a combination of them.

PAIR	CONGRUENT?	WHY
two squares of side 5	yes	same shape and size
a square of 5 and one of 6	no	different size
a triangle and its mirror image	yes	reflection is allowed
a shape and its enlargement	no	similar, not congruent

CONGRUENT IS “IDENTICAL”; SIMILAR IS “SAME SHAPE”

Two photographs of different sizes are similar; two copies of the same photograph are congruent. Grade 9 will separate the two ideas properly.

In the world

WHERE	SYMMETRY
a butterfly	one line
a snowflake	six lines, order 6
a wheel with 8 spokes	order 8
an Uzbek ornamental star	often order 8 or 16
the letter H	two lines, order 2
the letter S	no lines, order 2

REFLECTION AND ROTATION ARE INDEPENDENT

S has rotational symmetry but no line; an isosceles triangle has a line but no rotation. Having one says nothing about the other.

02 Worked examples

EXAMPLE 1 **How many lines of symmetry has a rectangle that is not a square? What is its order of rotational symmetry?**

1. Two lines: through the midpoints of opposite sides.
2. The diagonals are not lines of symmetry.
3. Rotating $180°$ leaves it unchanged, so the order is 2.

ANSWER

2 lines, order 2

EXAMPLE 2 **A regular octagon: how many lines of symmetry, and what order?**

1. A regular n-gon has n lines and order n.
2. Here $n = 8$.
3. Eight lines, order 8; the rotation angle is $45°$.

ANSWER

8 and 8

EXAMPLE 3 **Are a triangle and its mirror image congruent?**

1. Congruence allows reflection.
2. The two have the same sides and the same angles.
3. So yes.

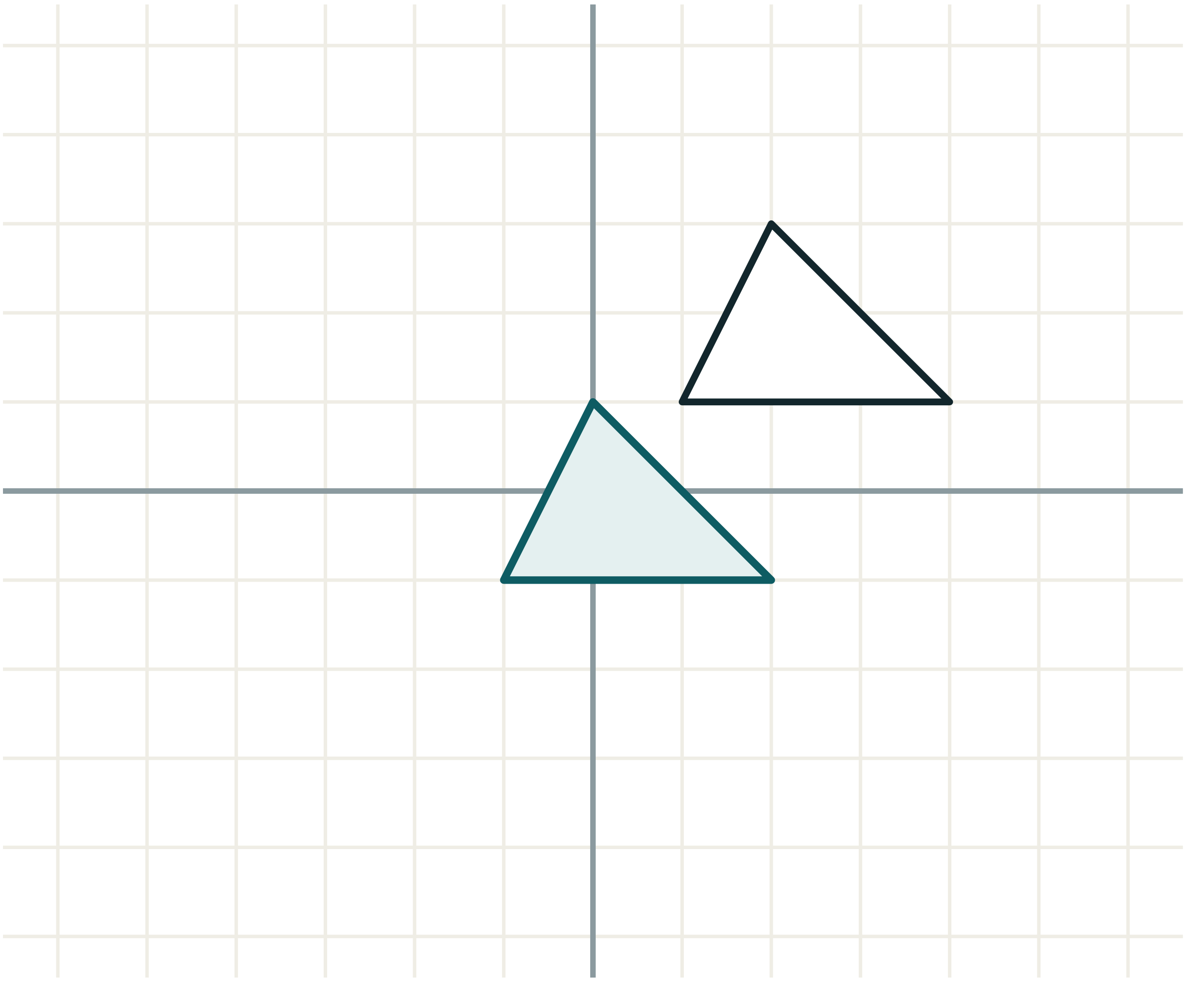
ANSWER

Yes

03 Interactive model

Fold paper shapes to find the lines and turn them on a pin to find the order; both symmetries become physical before they become vocabulary.

Reflect, rotate and translate a shape



transformation **translation** | lengths **unchanged** | angles **unchanged** | orientation **kept**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Symmetry	Simmetriya	Симметрия
Line of symmetry	Simmetriya o‘qi	Ось симметрии
Rotational symmetry	Aylanma simmetriya	Поворотная симметрия
Order of symmetry	Simmetriya tartibi	Порядок симметрии
Congruent	Teng (mos)	Равные (конгруэнтные)
Reflection	Simmetrik akslantirish	Отражение
Rotation	Burish	Поворот
Translation	Parallel ko‘chirish	Параллельный перенос

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A square has how many lines of symmetry?

- *2*
- *4*
- *6*
- *8*

A rectangle that is not a square:

- *1*
- *2*
- *4*
- *0*

An equilateral triangle has order:

- *1*
- *2*
- *3*
- *6*

The order of rotational symmetry is never:

- *0*
- *1*
- *2*
- *6*

Two shapes are congruent when:

- they look alike
- one fits exactly on the other
- they have the same area
- they are both regular

A shape and its enlargement are:

- congruent
- similar
- identical
- unrelated

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Lines of symmetry of a square

ANSWER *4*

2. Of a rectangle

ANSWER *2*

3. Of an equilateral triangle

ANSWER *3*

4. Of an isosceles triangle

ANSWER *1*

5. Of a scalene triangle

ANSWER *0*

6. Of a circle

ANSWER infinitely many

7. Order of a square

ANSWER *4*

Medium · the standard question

7 PROBLEMS

1. Order of a rectangle

ANSWER 2

2. Order of an equilateral triangle

ANSWER 3

3. Order of a regular hexagon

ANSWER 6

4. Order of a parallelogram

ANSWER 2

5. Lines of symmetry of a rhombus

ANSWER 2

6. Are two squares of side 5 congruent?

ANSWER Yes

7. Are a shape and its enlargement congruent?

ANSWER No — similar

Hard · stretch and reason

7 PROBLEMS

1. A regular octagon: lines and order

   ANSWER 8 and 8

2. A regular n-gon: lines and order

   ANSWER n and n

3. A shape with order 2 but no line of symmetry

   ANSWER The letter S, or a parallelogram

4. A shape with a line of symmetry but order 1

   ANSWER An isosceles triangle

5. The rotation angle for order 5

   ANSWER $72°$

6. Is a rotation of $360°$ counted?

   ANSWER Yes — it is the last of the order

7. Why is a rectangle's diagonal not a line of symmetry?

   ANSWER The folded corner does not land on a corner

07 Homework

Homework — 5 tasks

Fold to find lines; turn to find the order.

1. Find the lines of symmetry and the order for a square, a rectangle and a rhombus.
2. Find them for an equilateral, an isosceles and a scalene triangle.
3. Find the order of rotational symmetry of a regular pentagon and its rotation angle.
4. Write three capital letters with a line of symmetry and two with rotational symmetry.

5. Draw two congruent shapes in different positions and say which movement takes one to the other.

Recall — speed, distance and time

One formula, three arrangements, and the units that keep them honest.

LESSON 116 | 1 × 40 MIN | MATEMATIKA 6, TAKRORLASH | S7 12 RATES

LESSON CLOCK

10' | 12' | 12' | 6'

The formula	10 min
Rearranging	12 min
Units	12 min
Converting	6 min

LEARNING OBJECTIVES

- State the relation between speed, distance and time.
- Rearrange it to find any one of the three.
- Use consistent units.
- Convert between *km/h* and *m/s*.

TEXTBOOK

National	pp. 335–338
Cambridge	Stage 7, pp. 122–126
Workbook	Exercise 12.4

01 Explanation

The formula

$$s = vt$$

Distance equals speed times time — because a speed of *60* km/h means *60* km covered in each hour.

SPEED	TIME	DISTANCE
60 km/h	*3 h*	*180 km*
80 km/h	*2.5 h*	*200 km*
5 m/s	*12 s*	*60 m*
4 km/h	*45 min*	*3 km*

THE LAST ROW NEEDS THE TIME IN HOURS

45 minutes is 0.75 h, and $4 \cdot 0.75 = 3$ km. Using *45* with a speed in km/h gives *180* km, which is absurd — and the absurdity is the check.

Rearranging

$$s = vt \quad v = \frac{s}{t} \quad t = \frac{s}{v}$$

WANTED	GIVEN	WORKING	ANSWER
distance	$v = 60, t = 3$	$60 \cdot 3$	180 km
speed	$s = 180, t = 3$	$180 \div 3$	60 km/h
time	$s = 180, v = 60$	$180 \div 60$	3 h

ONE FORMULA, NOT THREE

$s = vt$ is all that has to be remembered; the other two are it rearranged. Writing the one you know and rearranging is safer than memorising a triangle.

Units

SPEED IN	DISTANCE IN	TIME IN
km/h	km	hours
m/s	m	seconds
m/min	m	minutes

TIME GIVEN	IN HOURS
30 min	0.5
45 min	0.75
20 min	$\frac{1}{3}$
1 h 30 min	1.5

THE UNITS OF THE ANSWER ARE BUILT FROM THE OTHERS

Kilometres divided by hours gives km/h automatically. Writing the units through the calculation makes the answer's unit appear by itself.

Converting

$$1\ km/h = \frac{1000}{3600}\ m/s = \frac{5}{18}\ m/s$$

KM/H	M/S
36	10
72	20
18	5
90	25

DIVIDE BY 3.6 TO GO FROM KM/H TO M/S

And multiply by 3.6 to come back. A sprinter at 10 m/s is running at 36 km/h, which is a useful pair of numbers to remember.

02 Worked examples

EXAMPLE 1 **A car travels at *80 km/h* for *2.5* hours. How far does it go?**

1. $s = vt$
2. $= 80 \cdot 2.5$
3. $= 200$ km.

ANSWER

$200\ km$

EXAMPLE 2 **A cyclist covers *3 km* in *45* minutes. Find the speed in km/h.**

1. 45 min = 0.75 h.

   Convert first.

2. $$v = \frac{3}{0.75}$$

3. = 4 km/h.

ANSWER

4 km/h

EXAMPLE 3 **Convert *72 km/h* to metres per second.**

1. $72 \div 3.6$

2. = 20 m/s.

3. Check: $20 \cdot 3.6 = 72$ ✓

ANSWER

20 m/s

03 Interactive model

Time a pupil walking a measured 20 m and compute the speed in both units; the numbers are their own and the units matter immediately.

Which of the three is wanted?

$v = 60$, $t = 3$: the distance is:

- *20*
- *63*
- *180*
- *180 km*

$s = 180$, $t = 3$: the speed is:

- *540*
- *60 km/h*
- *60 km*
- *3*

$s = 180$, $v = 60$: the time is:

- *3 h*
- *120*
- *240*
- *0.33*

45 minutes in hours:

- 0.45
- 0.75
- 4.5
- 45

20 minutes in hours:

- 0.2
- $\frac{1}{3}$
- 0.5
- 2

36 km/h in m/s:

- 10
- 36
- 100
- 3.6

20 m/s in km/h:

- *36*
- *72*
- *60*
- *7.2*

Using minutes with a speed in km/h gives:

- the right answer
- an absurd answer
- metres
- nothing

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Speed	Tezlik	Скорость
Distance	Masofa	Расстояние
Time	Vaqt	Время
Kilometres per hour	km/soat	км/ч
Metres per second	m/sek	м/с
To rearrange	O‘zgartirib yozish	Выразить
Consistent units	Mos birliklar	Согласованные единицы
Rate	Tezlik (me’yor)	Норма

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

s equals:

- vt
- $\frac{v}{t}$
- $\frac{t}{v}$
- $v + t$

v equals:

- st
- $\frac{s}{t}$
- $\frac{t}{s}$
- $s - t$

t equals:

- sv
- $\frac{v}{s}$
- $\frac{s}{v}$
- $s + v$

With a speed in km/h, the time must be in:

- minutes
- hours
- seconds
- any unit

72 km/h in m/s is:

- 20
- 26
- 7.2
- 259

A sprinter at 10 m/s runs at:

- 10 km/h
- 36 km/h
- 60 km/h
- 3.6 km/h

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $v = 60$, $t = 3$: the distance

   ANSWER 180 km

2. $v = 80$, $t = 2.5$

   ANSWER 200 km

3. $v = 5$ m/s, $t = 12$ s

   ANSWER 60 m

4. $s = 180$, $t = 3$: the speed

   ANSWER 60 km/h

5. $s = 180$, $v = 60$: the time

   ANSWER 3 h

6. 30 min in hours

   ANSWER 0.5

7. 45 min in hours

   ANSWER 0.75

Medium · the standard question

7 PROBLEMS

1. 3 km in 45 min: the speed

   ANSWER 4 km/h

2. 36 km/h in m/s

   ANSWER 10

3. 72 km/h in m/s

   ANSWER 20

4. 25 m/s in km/h

   ANSWER 90

5. $v = 4$ km/h, $t = 45$ min

   ANSWER 3 km

6. $s = 150$ km, $v = 60$ km/h

   ANSWER 2.5 h

7. 20 min in hours

   ANSWER $\frac{1}{3}$

Hard · stretch and reason

7 PROBLEMS

1. A train covers 210 km in $2\ h\ 30\ min$

   ANSWER $84\ km/h$

2. A walker at 5 km/h covers 4 km in

   ANSWER 48 min

3. A runner at 8 m/s in 90 s

   ANSWER $720\ m$

4. Convert 54 km/h to m/s

   ANSWER 15

5. Convert 12 m/s to km/h

   ANSWER 43.2

6. A car at 90 km/h covers 1 km in

   ANSWER 40 s

7. Why must units match?

   ANSWER The formula divides one by the other, so they must belong together

07 Homework

Homework — 5 tasks

Convert the time to hours before using a speed in km/h.

1. A bus travels at 70 km/h for 4 hours. Find the distance.
2. A car covers 240 km in 3 hours. Find its speed.
3. How long does 150 km take at 50 km/h?
4. A cyclist covers 9 km in 40 minutes. Find the speed in km/h.
5. Convert 108 km/h to m/s and 15 m/s to km/h.

Speed

What a speed measures, how it is written, and how fast things actually go.

LESSON 117–119 | 3 × 40 MIN | MATEMATIKA 6, §24 | S7 12 RATES

LESSON CLOCK

25' | 30' | 35' | 25' | 5

What a speed is	25 min
Finding a speed	30 min
Comparing speeds	35 min
Is it reasonable?	25 min
Homework	5 min

LEARNING OBJECTIVES

- Define speed as distance per unit of time.
- Compare speeds given in different units.
- Read a speed from a table or a description.
- Judge whether a speed is reasonable.

TEXTBOOK

National	pp. 339–348
Cambridge	Stage 7, pp. 122–126
Workbook	Exercise 12.5

01 Explanation

What a speed is

$$v = \frac{s}{t}$$

A speed of *60* km/h means *60* km in every hour — and therefore *30* km in half an hour, *1* km per minute, and *16.7* m each second.

SPEED	IN ONE HOUR	IN ONE MINUTE
60 km/h	*60 km*	*1 km*
90 km/h	*90 km*	*1.5 km*
5 km/h	*5 km*	*83 m*

A SPEED IS A RATE: SOMETHING PER SOMETHING

Price per kilogram, pupils per class, kilometres per hour — all are rates, and all are found by dividing one quantity by another.

Finding a speed

JOURNEY	WORKING	SPEED
180 km in 3 h	$180 \div 3$	60 km/h
240 km in 4 h	$240 \div 4$	60 km/h
100 m in 12.5 s	$100 \div 12.5$	8 m/s
9 km in 40 min	$9 \div \frac{2}{3}$	13.5 km/h

DIVIDE THE DISTANCE BY THE TIME, NEVER THE OTHER WAY ROUND

$3 \div 180$ gives 0.017, which is hours per kilometre — a real quantity, but not the one asked for. The units tell you which division you have done.

Comparing speeds

Two speeds can only be compared in the same units.

A	B	IN ONE UNIT	FASTER
72 km/h	18 m/s	20 and 18 m/s	A
54 km/h	16 m/s	15 and 16 m/s	B
4 km/h	70 m/min	66.7 and 70 m/min	B

CONVERT FIRST, COMPARE SECOND

The larger number is not the faster speed unless the units match. This is the same rule as comparing 50 cm with 2 m.

Is it reasonable?

MOVER	TYPICAL SPEED
a walker	*5 km/h*
a cyclist	*15–20 km/h*
a car in a town	*40–60 km/h*
a train	*100–160 km/h*
an aeroplane	*800–900 km/h*
sound in air	*340 m/s*

AN ANSWER OF *600* KM/H FOR A CYCLIST IS A SIGNAL, NOT A RESULT

Comparing an answer with the table catches unit errors and misplaced decimal points before the working is even reread.

02 Worked examples

EXAMPLE 1 **A runner covers *100* m in *12.5* seconds. Find the speed.**

1. $v = \frac{s}{t} = \frac{100}{12.5}$
2. $= 8$ m/s.
3. In km/h that is *28.8* — reasonable for a sprinter.

ANSWER

8 m/s

EXAMPLE 2 **Which is faster, *72* km/h or *18* m/s?**

1. $72 \div 3.6 = 20$ m/s.
2. $20 > 18$.
3. The first is faster.

ANSWER

72 km/h

EXAMPLE 3 **A cyclist covers *9* km in *40* minutes. Find the speed in km/h.**

1. 40 min = $\frac{2}{3}$ h.
2. $9 \div \frac{2}{3} = 9 \cdot \frac{3}{2}$
3. $= 13.5$ km/h.
   Reasonable for a cyclist ✓

ANSWER

$13.5\ km/h$

03 Interactive model

Have the class estimate the speed of a walk down the corridor, then measure it; being within 20% is a good result and makes the units real.

Speeds and their units

180 km in 3 h:

- 60 km/h
- 540 km/h
- 0.017
- 183

100 m in 12.5 s:

- 8 m/s
- 12.5 m/s
- 0.125 m/s
- 1250 m/s

9 km in 40 min:

- 13.5 km/h
- 0.225 km/h
- 360 km/h
- 22.5 km/h

Which is faster, 72 km/h or 18 m/s?

- 72 km/h
- 18 m/s
- equal
- cannot say

Which is faster, 54 km/h or 16 m/s?

- 54 km/h
- 16 m/s
- equal
- cannot say

A walker’s speed is about:

- 1 km/h
- 5 km/h
- 20 km/h
- 60 km/h

Sound travels at about:

- 34 m/s
- 340 m/s
- 3400 m/s
- 34 km/h

600 km/h for a cyclist means:

- a fast cyclist
- an error somewhere
- a downhill
- a record

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Speed	Tezlik	Скорость
Per hour	Bir soatda	В час
Per second	Bir sekundda	В секунду
Uniform speed	Bir tekis tezlik	Равномерная скорость
To compare	Taqqoslash	Сравнить
Reasonable	Maqbul	Разумный
Rate	Me'yor	Норма
Unit of speed	Tezlik birligi	Единица скорости

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Speed is:

- distance times time
- distance over time
- time over distance
- distance plus time

240 km in *4* h is:

- *60 km/h*
- *960 km/h*
- *0.017 km/h*
- *244 km/h*

To compare km/h with m/s you:

- add them
- convert one
- take the larger number
- divide them

4 km/h in metres per minute is about:

- *40*
- *67*
- *240*
- *4*

A train’s speed is about:

- *15* km/h
- *60* km/h
- *120* km/h
- *800* km/h

Hours per kilometre is:

- a speed
- the reciprocal of a speed
- nonsense
- the same thing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *180* km in *3* h

ANSWER *60 km/h*

2. *240* km in *4* h

ANSWER *60 km/h*

3. *100* m in *12.5* s

ANSWER *8 m/s*

4. *60* km in *1* h

ANSWER *60 km/h*

5. *20* m in *4* s

ANSWER *5 m/s*

6. A walker’s speed

ANSWER about *5 km/h*

7. Sound in air

ANSWER *340 m/s*

Medium · the standard question

7 PROBLEMS

1. 9 km in 40 min

ANSWER 13.5 km/h

2. 3 km in 45 min

ANSWER 4 km/h

3. Faster: 72 km/h or 18 m/s?

ANSWER 72 km/h

4. Faster: 54 km/h or 16 m/s?

ANSWER 16 m/s

5. 4 km/h in m/min

ANSWER 66.7

6. 90 km/h in m/s

ANSWER 25

7. A cyclist at 600 km/h means

ANSWER an error somewhere

Hard · stretch and reason

7 PROBLEMS

1. A train covers *420* km in *3 h 30 min*

   ANSWER *120 km/h*

2. An aeroplane covers *2 700* km in *3* h

   ANSWER *900 km/h*

3. Light takes *8* minutes from the Sun, *150* million km away: the speed

   ANSWER About *312 500* km/s

4. A *400* m lap in *50* s

   ANSWER *8 m/s*, or *28.8 km/h*

5. Which is faster: *1* km per minute or *60* km/h?

   ANSWER Equal

6. A snail at *1* cm/s in km/h

   ANSWER *0.036*

7. Why compare an answer with a table of typical speeds?

   ANSWER It catches unit errors at once

07 Homework

Homework — 5 tasks

Check every answer against a typical speed for that kind of mover.

1. A car covers *330* km in *5* hours. Find its speed.
2. A sprinter runs *200* m in *25* seconds. Find the speed in m/s and in km/h.
3. A walker covers *6* km in *1 h 20 min*. Find the speed.
4. Which is faster: *90* km/h or *26* m/s?
5. Write down a typical speed for a walker, a car and an aeroplane.

Finding speed, distance and time

Three questions from one formula, with the units chosen before the arithmetic.

LESSON 120–122 | 3 × 40 MIN | MATEMATIKA 6, §25 | S7 12 RATES

LESSON CLOCK

25' | 30' | 35' | 25' | 5

Finding the distance	25 min
Finding the speed	30 min
Finding the time	35 min
Hours and minutes	25 min
Homework	5 min

LEARNING OBJECTIVES

- Find any one of s, v, t from the other two.
- Convert times between hours, minutes and seconds.
- Give an answer in hours and minutes where that is natural.
- Check every answer against the sense of the situation.

TEXTBOOK

National	pp. 349–358
Cambridge	Stage 7, pp. 122–126
Workbook	Exercise 12.6

01 Explanation

Finding the distance

$$s = vt$$

SPEED	TIME	DISTANCE
65 km/h	*4 h*	*260 km*
50 km/h	*2 h 30 min*	*125 km*
12 km/h	*20 min*	*4 km*
15 m/s	*40 s*	*600 m*

WRITE THE TIME IN THE UNIT THE SPEED USES

2 h *30* min is *2.5* h; *20* min is $\frac{1}{3}$ h. Doing that conversion first turns every problem into one multiplication.

Finding the speed

$$v = \frac{s}{t}$$

DISTANCE	TIME	SPEED
260 km	*4 h*	*65 km/h*
125 km	*2.5 h*	*50 km/h*
4 km	*20 min*	*12 km/h*
600 m	*40 s*	*15 m/s*

A SPEED IN KM/H NEEDS THE TIME IN HOURS

4 km in *20* minutes is *12* km/h, not *0.2* km/h. Dividing by *20* answers a different question.

Finding the time

$$t = \frac{s}{v}$$

DISTANCE	SPEED	TIME	IN HOURS AND MINUTES
260 km	*65 km/h*	*4*	*4 h*
150 km	*60 km/h*	*2.5*	*2 h 30 min*
90 km	*80 km/h*	*1.125*	*1 h 7.5 min*
7 km	*5 km/h*	*1.4*	*1 h 24 min*

TURNING A DECIMAL HOUR INTO MINUTES

Multiply the decimal part by *60*: *0.4* h is *24* min, *0.25* h is *15* min. Writing *1.4* h as “one hour forty minutes” is a real and common error.

Hours and minutes

PROBLEM	WORKING	ANSWER
a bus leaves at *8:00* and travels *150* km at *60* km/h	*2.5 h*	arrives *10:30*
a train leaves at *7:20* and takes *1 h 50 min*	*7:20 + 1:50*	arrives *9:10*
a journey of *7* km at *5* km/h	*1.4 h*	*1 h 24 min*
a walk from *9:15* to *11:00* covering *7* km	*1.75 h*	*4 km/h*

ADDING TIMES IS NOT DECIMAL ADDITION

7:20 plus *1* h *50* min is *9:10*, because *70* minutes is an hour and ten. Times carry at *60*, not at *100*.

02 Worked examples

EXAMPLE 1 **A cyclist rides at *12* km/h for *20* minutes. How far?**

1. *20* min = $\frac{1}{3}$ h.
2. $s = 12 \cdot \frac{1}{3}$
3. $= 4$ km.

ANSWER

4 km

EXAMPLE 2 **A journey of *7* km is walked at *5* km/h. How long does it take?**

1. $t = \dfrac{7}{5} = 1.4$ h.

2. $0.4 \cdot 60 = 24$ min.

3. 1 h 24 min.

   Not one hour forty.

ANSWER

1 h 24 min

EXAMPLE 3 **A bus leaves at *8:00* and covers *150* km at *60* km/h. When does it arrive?**

1. $t = \dfrac{150}{60} = 2.5$ h.

2. 2.5 h is 2 h 30 min.

3. $8{:}00 + 2{:}30 = 10{:}30$.

ANSWER

10:30

03 Interactive model

Use a real bus timetable: pupils compute the average speed between two stops and compare it with the road speed limit, which is a genuinely interesting comparison.

Which one is wanted?

65 km/h for *4* h:

- *16.25* km
- *260* km
- *69* km
- *260* km/h

50 km/h for *2 h 30 min*:

- *100*
- *125*
- *150*
- *250*

12 km/h for *20* min:

- *240*
- *4*
- *0.6*
- *36*

4 km in 20 min:

- 0.2 km/h
- 12 km/h
- 80 km/h
- 4 km/h

150 km at 60 km/h takes:

- 2.5 h
- 90 h
- 2.3 h
- 9000 h

1.4 h in hours and minutes:

- $1\ h\ 40\ min$
- $1\ h\ 24\ min$
- $1\ h\ 4\ min$
- $14\ min$

$7{:}20$ plus $1\ h\ 50\ min$:

- $8{:}70$
- $9{:}10$
- $8{:}10$
- $9{:}70$

7 km from $9{:}15$ to $11{:}00$:

- 4 km/h
- 7 km/h
- 1.75 km/h
- 12.25 km/h

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	OʻZBEKCHA	РУССКИЙ
Formula	Formula	Формула
To rearrange	Oʻzgartirib yozish	Выразить
Hours and minutes	Soat va daqiqa	Часы и минуты
Decimal hours	Oʻnli soat	Часы в десятичной форме
Arrival time	Yetib borish vaqti	Время прибытия
Departure	Joʻnash	Отправление
Journey	Sayohat	Поездка
Sense check	Mantiqiy tekshirish	Проверка по смыслу

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

To find the distance you:

- divide
- multiply
- subtract
- add

To find the time you:

- $s \cdot v$
- $s \div v$
- $v \div s$
- $s + v$

20 minutes in hours is:

- 0.2
- $\frac{1}{3}$
- 0.5
- 2

0.25 h in minutes is:

- 25
- 15
- 20
- 45

90 km at 80 km/h takes:

- 1.125 h
- 1.9 h
- 0.9 h
- 7200 h

A bus leaving at $8{:}00$ and taking 2 h 30 min arrives at:

- $10{:}00$
- $10{:}30$
- $10{:}50$
- $11{:}30$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *65* km/h for *4* h

   ANSWER *260 km*

2. *50* km/h for *2.5* h

   ANSWER *125 km*

3. *15* m/s for *40* s

   ANSWER *600 m*

4. *260* km in *4* h

   ANSWER *65 km/h*

5. *125* km in *2.5* h

   ANSWER *50 km/h*

6. *260* km at *65* km/h

   ANSWER *4 h*

7. *150* km at *60* km/h

   ANSWER *2.5 h*

Medium · the standard question

7 PROBLEMS

1. *12* km/h for *20* min

ANSWER *4 km*

2. *4* km in *20* min

ANSWER *12 km/h*

3. *7* km at *5* km/h

ANSWER *1 h 24 min*

4. *90* km at *80* km/h

ANSWER *1 h 7.5 min*

5. A bus leaves *8:00*, *150* km at *60* km/h: arrival

ANSWER *10:30*

6. *7:20* plus *1 h 50 min*

ANSWER *9:10*

7. *7* km from *9:15* to *11:00*

ANSWER *4 km/h*

Hard · stretch and reason

7 PROBLEMS

1. A train leaves $6{:}45$ and takes $3\ h\ 40\ min$: arrival

   ANSWER $10{:}25$

2. A journey of 340 km at 85 km/h

   ANSWER $4\ h$

3. A journey of 200 km taking $2\ h\ 40\ min$

   ANSWER $75\ km/h$

4. How far in $1\ h\ 45\ min$ at 60 km/h?

   ANSWER $105\ km$

5. A car covering 45 km in 30 min

   ANSWER $90\ km/h$

6. A walk of 11 km at 4 km/h

   ANSWER $2\ h\ 45\ min$

7. Why is 1.4 h not 1 h 40 min?

   ANSWER The decimal part is 0.4 of an hour, which is 24 minutes

07 Homework

Homework — 5 tasks

Convert to hours before dividing, and back to minutes at the end.

1. A car travels $3\ h\ 15\ min$ at 80 km/h. Find the distance.
2. A journey of 96 km takes $1\ h\ 20\ min$. Find the speed.
3. How long does 220 km take at 55 km/h?
4. A bus leaves at $7{:}40$ and takes $2\ h\ 35\ min$. When does it arrive?
5. A walker covers 9 km at 4 km/h. Give the time in hours and minutes.

Word problems on motion with two speeds

A journey in two parts, and the current or wind that changes the speed.

LESSON 123–125 | 3 × 40 MIN | MATEMATIKA 6, §26 | S7 12 RATES

LESSON CLOCK

25' | 30' | 35' | 25' | 5

Two stages	25 min
Setting out a table	30 min
Downstream and upstream	35 min
Harder problems	25 min
Homework	5 min

LEARNING OBJECTIVES

- Handle a journey made in two stages at different speeds.
- Use downstream and upstream speeds.
- Set out the working in a table.
- Check each stage separately as well as the whole.

TEXTBOOK

National	pp. 359–368
Cambridge	Stage 7, pp. 122–126
Workbook	Exercise 12.7

01 Explanation

Two stages

Work each stage separately, then add the distances and add the times. Never add the speeds.

STAGE	SPEED	TIME	DISTANCE
first	*60 km/h*	*2 h*	*120 km*
second	*80 km/h*	*3 h*	*240 km*
total	—	*5 h*	*360 km*

SPEEDS ARE NEVER ADDED

The two stages above do not make a journey at *140* km/h. Distances add, times add; a speed is a ratio between them and has to be worked out at the end.

Setting out a table

PROBLEM	TABLE	ANSWER
2 h at 60 then 3 h at 80	$120 + 240$	360 km
90 km at 45 then 60 km at 60	$2 + 1$	3 h
1.5 h at 50 then 30 min at 70	$75 + 35$	110 km

THREE COLUMNS, ONE ROW PER STAGE

Speed, time, distance — with a totals row underneath. The table does the organising, and the arithmetic becomes obvious.

Downstream and upstream

$$\text{downstream} = v + u \quad \text{upstream} = v - u$$

with v the speed of the boat in still water and u the speed of the current.

BOAT	CURRENT	DOWNSTREAM	UPSTREAM
16	4	20	12
18	3	21	15
10	2	12	8

THE SAME IDEA WORKS FOR WIND

An aeroplane with a tailwind flies at $v + u$ and against it at $v - u$. The two speeds add and subtract exactly as with a river.

Harder problems

PROBLEM	WORKING	ANSWER
a boat at 16 in still water, current 4: 60 km downstream	$60 \div 20$	$3\ h$
the same boat coming back	$60 \div 12$	$5\ h$
the whole trip	$3 + 5$	$8\ h$
a boat covering 20 km down in 1 h and 12 km up in 1 h	$(20 + 12) \div 2$	boat 16, current 4

HALF THE SUM AND HALF THE DIFFERENCE

Given the two speeds, the boat's own speed is their average and the current is half their difference. Grade 7 will write that as a system of two equations.

02 Worked examples

EXAMPLE 1 **A car drives 2 hours at 60 km/h and then 3 hours at 80 km/h. Find the total distance.**

1. First stage: 120 km.
2. Second stage: 240 km.
3. $120 + 240 = 360$ km.
   The speeds are not added.

ANSWER

$360\ km$

EXAMPLE 2

A boat travels at *16* km/h in still water. The current is *4* km/h. How long to go *60* km downstream and back?

1. Downstream: 20 km/h, so 3 h.
2. Upstream: 12 km/h, so 5 h.
3. $3 + 5 = 8$ h.

   Not $60 \div 16$ doubled.

ANSWER

8 hours

EXAMPLE 3

A boat covers *20* km downstream and *12* km upstream, each in one hour. Find the speed of the boat and of the current.

1. Downstream speed 20, upstream 12.
2. Boat: $(20 + 12) \div 2 = 16$ km/h.
3. Current: $(20 - 12) \div 2 = 4$ km/h.

   Check: $16 + 4 = 20$ ✓

ANSWER

Boat 16, current 4

03 Interactive model

Draw the table on the board and fill it in with the class before any arithmetic; the organisation is what makes these problems easy.

Two stages, two speeds

2 h at 60 then 3 h at 80: the distance is:

- 140
- 360
- 280
- 700

The total time is:

- 5 h
- 6 h
- 2.5 h
- 140 h

Adding the two speeds gives:

- the average speed
- nothing useful
- the total
- the distance

A boat at 16 with a 4 current goes downstream at:

- 12
- 16
- 20
- 64

And upstream at:

- 12
- 16
- 20
- 4

60 km downstream at 20 takes:

- 3 h
- 5 h
- 4 h
- 12 h

And back upstream:

- 3 h
- 5 h
- 4 h
- 15 h

Down 20, up 12: the current is:

- 4
- 8
- 16
- 32

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Stage of a journey	Yo‘l bosqichi	Участок пути
Downstream	Oqim bo‘ylab	По течению
Upstream	Oqimga qarshi	Против течения
Current	Oqim	Течение
Still water	Turg‘un suv	Стоячая вода
Total distance	Umumiy masofa	Общее расстояние
Total time	Umumiy vaqt	Общее время
Table of working	Jadval	Таблица

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

In a two-stage journey you add:

- the speeds
- the distances and the times
- the speeds and times
- nothing

Downstream speed is:

- $v - u$
- $v + u$
- vu
- v

Upstream speed is:

- $v - u$
- $v + u$
- $\frac{\mathrm{v}}{\mathrm{u}}$
- u

A boat at *18* with a *3* current goes upstream at:

- *15*
- *21*
- *18*
- *6*

Given both speeds, the boat's own speed is:

- their sum
- their average
- their difference
- the larger

And the current is:

- their average
- half their difference
- their difference
- the smaller

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. 2 h at 60 then 3 h at 80: the distance

   ANSWER *360 km*

2. And the total time

   ANSWER *5 h*

3. A boat at 16, current 4: downstream

   ANSWER *20 km/h*

4. And upstream

   ANSWER *12 km/h*

5. A boat at 18, current 3: downstream

   ANSWER *21 km/h*

6. And upstream

   ANSWER *15 km/h*

7. 60 km at 20 km/h

   ANSWER *3 h*

Medium · the standard question

7 PROBLEMS

1. *90* km at *45* then *60* km at *60*: the time

ANSWER *3 h*

2. *1.5* h at *50* then *30* min at *70*

ANSWER *110 km*

3. A boat, *60* km down at *20* and back at *12*

ANSWER *8 h*

4. Down *20*, up *12*: the boat

ANSWER *16 km/h*

5. And the current

ANSWER *4 km/h*

6. Down *24*, up *16*: the boat and current

ANSWER *20* and *4*

7. A plane at *500* with a *50* tailwind

ANSWER *550 km/h*

Hard · stretch and reason

7 PROBLEMS

1. A boat at 12 in still water, current 3: 45 km each way

   ANSWER 8 h

2. A car does 120 km at 60 and 120 km at 40: the total time

   ANSWER 5 h

3. A plane flies $1\ 800$ km out at 600 and back at 450

   ANSWER 7 h

4. A boat takes 2 h down and 3 h up for the same 36 km

   ANSWER Boat 15, current 3

5. A journey of 300 km, half at 50 and half at 75

   ANSWER 5 h

6. Why can the two speeds not be added?

   ANSWER A speed is a ratio, not an amount

7. A boat's downstream speed is twice its upstream speed and the current is 3: the boat

   ANSWER 9 km/h

07 Homework

Homework — 5 tasks

Draw the three-column table before writing any arithmetic.

1. A car drives 3 h at 70 km/h then 2 h at 90. Find the distance and the total time.
2. A boat travels at 14 km/h in still water with a current of 2. Find its speed each way.
3. How long does that boat take to go 48 km downstream and back?
4. A boat covers 30 km downstream in 1.5 h and 30 km upstream in 3 h. Find the boat and the current.

5. A journey of 180 km is half at 45 km/h and half at 90. Find the total time.

Average speed

Total distance over total time — and why it is not the average of the two speeds.

LESSON 126–128 | 3 × 40 MIN | MATEMATIKA 6, §27 | S7 12 RATES

LESSON CLOCK

25' | 30' | 35' | 25' | 5

The definition	25 min
Journeys in stages	30 min
Why not the mean of the speeds	35 min
Stops	25 min
Homework	5 min

LEARNING OBJECTIVES

- Define average speed as total distance over total time.
- Compute it for a journey in stages.
- Explain why the mean of the speeds is usually wrong.
- Include stops in the total time when the question requires it.

TEXTBOOK

National	pp. 369–378
Cambridge	Stage 7, pp. 122–126
Workbook	Exercise 12.8

01 Explanation

The definition

$$\text{average speed} = \frac{\text{total distance}}{\text{total time}}$$

JOURNEY	TOTAL DISTANCE	TOTAL TIME	AVERAGE SPEED
2 h at 60, 3 h at 80	360 km	5 h	72 km/h
120 km at 60, 120 km at 40	240 km	5 h	48 km/h
90 km at 45, 60 km at 60	150 km	3 h	50 km/h

ONE DIVISION, AT THE VERY END

Find the whole distance, find the whole time, divide once. Every average-speed problem is that sentence.

Journeys in stages

STAGE	SPEED	TIME	DISTANCE
first	*45 km/h*	*2 h*	*90 km*
second	*60 km/h*	*1 h*	*60 km*
total	*50 km/h*	*3 h*	*150 km*

THE AVERAGE LIES BETWEEN THE TWO SPEEDS

Here between *45* and *60*, and nearer the one that took longer. If your answer falls outside the two speeds, something has gone wrong.

Why not the mean of the speeds

A car goes *120* km at *60* km/h and comes back at *40* km/h. The mean of the speeds is *50*; the average speed is not.

STAGE	DISTANCE	SPEED	TIME
out	*120*	*60*	*2 h*
back	*120*	*40*	*3 h*
total	*240*	—	*5 h*

$$\text{average} = \frac{240}{5} = 48 \text{ km/h, not } 50$$

MORE TIME IS SPENT AT THE SLOWER SPEED

That is why the average is pulled below the mean. Only when the two *times* are equal does the mean of the speeds happen to be right.

Stops

QUESTION	TIME USED	ANSWER
150 km driven in *2* h with a *30* min stop: average driving speed	*2 h*	*75 km/h*
the same journey: average speed for the whole trip	*2.5 h*	*60 km/h*

READ WHETHER THE STOP IS INCLUDED

"Average speed for the journey" includes the stop; "average driving speed" does not. The same numbers give *60* or *75* depending on which was asked.

02 Worked examples

EXAMPLE 1 **A car drives *2* h at *60* km/h and *3* h at *80*. Find the average speed.**

1. Distance: $120 + 240 = 360$ km.
2. Time: *5* h.
3. $360 \div 5 = 72$ km/h.
   Between *60* and *80* ✓

ANSWER

72 km/h

EXAMPLE 2 **A car goes *120* km at *60* km/h and returns at *40*. Find the average speed for the whole trip.**

1. Out: *2* h. Back: *3* h.
2. Total: *240* km in *5* h.
3. *48* km/h.
   Not *50* — more time at the slower speed.

ANSWER

48 km/h

EXAMPLE 3 **A bus covers *150* km in *2* hours of driving, with a *30*-minute stop. Find both averages.**

1. Driving: $150 \div 2 = 75$ km/h.
2. Whole trip: 2.5 h.
3. $150 \div 2.5 = 60$ km/h.

ANSWER

75 and 60 km/h

03 Interactive model

Ask for the average of 60 and 40 first; almost everyone says 50, and the table that follows is then genuinely surprising.

Total over total

Average speed is:

- the mean of the speeds
- total distance over total time
- the largest speed
- the smallest

2 h at 60 and 3 h at 80 gives:

- 70
- 72
- 140
- 68

120 km at 60 and back at 40 gives:

- 50
- 48
- 52
- 100

The answer is below 50 because:

- the distances differ
- more time is spent slower
- of rounding
- of the stop

90 km at 45 and 60 km at 60 gives:

- 50
- 52.5
- 55
- 105

An average speed must lie:

- above both speeds
- between them
- below both
- anywhere

150 km in 2 h driving plus a 30 min stop, whole trip:

- 75
- 60
- 50
- 120

The mean of the speeds is right only when:

- the distances are equal
- the times are equal
- always
- never

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Average speed	O‘rtacha tezlik	Средняя скорость
Total distance	Umumiy masofa	Общее расстояние
Total time	Umumiy vaqt	Общее время
Stop	To‘xtash	Остановка
Stage	Bosqich	Участок
Mean of speeds	Tezliklarning o‘rtachasi	Среднее скоростей
Weighted	Vaznli	Взвешенный
Journey	Sayohat	Поездка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Average speed uses:

- the mean of the speeds
- total distance and total time
- the first speed
- the last speed

240 km in 5 h is:

- 48
- 50
- 1200
- 4.8

The average of a journey out at 60 and back at 40 is:

- exactly 50
- less than 50
- more than 50
- 100

An answer of *90* for stages at *45* and *60* is:

- possible
- impossible
- exact
- the mean

“Average driving speed” excludes:

- the distance
- the stops
- the return
- nothing

Which weights the average?

- the distances
- the times
- the speeds
- nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. 360 km in 5 h

   ANSWER 72 km/h

2. 240 km in 5 h

   ANSWER 48 km/h

3. 150 km in 3 h

   ANSWER 50 km/h

4. 150 km in 2 h

   ANSWER 75 km/h

5. 150 km in 2.5 h

   ANSWER 60 km/h

6. Average speed is total distance over

   ANSWER total time

7. An average must lie

   ANSWER between the two speeds

Medium · the standard question

7 PROBLEMS

1. 2 h at 60 and 3 h at 80

ANSWER 72 km/h

2. 120 km at 60 and back at 40

ANSWER 48 km/h

3. 90 km at 45 and 60 km at 60

ANSWER 50 km/h

4. 1.5 h at 50 and 0.5 h at 70

ANSWER 55 km/h

5. Why is the return trip average below 50?

ANSWER More time is spent at the slower speed

6. With a 30 min stop in a 2 h drive of 150 km

ANSWER 60 km/h

7. Without the stop

ANSWER 75 km/h

Hard · stretch and reason

7 PROBLEMS

1. Out at *80* and back at *20* over *80* km each way

   ANSWER *32 km/h*

2. Out at *30* and back at *30*

   ANSWER *30 km/h*

3. *100* km at *50* then *100* km at *100*

   ANSWER *66.7 km/h*

4. A journey of *3* equal parts at *30*, *60* and *60*

   ANSWER *45 km/h*

5. When does the mean of two speeds give the right answer?

   ANSWER When the two times are equal

6. A walk of *2* km at *4* km/h and *2* km at *6*

   ANSWER *4.8 km/h*

7. Is the average speed ever above both speeds?

   ANSWER No

07 Homework

Homework — 5 tasks

Find the total distance and the total time before dividing once.

1. A car drives *3* h at *50* km/h and *2* h at *75*. Find the average speed.
2. A journey of *60* km out at *30* km/h and back at *60*. Find the average speed.
3. Explain in one sentence why the answer to task 2 is not *45*.
4. A bus covers *180* km in *3* hours of driving with a *45*-minute stop. Find both averages.
5. A walk of *3* km at *6* km/h and *3* km at *3*. Find the average speed.

Word problems on the motion of two bodies

Towards each other, in the same direction, and the closing speed that decides both.

LESSON 129–131 | 3 × 40 MIN | MATEMATIKA 6, §28 | S7 12 RATES

LESSON CLOCK

25' | 30' | 35' | 25' | 5

Towards each other	25 min
The place of meeting	30 min
Overtaking	35 min
Head starts	25 min
Homework	5 min

LEARNING OBJECTIVES

- Use the closing speed for two bodies approaching.
- Use the difference of speeds for one overtaking another.
- Find the time and place of meeting.
- Draw a diagram before writing any arithmetic.

TEXTBOOK

National	pp. 379–388
Cambridge	Stage 7, pp. 122–126
Workbook	Exercise 12.9

01 Explanation

Towards each other

Two bodies approaching close the gap at the **sum** of their speeds.

$$\text{time to meet} = \frac{\text{distance apart}}{v_1 + v_2}$$

APART	SPEEDS	CLOSING SPEED	TIME TO MEET
300 km	*60* and *40*	*100 km/h*	*3 h*
240 km	*70* and *50*	*120 km/h*	*2 h*
12 km	*5* and *3*	*8 km/h*	*1.5 h*

HERE THE SPEEDS REALLY DO ADD

Not because speeds can be added in general, but because each hour the gap shrinks by both distances at once. The sum is the speed of the *gap*, not of either body.

The place of meeting

STEP	FOR *300* KM APART AT *60* AND *40*
closing speed	*100 km/h*
time to meet	*3 h*
first travels	*180 km*
second travels	*120 km*
check	*180 + 120 = 300* ✓

THE FASTER ONE COVERS MORE OF THE GAP

In the same time, distances are in the ratio of the speeds — here *60 : 40 = 3 : 2*, so the *300* km splits as *180* and *120*.

Overtaking

Two bodies going the same way close the gap at the **difference** of their speeds.

$$\text{time to catch} = \frac{\text{head start}}{v_1 - v_2}$$

HEAD START	SPEEDS	DIFFERENCE	TIME TO CATCH
30 km	*80* and *60*	*20 km/h*	*1.5 h*
12 km	*15* and *12*	*3 km/h*	*4 h*
100 m	*8* and *6* m/s	*2 m/s*	*50 s*

IF THE SPEEDS ARE EQUAL, THE GAP NEVER CLOSES

A difference of zero means no catching up, however long the chase. The formula says so by dividing by zero, which is the mathematics agreeing with common sense.

Head starts

PROBLEM	HEAD START	ANSWER
a cyclist at *12* leaves *1* h before a car at *60*	*12 km*	caught after *0.25* h
a walker at *5* leaves *2* h before a cyclist at *15*	*10 km*	caught after *1* h
a bus at *60* leaves *30* min before a car at *80*	*30 km*	caught after *1.5* h

TURN THE TIME HEAD START INTO A DISTANCE HEAD START

An hour's start at *12* km/h is a *12* km gap. Once the head start is a distance, the chase formula applies directly.

02 Worked examples

EXAMPLE 1

Two cars *300* km apart drive towards each other at *60* and *40* km/h. When do they meet, and where?

1. Closing speed: *100* km/h.
2. Time: $300 \div 100 = 3$ h.
3. First: *180* km; second: *120* km.
   Check: they total *300* ✓

ANSWER

After *3* h, *180* km from the first

EXAMPLE 2 **A bus leaves at 60 km/h. Half an hour later a car follows at 80. When does it catch up?**

1. Head start: $60 \cdot 0.5 = 30$ km.
2. Difference of speeds: 20 km/h.
3. $30 \div 20 = 1.5$ h after the car starts.

ANSWER

After 1.5 hours

EXAMPLE 3 **Two walkers 12 km apart walk towards each other at 5 and 3 km/h. When do they meet?**

1. Closing speed: 8 km/h.
2. $12 \div 8 = 1.5$ h.
3. They meet 7.5 km from the first.

   $5 \cdot 1.5$.

ANSWER

After 1.5 hours

03 Interactive model

Two pupils walking towards each other along the corridor, timed, is a five-minute experiment that makes the closing speed obvious.

Sum or difference?

Two bodies approaching close the gap at:

- the sum of the speeds
- the difference
- the average
- the faster speed

One overtaking another closes at:

- the sum
- the difference
- the average
- the slower speed

300 km apart at 60 and 40: they meet after:

- 3 h
- 5 h
- 7.5 h
- 30 h

The first car has then travelled:

- 120 km
- 150 km
- 180 km
- 300 km

A 30 km head start with speeds 80 and 60: caught after:

- 0.375 h
- 1.5 h
- 2 h
- 3 h

A bus at 60 leaving 30 min early has a head start of:

- 30 km
- 60 km
- 0.5 km
- 120 km

If both speeds are equal, the chaser:

- catches up slowly
- never catches up
- catches up at once
- falls behind

Two walkers 12 km apart at 5 and 3 meet after:

- 1.5 h
- 2 h
- 4 h
- 6 h

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
To approach	Yaqinlashmoq	Сближаться
To meet	Uchrashmoq	Встретиться
Closing speed	Yaqinlashish tezligi	Скорость сближения
To overtake	Quvib yetmoq	Догонять
Same direction	Bir yo‘nalishda	В одном направлении
Opposite directions	Qarama-qarshi	Навстречу
Meeting point	Uchrashuv nuqtasi	Точка встречи
Head start	Oldindan boshlash	Фора

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Towards each other, the closing speed is:

- $v_1 + v_2$
- $v_1 - v_2$
- $v_1 v_2$
- v_1

In the same direction it is:

- $v_1 + v_2$
- $v_1 - v_2$
- the average
- the larger

240 km apart at 70 and 50: the time to meet is:

- 2 h
- 4 h
- 12 h
- 1.2 h

The distances covered are in the ratio of:

- the times
- the speeds
- the total
- nothing

A 12 km head start with speeds 15 and 12:

- 0.8 h
- 4 h
- 1 h
- 12 h

The first step in these problems is:

- a formula
- a diagram
- a calculation
- a guess

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *300* km apart at *60* and *40*: the closing speed

   ANSWER *100 km/h*

2. And the time to meet

   ANSWER *3 h*

3. *240* km apart at *70* and *50*

   ANSWER *2 h*

4. *12* km apart at *5* and *3*

   ANSWER *1.5 h*

5. A *30* km head start, speeds *80* and *60*

   ANSWER *1.5 h*

6. A *12* km head start, speeds *15* and *12*

   ANSWER *4 h*

7. Equal speeds in the same direction

   ANSWER The gap never closes

Medium · the standard question

7 PROBLEMS

1. Where do the *300* km cars meet?

   ANSWER *180* km from the first

2. A bus at *60* leaves *30* min before a car at *80*

   ANSWER Caught after *1.5 h*

3. A walker at *5* leaves *2* h before a cyclist at *15*

   ANSWER Caught after *1 h*

4. A cyclist at *12* leaves *1* h before a car at *60*

   ANSWER Caught after *0.25 h*

5. Runners at *8* and *6* m/s with a *100* m start

   ANSWER *50 s*

6. Two trains *420* km apart at *90* and *120*

   ANSWER *2 h*

7. Where do they meet?

   ANSWER *180* km from the first

Hard · stretch and reason

7 PROBLEMS

1. Two cars 450 km apart meet after 3 h; one does 80: the other

   ANSWER 70 km/h

2. A car catches a bus 40 km ahead in 2 h: the difference in speeds

   ANSWER 20 km/h

3. Two walkers 9 km apart at 4 and 5: where do they meet?

   ANSWER 4 km from the first

4. A boat and a raft 24 km apart moving towards each other at 10 and 2

   ANSWER 2 h

5. A train 200 m long passing a pole at 20 m/s

   ANSWER 10 s

6. Why does the gap close at the sum when they approach?

   ANSWER Each covers part of it in the same time

7. Two cyclists start together at 12 and 18: the gap after 2 h

   ANSWER 12 km

07 Homework

Homework — 5 tasks

Draw the two bodies and an arrow each before choosing sum or difference.

1. Two cars 420 km apart drive towards each other at 90 and 120 km/h. When do they meet?
2. How far has each travelled by then?
3. A bus at 50 km/h leaves 1 hour before a car at 75. When does the car catch it?
4. Two runners 120 m apart run in the same direction at 7 and 5 m/s. When is the first caught?
5. Explain why the closing speed is the sum in one case and the difference in the other.

Control work 7 — speed and motion

The formula, its rearrangements, average speed and two-body problems.

LESSON 132 | 1 × 40 MIN | MATEMATIKA 6, NAZORAT ISHI 7 | S7 12 REVIEW

LESSON CLOCK

2' | 28' | 8' | 2'

Instructions	2 min
The paper	28 min
Answers and diagnosis	8 min
What comes next	2 min

LEARNING OBJECTIVES

- Use $s = vt$ in all three directions with correct units.
- Find an average speed over a journey in stages.
- Solve a meeting or overtaking problem.
- Classify each lost mark and rewrite the whole solution.

TEXTBOOK

National	pp. 335–388
Cambridge	Stage 7, pp. 122–126
Workbook	Control paper 7

01 Explanation

The paper — 20 marks, 28 minutes

Q	TASK	MARKS	FROM
1	A car does *75* km/h for *4* h: the distance	*2*	L117–119
2	*4* km in *20* min: the speed in km/h	*3*	L120–122
3	*7* km at *5* km/h: the time in hours and minutes	*3*	L120–122
4	Convert *72* km/h to m/s	*2*	L116
5	*120* km at *60* and back at *40*: the average speed	*4*	L126–128
6	Two cars *300* km apart at *60* and *40*: when and where do they meet?	*4*	L129–131
7	A bus at *60* leaves *30* min before a car at *80*: when is it caught?	*2*	L129–131

THE ANSWERS

300 km; *12* km/h; *1* h *24* min; *20* m/s; *48* km/h; after *3* h, *180* km from the first; after *1.5* h.

Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
minutes used with km/h	$4 \div 20$	$4 \div \frac{1}{3}$
decimal hours read as minutes	*1 h 40 min*	*1 h 24 min*
multiplied instead of divided	$72 \cdot 3.6$	$72 \div 3.6$
speeds averaged	$(60 + 40) \div 2$	$240 \div 5$
closing speed subtracted for an approach	$60 - 40$	$60 + 40$
head start left as a time	$0.5 \div 20$	$30 \div 20$
units missing	*300*	*300 km*

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

What comes next

IF YOU LOST MARKS ON	REVISE
Q1–Q3	the formula and time conversion, L116–122
Q4	unit conversion, L116
Q5	average speed, L126–128
Q6–Q7	two-body problems, L129–131

LOOKING FORWARD

The rest of Quarter III is a Cambridge project, two lessons on fractions and sequences, and then the volume of the cuboid — where a formula again does the work.

02 Worked examples

EXAMPLE 1 **Model answer, Q2: *4* km in *20* minutes.**

1. 20 min $= \frac{1}{3}$ h.

   Convert first.

2. $v = 4 \div \frac{1}{3} = 4 \cdot 3$

3. $= 12$ km/h.

   Reasonable for a cyclist ✓

ANSWER

12 km/h

EXAMPLE 2 **Model answer, Q5: *120* km at *60* and back at *40*.**

1. Out: 2 h. Back: 3 h.

2. Total 240 km in 5 h.

3. 48 km/h.

   Not the mean of the speeds.

ANSWER

48 km/h

EXAMPLE 3 **Model answer, Q6: two cars *300* km apart.**

1. Closing speed 100 km/h.
   They approach, so add.

2. Time 3 h.

3. First travels 180 km, second 120 km.
   Check: 300 ✓

ANSWER

After 3 h, 180 km from the first

03 Interactive model

Return Q5 first and ask how many wrote 50; naming the error as “averaging the speeds” makes it memorable.

The chapter in eight questions

75 km/h for 4 h:

- 18.75
- 300
- 79
- 300 km/h

4 km in 20 min:

- 0.2
- 12
- 80
- 4

7 km at 5 km/h:

- 1.4 h
- 1 h 40 min
- 1 h 24 min
- 35 h

72 km/h in m/s:

- 20
- 259
- 26
- 7.2

120 km at 60 and back at 40:

- 50
- 48
- 100
- 240

Two cars approaching close the gap at:

- the sum
- the difference
- the average
- the larger

300 km apart at 60 and 40: they meet after:

- 3 h
- 5 h
- 7.5 h
- 2 h

A *30* min head start at *60* is:

- *0.5* km
- *30* km
- *60* km
- *120* km

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Speed	Tezlik	Скорость
Distance	Masofa	Расстояние
Time	Vaqt	Время
Average speed	O‘rtacha tezlik	Средняя скорость
Closing speed	Yaqinlashish tezligi	Скорость сближения
Units	O‘lchov birligi	Единицы
Diagnosis	Tashxis	Диагностика

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Q2 needs the time in:

- minutes
- hours
- seconds
- any unit

Q3 answer in hours and minutes is:

- 1.4 h
- $1\ h\ 40\ min$
- $1\ h\ 24\ min$
- 84 min only

Q4 divides by:

- 3.6
- 1000
- 60
- 36

Q5 must not:

- add the distances
- average the speeds
- add the times
- divide once

Q6 adds the speeds because:

- they travel together
- they approach
- they are equal
- of the units

Q7 turns the head start into:

- a time
- a distance
- a speed
- nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *75* km/h for *4* h

   ANSWER *300 km*

2. *4* km in *20* min

   ANSWER *12 km/h*

3. *7* km at *5* km/h

   ANSWER *1 h 24 min*

4. *72* km/h in m/s

   ANSWER *20*

5. *240* km in *5* h

   ANSWER *48 km/h*

6. Closing speed of *60* and *40* approaching

   ANSWER *100 km/h*

7. A *30* min head start at *60*

   ANSWER *30 km*

Medium · the standard question

7 PROBLEMS

1. *120* km at *60* and back at *40*

   ANSWER *48 km/h*

2. Two cars *300* km apart at *60* and *40*

   ANSWER Meet after *3 h*

3. Where?

   ANSWER *180* km from the first

4. A bus at *60* caught by a car at *80* after a *30* min start

   ANSWER *1.5 h*

5. *54* km/h in m/s

   ANSWER *15*

6. *9* km at *4* km/h

   ANSWER *2 h 15 min*

7. *150* km in *2 h 30 min*

   ANSWER *60 km/h*

Hard · stretch and reason

7 PROBLEMS

1. A journey of *200* km, half at *40* and half at *60*

   ANSWER *48 km/h*

2. Two trains *540* km apart at *100* and *80*

   ANSWER Meet after *3 h*

3. A car catches a lorry *45* km ahead in *1.5* h: the speed difference

   ANSWER *30 km/h*

4. A cyclist at *15* km/h for *40* min then *20* km/h for *30* min: the average

   ANSWER *17.1 km/h*

5. A train *180* m long passing a pole at *15* m/s

   ANSWER *12 s*

6. Why is the average of *60* and *40* not *50* here?

   ANSWER More time is spent at *40*

7. *12* m/s in km/h

   ANSWER *43.2*

07 Homework

Homework — 5 tasks

Rewrite every question you lost a mark on, with its units.

1. Rewrite in full every question on which you lost a mark.
2. A car covers *294* km in *3 h 30 min*. Find its speed.
3. Find the average speed for *60* km at *30* km/h and *60* km at *60*.
4. Two cyclists *45* km apart ride towards each other at *12* and *18* km/h. When do they meet?
5. Convert *90* km/h to m/s and *25* m/s to km/h.

Think — Project 3: fraction averages

A Cambridge project: the fraction that lies between two others, and where it always lands.

LESSON 133 | 1 × 40 MIN | MATEMATIKA 6, O'YLAB KO'R | S7 PROJECT 3

LESSON CLOCK

8' | 12' | 14' | 6'

The mean of two fractions	8 min
A strange rule	12 min
Testing the conjecture	14 min
Presenting it	6 min

LEARNING OBJECTIVES

- Find the mean of two fractions.
- Investigate what happens when numerators and denominators are added instead.
- State a conjecture and test it.
- Present the investigation clearly.

TEXTBOOK

National	pp. 389–391
Cambridge	Stage 7, project pages
Workbook	Project sheet 3

01 Explanation

The mean of two fractions

$$\text{mean} = \frac{a+b}{2}$$

TWO FRACTIONS	SUM	MEAN
$\frac{1}{2}, \frac{1}{4}$	$\frac{3}{4}$	$\frac{3}{8}$
$\frac{1}{3}, \frac{1}{2}$	$\frac{5}{6}$	$\frac{5}{12}$
$\frac{2}{5}, \frac{3}{5}$	1	$\frac{1}{2}$

THE MEAN ALWAYS LIES BETWEEN THE TWO

That is the point of an average, and it is the check to make on every answer in this investigation.

A strange rule

Now try something that looks wrong: add the numerators and add the denominators.

$$\frac{a}{b} \oplus \frac{c}{d} = \frac{a+c}{b+d}$$

TWO FRACTIONS	THE STRANGE RULE	THE TRUE MEAN	BETWEEN THEM?
$\frac{1}{2}, \frac{1}{4}$	$\frac{2}{6} = \frac{1}{3}$	$\frac{3}{8}$	yes
$\frac{1}{3}, \frac{1}{2}$	$\frac{2}{5}$	$\frac{5}{12}$	yes
$\frac{1}{4}, \frac{3}{4}$	$\frac{4}{8} = \frac{1}{2}$	$\frac{1}{2}$	yes

THIS IS NOT HOW FRACTIONS ARE ADDED

$\frac{1}{2} + \frac{1}{4}$ is $\frac{3}{4}$, not $\frac{2}{6}$. The rule above is a different operation, worth investigating precisely because it is the mistake everyone makes.

Testing the conjecture

Conjecture: the result of the strange rule always lies between the two fractions.

TEST	RESULT	BETWEEN?
$\frac{1}{5}, \frac{4}{5}$	$\frac{5}{10} = \frac{1}{2}$	yes
$\frac{2}{3}, \frac{5}{7}$	$\frac{7}{10}$	yes — $0.667 < 0.7 < 0.714$
$\frac{1}{2}, \frac{1}{2}$	$\frac{2}{4} = \frac{1}{2}$	equal to both

THE CONJECTURE SURVIVES EVERY TEST

It is true, and it has a name: the **mediant** of two fractions. It always lies between them — but it is not usually the mean, as the first table showed.

Presenting it

PART	WHAT IT CONTAINS
the question	where does $\dfrac{a + c}{b + d}$ lie?
the data	at least six pairs, chosen systematically
the pattern	always between, but not the mean
the comparison	the mean against the mediant, in decimals
the conclusion	two sentences, with one worked example

WHERE THE MEDIANT IS USED

Two batsmen scoring *40* runs off *50* balls and *30* off *40* have a combined rate of $\dfrac{70}{90}$ — the mediant, not the mean of their two rates. Adding the tops and the bottoms is exactly right for combining rates.

02 Worked examples

EXAMPLE 1 **Find the mean of $\dfrac{1}{2}$ and $\dfrac{1}{4}$.**

1. $\dfrac{1}{2} + \dfrac{1}{4} = \dfrac{3}{4}$
2. Divide by *2*.
3. $\dfrac{3}{8}$

   Between $\dfrac{1}{4}$ and $\dfrac{1}{2}$ ✓

ANSWER

$\dfrac{3}{8}$

EXAMPLE 2 **Apply the strange rule to $\frac{1}{3}$ and $\frac{1}{2}$, and compare with the mean.**

1. Strange rule: $\frac{2}{5} = 0.4$.
2. Mean: $\frac{5}{12} \approx 0.417$.
3. Both lie between 0.333 and 0.5, but they differ.

ANSWER

$\frac{2}{5}$ against $\frac{5}{12}$

EXAMPLE 3 **A batsman scores *40* off *50* balls and then *30* off *40*. Find the combined rate.**

1. Total runs: 70. Total balls: 90.
2. $\frac{70}{90} \approx 0.78$ runs a ball.
3. That is the mediant of $\frac{40}{50}$ and $\frac{30}{40}$.

   Adding tops and bottoms is right here.

ANSWER

$\frac{7}{9}$

03 Interactive model

Point out that the “wrong” way to add fractions is exactly the right way to combine rates; the class remembers both rules better for seeing where each belongs.

Mean or mediant?

The mean of $\frac{1}{2}$ and $\frac{1}{4}$ is:

- $\frac{1}{3}$
- $\frac{3}{8}$
- $\frac{2}{6}$
- $\frac{3}{4}$

The mediant of the same two is:

- $\frac{1}{3}$
- $\frac{3}{8}$
- $\frac{3}{4}$
- $\frac{1}{6}$

$\frac{1}{2} + \frac{1}{4}$ equals:

- $\frac{2}{6}$
- $\frac{3}{4}$
- $\frac{1}{3}$
- $\frac{1}{8}$

The mediant always lies:

- above both
- below both
- between them
- anywhere

Is the mediant the mean?

- always
- sometimes
- never
- only for equal fractions

Two rates $\frac{40}{50}$ and $\frac{30}{40}$ combine to:

- $\frac{70}{90}$
- $\frac{70}{45}$
- $\frac{35}{45}$
- $\frac{1200}{2000}$

Adding tops and bottoms is right for:

- adding fractions
- combining rates
- both
- neither

A conjecture becomes a fact when it is:

- tested twice
- proved
- written down
- believed

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Mean	O‘rta arifmetik	Среднее арифметическое
Between	Orasida	Между
Conjecture	Faraz	Гипотеза
To test	Sinab ko‘rmoq	Проверить
Numerator	Surat	Числитель
Denominator	Maxraj	Знаменатель
Investigation	Tadqiqot	Исследование
Counter-example	Qarshi misol	Контрпример

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The mean of two numbers is:

- their sum
- half their sum
- their difference
- their product

The mediant of $\frac{a}{b}$ and $\frac{c}{d}$ is:

- $\frac{a+c}{b+d}$
- $\frac{a+c}{2}$
- $\frac{ac}{bd}$
- $\frac{a}{b} + \frac{c}{d}$

The mediant of $\frac{1}{4}$ and $\frac{3}{4}$ is:

- $\frac{1}{2}$
- $\frac{4}{4}$
- $\frac{3}{16}$
- $\frac{1}{8}$

Is that also the mean here?

- yes
- no
- sometimes
- never

Adding fractions needs:

- a common denominator
- adding the denominators
- the mediant
- nothing

The strange rule is right for:

- fractions
- rates
- both
- neither

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. The mean of $\frac{1}{2}$ and $\frac{1}{4}$

ANSWER $\frac{3}{8}$

2. The mean of $\frac{1}{3}$ and $\frac{1}{2}$

ANSWER $\frac{5}{12}$

3. The mean of $\frac{2}{5}$ and $\frac{3}{5}$

ANSWER $\frac{1}{2}$

4. The mediant of $\frac{1}{2}$ and $\frac{1}{4}$

ANSWER $\frac{1}{3}$

5. The mediant of $\frac{1}{3}$ and $\frac{1}{2}$

ANSWER $\frac{2}{5}$

6. The mediant of $\frac{1}{4}$ and $\frac{3}{4}$

ANSWER $\frac{1}{2}$

7. $\frac{1}{2} + \frac{1}{4}$

ANSWER $\frac{3}{4}$

Medium · the standard question

7 PROBLEMS

1. Is the mediant always between the two fractions?

ANSWER Yes

2. Is the mediant always the mean?

ANSWER No

3. When are they equal?

ANSWER When the two fractions are equal, or the denominators match

4. The mediant of $\frac{1}{5}$ and $\frac{4}{5}$

ANSWER $\frac{1}{2}$

5. The mediant of $\frac{2}{3}$ and $\frac{5}{7}$

ANSWER $\frac{7}{10}$

6. Combine 40 off 50 and 30 off 40

ANSWER $\frac{7}{9}$

7. Which rule adds fractions?

ANSWER The common denominator

Hard · stretch and reason

7 PROBLEMS

1. Show that $\frac{7}{10}$ lies between $\frac{2}{3}$ and $\frac{5}{7}$

   ANSWER $0.667 < 0.7 < 0.714$

2. The mean of $\frac{2}{3}$ and $\frac{5}{7}$

   ANSWER $\frac{29}{42}$

3. Which is larger, the mean or the mediant, for those two?

   ANSWER The mean, at 0.690 against 0.7 — the mediant

4. A pupil scores $18/25$ and $27/35$: the combined mark

   ANSWER $\frac{45}{60} = 75\%$

5. Why is the mediant right for combining marks?

   ANSWER Totals over totals is what a combined mark means

6. Find a fraction between $\frac{3}{7}$ and $\frac{4}{7}$

   ANSWER $\frac{7}{14} = \frac{1}{2}$

7. Find a fraction between $\frac{5}{8}$ and $\frac{2}{3}$

   ANSWER $\frac{7}{11}$

07 Homework

Homework — the project

One page: six tested pairs, the pattern, and where the rule is genuinely useful.

1. Find the mean and the mediant of six pairs of fractions of your own.

2. Record both in a table, with decimals for comparison.
3. State in one sentence where the mediant always lies.
4. Give one example where adding tops and bottoms is the right thing to do.
5. Explain why it is nevertheless the wrong way to add two fractions.

Ordering fractions; adding mixed numbers

One common denominator settles both the order and the sum. [Cambridge insert]

LESSON 134–136 | 3 × 40 MIN | MATEMATIKA 6, TAKRORLASH | S7 7.1–7.2 FRACTIONS

LESSON CLOCK

25' | 30' | 35' | 25' | 5

Comparing two fractions	25 min
Ordering a list	30 min
Adding mixed numbers	35 min
Subtracting with a regroup	25 min
Homework	5 min

LEARNING OBJECTIVES

- Compare two fractions by rewriting them over a common denominator.
- Order a list of fractions and place them on a number line.
- Add and subtract mixed numbers, converting the improper part back.
- Regroup one whole when the subtraction of the fraction parts will not go.

TEXTBOOK

National	pp. 389–394
Cambridge	Stage 7, pp. 68–75
Workbook	Exercise 7.1–7.2

01 Explanation

Comparing two fractions

Two fractions can be compared the moment they share a denominator: the one with the larger numerator is larger. Everything in this topic is that one sentence, applied twice.

PAIR	COMMON DENOMINATOR	REWRITTEN	LARGER
$\frac{3}{4}$ *and* $\frac{5}{7}$	*28*	$\frac{21}{28}$ *and* $\frac{20}{28}$	$\frac{3}{4}$
$\frac{5}{8}$ *and* $\frac{7}{12}$	*24*	$\frac{15}{24}$ *and* $\frac{14}{24}$	$\frac{5}{8}$
$\frac{4}{9}$ *and* $\frac{3}{7}$	*63*	$\frac{28}{63}$ *and* $\frac{27}{63}$	$\frac{4}{9}$
$\frac{2}{3}$ *and* $\frac{5}{8}$	*24*	$\frac{16}{24}$ *and* $\frac{15}{24}$	$\frac{2}{3}$

THE CROSS CHECK, FOR TWO FRACTIONS ONLY

For $\frac{a}{b}$ against $\frac{c}{d}$, compare ad with bc. For $\frac{3}{4}$ and $\frac{5}{7}$: $3 \cdot 7 = 21$ against $4 \cdot 5 = 20$, so the first is larger. It is the same arithmetic as the table, with the denominator left unwritten.

Ordering a list

For three or more fractions the cross check is no help — one denominator for the whole list is quicker. Order $\frac{2}{3}$, $\frac{3}{5}$, $\frac{7}{10}$, $\frac{5}{8}$.

FRACTION	OVER 120	AS A DECIMAL	PLACE
$\frac{3}{5}$	$\frac{72}{120}$	*0.600*	1st
$\frac{5}{8}$	$\frac{75}{120}$	*0.625*	2nd
$\frac{2}{3}$	$\frac{80}{120}$	*0.667*	3rd
$\frac{7}{10}$	$\frac{84}{120}$	*0.700*	4th

A BIGGER DENOMINATOR DOES NOT MEAN A BIGGER FRACTION

$\frac{7}{10}$ beats $\frac{5}{8}$ and $\frac{3}{5}$ beats nothing at all: the denominator alone decides nothing. Rewrite, then compare.

With directed numbers the order reverses on the negative side: $\frac{2}{3}$ is larger than $\frac{3}{5}$, so $-\frac{2}{3}$ is **smaller** than $-\frac{3}{5}$.

Adding mixed numbers

Add the whole parts, add the fraction parts over a common denominator, then carry any whole that the fraction part has produced.

$$2\frac{3}{4} + 1\frac{5}{6} = 3 + \frac{9}{12} + \frac{10}{12} = 3\frac{19}{12} = 4\frac{7}{12}$$

SUM	WHOLES	FRACTIONS	ANSWER
$3\frac{2}{5} + 2\frac{1}{3}$	5	$\frac{6}{15} + \frac{5}{15} = \frac{11}{15}$	$5\frac{11}{15}$
$1\frac{1}{2} + 2\frac{2}{3}$	3	$\frac{3}{6} + \frac{4}{6} = \frac{7}{6}$	$4\frac{1}{6}$
$3\frac{5}{6} + 1\frac{3}{4}$	4	$\frac{10}{12} + \frac{9}{12} = \frac{19}{12}$	$5\frac{7}{12}$
$2\frac{7}{10} + 3\frac{4}{5}$	5	$\frac{7}{10} + \frac{8}{10} = \frac{15}{10}$	$6\frac{1}{2}$

NEVER LEAVE AN IMPROPER FRACTION IN THE ANSWER

$3\frac{19}{12}$ is a correct value but not a finished answer: $\frac{19}{12} = 1\frac{7}{12}$, so the whole part becomes 4.

Subtracting with a regroup

Subtraction is the same, until the fraction parts will not go. Then one whole is exchanged for a fraction of the same denominator — exactly the borrowing of column subtraction.

$$4\frac{1}{3} - 1\frac{5}{6} = 4\frac{2}{6} = 3\frac{8}{6}, \text{ so } 3\frac{8}{6} - 1\frac{5}{6} = 2\frac{3}{6} = 2\frac{1}{2}$$

DIFFERENCE	REGROUP	ANSWER
$5\frac{1}{4} - 2\frac{3}{4}$	$4\frac{5}{4} - 2\frac{3}{4}$	$2\frac{1}{2}$
$6 - 2\frac{3}{8}$	$5\frac{8}{8} - 2\frac{3}{8}$	$3\frac{5}{8}$
$7\frac{2}{5} - 3\frac{4}{5}$	$6\frac{7}{5} - 3\frac{4}{5}$	$3\frac{3}{5}$
$5\frac{1}{6} - 2\frac{3}{4}$	$4\frac{14}{12} - 2\frac{9}{12}$	$2\frac{5}{12}$

TAKE THE WHOLE ACROSS IN THE RIGHT PIECES

One whole is $\frac{4}{4}$ when the denominator is 4 and $\frac{12}{12}$ when it is 12. Change the denominators first, regroup second, or the exchange will be the wrong size.

02 Worked examples

EXAMPLE 1 **Which is larger, $\frac{4}{9}$ or $\frac{3}{7}$?**

1. Common denominator 63.
2. $\frac{4}{9} = \frac{28}{63}$ and $\frac{3}{7} = \frac{27}{63}$.
3. $28 > 27$.

   Cross check: $4 \cdot 7 = 28 > 27 = 9 \cdot 3$ ✓

ANSWER

$\frac{4}{9}$ is larger

EXAMPLE 2 **Work out $2\frac{3}{4} + 1\frac{5}{6}$.**

1. Wholes: $2 + 1 = 3$.

2. Fractions over 12: $\frac{9}{12} + \frac{10}{12} = \frac{19}{12}$.

3. $\frac{19}{12} = 1\frac{7}{12}$, so the answer is $4\frac{7}{12}$.

   Estimate: $2.75 + 1.83 \approx 4.6$ ✓

ANSWER

$4\frac{7}{12}$

EXAMPLE 3 **Work out $4\frac{1}{3} - 1\frac{5}{6}$.**

1. Over 6: $4\frac{2}{6} - 1\frac{5}{6}$.

2. $\frac{2}{6} < \frac{5}{6}$, so regroup: $3\frac{8}{6}$.

3. $3\frac{8}{6} - 1\frac{5}{6} = 2\frac{3}{6} = 2\frac{1}{2}$.

ANSWER

$2\frac{1}{2}$

03 Interactive model

Run the model on the pair 9 and 7 last: the extra factors are the other denominator, and the class sees why unrelated denominators multiply.

Building the common denominator

DENOMINATOR	FACTORISED	EXTRA FACTOR NEEDED
4	$2 \cdot 2$	$\times\ 3$
6	$2 \cdot 3$	$\times\ 2$

LOWEST COMMON DENOMINATOR

$$2 \cdot 2 \cdot 3 = 12$$

The shared factor 2 is counted once, not twice.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Mixed number	Aralash son	Смешанное число
Improper fraction	Noto‘g‘ri kasr	Неправильная дробь
Proper fraction	To‘g‘ri kasr	Правильная дробь
Common denominator	Umumiy maxraj	Общий знаменатель
Lowest common denominator	Eng kichik umumiy maxraj	Наименьший общий знаменатель
To order	Tartiblamoq	Упорядочить
Ascending order	O‘sish tartibida	По возрастанию
To regroup a whole	Butundan olmoq	Занимать из целой части

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

To compare two fractions you first make equal the:

- numerators
- denominators
- wholes
- decimals

The larger of $\frac{5}{8}$ and $\frac{7}{12}$ is:

- $\frac{5}{8}$
- $\frac{7}{12}$
- they are equal
- cannot tell

$2\frac{3}{4} + 1\frac{5}{6}$ equals:

- $3\frac{19}{12}$
- $4\frac{7}{12}$
- $3\frac{8}{10}$
- $4\frac{1}{2}$

In $6 - 2\frac{3}{8}$ the whole 6 becomes:

- $5\frac{8}{8}$
- $6\frac{8}{8}$
- $5\frac{1}{8}$
- 5

Of $-\frac{2}{3}$ and $-\frac{3}{5}$ the smaller is:

- $-\frac{3}{5}$
- $-\frac{2}{3}$
- equal
- neither

The lowest common denominator of 9 and 7 is:

- 16
- 63
- 9
- 126

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Which is larger, $\frac{3}{4}$ or $\frac{5}{7}$?

ANSWER $\frac{3}{4}$

2. Which is larger, $\frac{5}{8}$ or $\frac{7}{12}$?

ANSWER $\frac{5}{8}$

3. Which is larger, $\frac{2}{3}$ or $\frac{5}{8}$?

ANSWER $\frac{2}{3}$

4. $1\frac{1}{4} + 2\frac{1}{4}$

ANSWER $3\frac{1}{2}$

5. $3\frac{2}{5} + 1\frac{1}{5}$

ANSWER $4\frac{3}{5}$

6. $5\frac{3}{4} - 2\frac{1}{4}$

ANSWER $3\frac{1}{2}$

7. The lowest common denominator of 6 and 8

ANSWER 24

Medium · the standard question 7 PROBLEMS

1. $2\frac{3}{4} + 1\frac{5}{6}$

ANSWER $4\frac{7}{12}$

2. $3\frac{5}{6} + 1\frac{3}{4}$

ANSWER $5\frac{7}{12}$

3. $4\frac{1}{3} - 1\frac{5}{6}$

ANSWER $2\frac{1}{2}$

4. $6 - 2\frac{3}{8}$

ANSWER $3\frac{5}{8}$

5. $7\frac{2}{5} - 3\frac{4}{5}$

ANSWER $3\frac{3}{5}$

6. Order $\frac{3}{5}, \frac{2}{3}, \frac{7}{10}, \frac{5}{8}$ upwards

ANSWER $\frac{3}{5}, \frac{5}{8}, \frac{2}{3}, \frac{7}{10}$

7. Which is larger, $-\frac{2}{3}$ or $-\frac{3}{5}$?

ANSWER $-\frac{3}{5}$

Hard · stretch and reason

7 PROBLEMS

1. $2\frac{7}{10} + 3\frac{4}{5}$

ANSWER $6\frac{1}{2}$

2. $5\frac{1}{6} - 2\frac{3}{4}$

ANSWER $2\frac{5}{12}$

3. $1\frac{3}{8} + 2\frac{1}{6} + 1\frac{1}{4}$

ANSWER $4\frac{19}{24}$

4. $10 - 3\frac{2}{7} - 2\frac{5}{7}$

ANSWER 4

5. Find a fraction between $\frac{5}{9}$ and $\frac{4}{7}$

ANSWER $\frac{71}{126}$

6. A recipe needs $2\frac{1}{3}$ cups and $1\frac{3}{4}$ cups: the total

ANSWER $4\frac{1}{12}$ cups

7. Which is larger, $\frac{12}{17}$ or $\frac{7}{10}$?

ANSWER $\frac{12}{17}$, since $120 > 119$

07 Homework

Homework — 5 tasks

Every answer as a mixed number in lowest terms, with the common denominator shown.

1. Which is larger, $\frac{7}{9}$ or $\frac{5}{6}$? Show the rewriting.
2. Order $\frac{1}{2}$, $\frac{4}{9}$, $\frac{5}{12}$ and $\frac{7}{18}$ in ascending order.
3. Work out $3\frac{2}{3} + 2\frac{3}{4}$.
4. Work out $5\frac{1}{4} - 2\frac{2}{3}$.
5. A plank 4 m long has $1\frac{5}{8}$ m sawn off. How much is left?

Generating sequences and finding the nth term

From “add four each time” to a rule that reaches the hundredth term at once. [Cambridge insert]

LESSON 137–138 | 2 × 40 MIN | MATEMATIKA 6, QO‘SHIMCHA MAVZU | S7 9 SEQUENCES AND FUNCTIONS

LESSON CLOCK

20' | 20' | 25' | 10' | 5'

Term-to-term rules	20 min
From position to term	20 min
Finding the nth term	25 min
Using the rule	10 min
Homework	5 min

LEARNING OBJECTIVES

- Continue a sequence from its term-to-term rule.
- Generate terms from a position-to-term rule such as $3n + 2$.
- Find the nth term of a linear sequence from its common difference.
- Decide whether a given number is a term, and say which one.

TEXTBOOK

National	pp. 395–399
Cambridge	Stage 7, pp. 90–97
Workbook	Exercise 9.1–9.3

01 Explanation

Term-to-term rules

A sequence is a list of numbers in a fixed order. Each number is a **term**, and its **position** is $n = 1, 2, 3, \ldots$. The easiest rule says what to do to one term to get the next.

SEQUENCE	TERM-TO-TERM RULE	NEXT TWO TERMS
$3, 7, 11, 15, \ldots$	add 4	$19, 23$
$5, 8, 11, 14, \ldots$	add 3	$17, 20$
$20, 17, 14, 11, \ldots$	subtract 3	$8, 5$
$2, 4, 8, 16, \ldots$	multiply by 2	$32, 64$

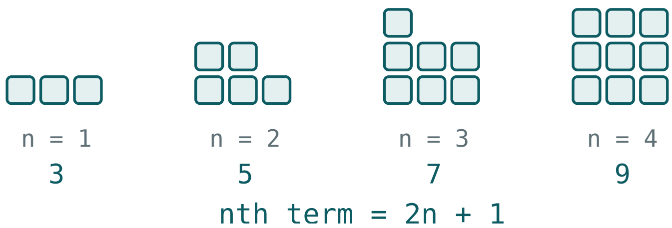
A TERM-TO-TERM RULE CANNOT JUMP

To reach the 100th term of the first sequence this way you would add 4 ninety-nine times. That is why the next rule is worth having.

From position to term

A **position-to-term** rule — the nth term — takes the position straight to the term. Substitute $n = 1, 2, 3, \dots$ and the sequence appears.

NTH TERM	$N = 1$	$N = 2$	$N = 3$	$N = 4$
$3n + 2$	5	8	11	14
$4n - 1$	3	7	11	15
$23 - 3n$	20	17	14	11
n^2	1	4	9	16



A pattern of squares — the count rises by two each time, so the nth term is 2n + 1

THE RULE REACHES ANY TERM IN ONE STEP

The 100th term of $4n - 1$ is $4 \cdot 100 - 1 = 399$. No list, no adding ninety-nine times.

Finding the nth term

A sequence that goes up (or down) by the same amount every time is **linear**. Call that amount the common difference d, and the first term a.

$$n\text{th term} = dn + (a - d)$$

The dn builds the steady climb; the bracket is the adjustment that makes the first term come out right. In practice: write dn, work out the first term it gives, and correct it.

SEQUENCE	D	DN GIVES AT $N = 1$	CORRECTION	NTH TERM
$5, 8, 11, 14$	3	3	$+2$	$3n + 2$
$2, 9, 16, 23$	7	7	-5	$7n - 5$
$6, 10, 14, 18$	4	4	$+2$	$4n + 2$
$20, 17, 14, 11$	-3	-3	$+23$	$23 - 3n$

CHECK THE RULE ON THE FIRST TWO TERMS

A rule that gives the right first term but the wrong second one has the wrong d; one that climbs correctly but starts wrong needs a different correction. Both take five seconds to test.

Using the rule

Once the nth term is known, two questions can be answered that a list cannot answer at all.

QUESTION	WORKING	ANSWER
the 50th term of $3n + 2$	$3 \cdot 50 + 2$	152
is 47 a term of $3n + 2$?	$3n = 45, n = 15$	yes, the 15th
is 60 a term of $4n - 1$?	$4n = 61$	no — n is not whole
the first negative term of $23 - 3n$	$23 - 3n < 0, n > 7.67$	$n = 8$, term -1

THE POSITION MUST BE A WHOLE NUMBER

That is the whole test for membership. Solve for n; if it comes out a whole number the value is a term, and n says which one.

02 Worked examples

EXAMPLE 1 **Find the nth term of $5, 8, 11, 14, \dots$ and its 20th term.**

1. Common difference $d = 3$, so start from $3n$.
2. $3n$ gives 3 at $n = 1$, but the term is 5: add 2.
3. nth term $= 3n + 2$; at $n = 20$, $3 \cdot 20 + 2 = 62$.
   Check $n = 2$: 8 ✓

ANSWER

$3n + 2$, and 62

EXAMPLE 2 **Find the nth term of $20, 17, 14, 11, \dots$ and its first negative term.**

1. $d = -3$, so start from $-3n$.
2. $-3n$ gives -3 at $n = 1$, but the term is 20: add 23.
3. nth term $= 23 - 3n$. It is negative when $n > 7\frac{2}{3}$, so $n = 8$ gives -1.

ANSWER

$23 - 3n$; the 8th term, -1

EXAMPLE 3 **Is 47 a term of the sequence with nth term $3n + 2$?**

1. Set $3n + 2 = 47$.
2. $3n = 45$.
3. $n = 15$ — a whole number, so yes.

   Check: $3 \cdot 15 + 2 = 47$ ✓

ANSWER

Yes, the 15th term

03 Interactive model

Ask for the 100th term before giving the nth-term rule; the impatience the class feels is exactly the reason the rule exists.

Position, term and rule

In $3, 7, 11, 15$ the common difference is:

- 3
- 4
- 7
- 11

The nth term of $5, 8, 11, 14$ is:

- $3n$
- $3n + 2$
- $5n$
- $n + 3$

The 10th term of $4n - 1$ is:

- 39
- 40
- 41
- 36

The nth term of $20, 17, 14$ is:

- $3n + 20$
- $23 - 3n$
- $20 - 3n$
- $17n$

Is 60 a term of $4n - 1$?

- yes, the 15th
- no
- yes, the 61st
- cannot tell

$1, 4, 9, 16$ is generated by:

- $4n$
- n^2
- $3n + 1$
- $n + 3$

A term-to-term rule of “multiply by 2” makes the sequence:

- linear
- not linear
- decreasing
- constant

The nth term of $6, 10, 14, 18$ is:

- $4n + 2$
- $4n$
- $6n$
- $2n + 4$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Sequence	Ketma-ketlik	Последовательность
Term	Had	Член последовательности
Position	O'rin (tartib raqami)	Номер (позиция)
Term-to-term rule	Haddan hadga o'tish qoidasi	Рекуррентное правило
nth term	n-chi had	n-й член
Common difference	Ayirma	Разность
Linear sequence	Chiziqli ketma-ketlik	Линейная последовательность
Square numbers	Kvadrat sonlar	Квадратные числа

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The common difference of $2, 9, 16, 23$ is:

- 5
- 7
- 9
- 2

The nth term of $2, 9, 16, 23$ is:

- $7n$
- $7n - 5$
- $7n + 2$
- $2n + 7$

A position-to-term rule lets you:

- only continue the list
- reach any term at once
- find the difference
- nothing new

The 100th term of $4n - 1$ is:

- 399
- 400
- 401
- 403

A number is a term only when n comes out:

- positive
- a whole number
- even
- small

In $dn + (a - d)$ the letter d is:

- the first term
- the common difference
- the position
- the last term

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. The next two terms of $3, 7, 11, 15$

   ANSWER $19, 23$

2. The next two terms of $20, 17, 14$

   ANSWER $11, 8$

3. The first three terms of $5n$

   ANSWER $5, 10, 15$

4. The first three terms of $2n + 3$

   ANSWER $5, 7, 9$

5. The 10th term of $4n - 1$

   ANSWER 39

6. The term-to-term rule of $5, 8, 11$

   ANSWER Add 3

7. The first term of $3n + 2$

   ANSWER 5

Medium · the standard question

7 PROBLEMS

1. The nth term of $5, 8, 11, 14$

   ANSWER $3n + 2$

2. The nth term of $2, 9, 16, 23$

   ANSWER $7n - 5$

3. The nth term of $6, 10, 14, 18$

   ANSWER $4n + 2$

4. The nth term of $20, 17, 14, 11$

   ANSWER $23 - 3n$

5. The 100th term of $4n - 1$

   ANSWER 399

6. The 50th term of $3n + 2$

   ANSWER 152

7. Is 47 a term of $3n + 2$?

   ANSWER Yes, the 15th

Hard · stretch and reason

7 PROBLEMS

1. Is 60 a term of $4n - 1$?

ANSWER No — $4n = 61$ is not whole

2. Which term of $7n - 5$ equals 100?

ANSWER The 15th

3. The nth term of $1, 4, 9, 16$

ANSWER n^2

4. The nth term of $1, 3, 6, 10$

ANSWER $n(n + 1) \div 2$

5. The first term of $4n + 2$ above 100

ANSWER 102, at $n = 25$

6. A sequence starts at 4 and adds 6: its nth term

ANSWER $6n - 2$

7. The first negative term of $23 - 3n$

ANSWER -1, at $n = 8$

07 Homework

Homework — 5 tasks

Test every nth term you write on the first two terms before moving on.

1. Write the first five terms of the sequence with nth term $6n - 1$.
2. Find the nth term of $7, 12, 17, 22, \ldots$.
3. Find the nth term of $30, 26, 22, 18, \ldots$.
4. Is 82 a term of the sequence with nth term $5n + 2$? Say which term, or why not.
5. A pattern of squares uses $4, 7, 10, \ldots$ tiles. Find the nth term and the 50th figure.

Recall — the cube and the cuboid

Faces, edges and vertices; volume as a product of three lengths; the litre.

LESSON 139 | 1 × 40 MIN | MATEMATIKA 6, TAKRORLASH | S7 15 DISTANCE, AREA AND VOLUME

LESSON CLOCK

10' | 12' | 12' | 6'

The two solids	10 min
Volume	12 min
Surface area	12 min
Units and capacity	6 min

LEARNING OBJECTIVES

- Name the faces, edges and vertices of a cuboid.
- Find the volume of a cuboid and of a cube.
- Find the total surface area from the three pairs of faces.
- Convert between cm^3, m^3 and litres.

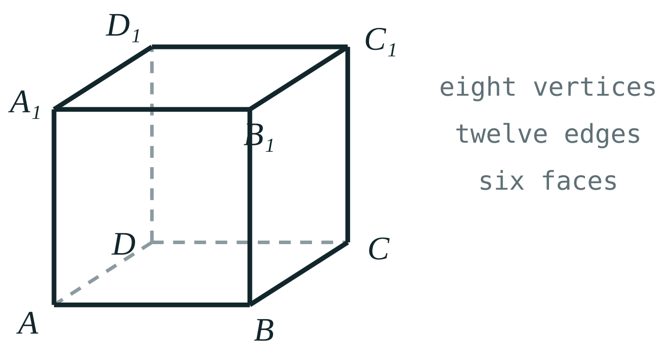
TEXTBOOK

National	pp. 400–403
Cambridge	Stage 7, pp. 156–160
Workbook	Exercise 15.1

01 Explanation

The two solids

A cuboid has six rectangular faces, meeting in twelve edges at eight vertices. A cube is the cuboid whose twelve edges are all the same length.



A cube — six faces, twelve edges, eight vertices

SOLID	FACES	EDGES	VERTICES	FACES ARE
cuboid	*6*	*12*	*8*	rectangles, equal in opposite pairs
cube	*6*	*12*	*8*	six equal squares

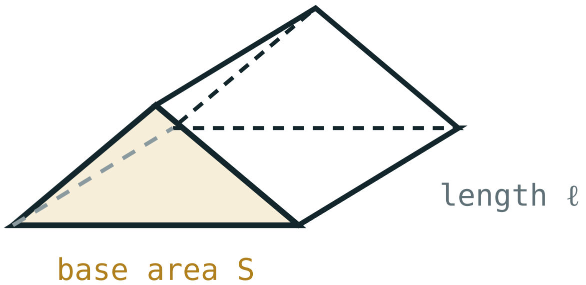
THREE LENGTHS DESCRIBE A CUBOID COMPLETELY

Length, width and height — usually written a, b, c. For a cube all three are the same, so one number is enough.

Volume

$V = abc$ and for a cube $V = a^3$

The bottom layer holds ab unit cubes, and there are c such layers.



$V = S \cdot \ell$

Area of the base multiplied by the height — the rule behind every prism

SOLID	DIMENSIONS	WORKING	VOLUME
cuboid	$5 \times 4 \times 3$ cm	$5 \cdot 4 \cdot 3$	60 cm^3
cuboid	$8 \times 5 \times 2$ cm	$8 \cdot 5 \cdot 2$	80 cm^3
cuboid	$10 \times 6 \times 4$ cm	$10 \cdot 6 \cdot 4$	240 cm^3
cube	4 cm edge	4^3	64 cm^3

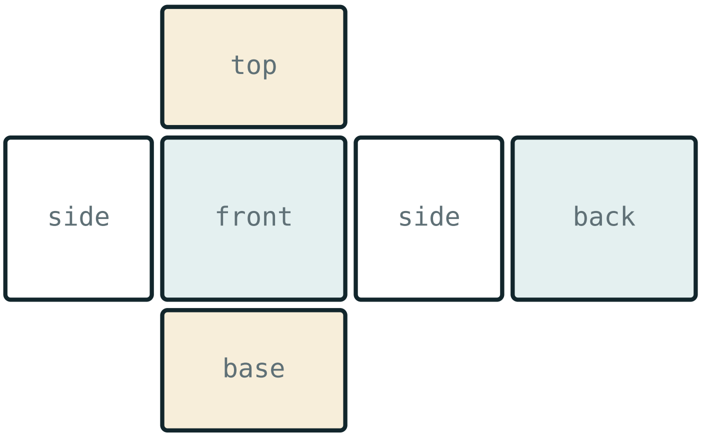
ALL THREE LENGTHS IN THE SAME UNIT

A box 1 m by 50 cm by 40 cm is $100 \times 50 \times 40 = 200\,000$ cm³, not $1 \times 50 \times 40$. Convert first, multiply second.

Surface area

The six faces come in three equal pairs, so the total is twice the sum of three rectangles.

$S = 2(ab + bc + ac)$ and for a cube $S = 6a^2$



The net — the six faces laid flat, which is what the formula adds up

SOLID	THE THREE RECTANGLES	SUM	SURFACE AREA
$5 \times 4 \times 3$	$20, 12, 15$	47	94 cm²
$8 \times 5 \times 2$	$40, 10, 16$	66	132 cm²
$10 \times 6 \times 4$	$60, 24, 40$	124	248 cm²
cube, edge 4	$16, 16, 16$	48	96 cm²

AREA IS SQUARED, VOLUME IS CUBED

Two lengths multiplied give cm^2; three give cm^3. An answer in the wrong unit is almost always the wrong quantity.

Units and capacity

UNIT	EQUALS	EVERYDAY SIZE
$1\ cm^3$	$1\ ml$	a small sugar cube
$1000\ cm^3$	1 litre	a cube of edge 10 cm
$1\ m^3$	$1\ 000\ 000\ cm^3 = 1000$ litres	a cube of edge 1 m

1 M³ IS A MILLION CM³, NOT A HUNDRED

Each of the three lengths is multiplied by 100, so the volume is multiplied by $100^3 = 1\ 000\ 000$. The same trap turns 1 m² into $10\ 000$ cm², not 100.

02 Worked examples

EXAMPLE 1 **Find the volume and the surface area of a cuboid 5 cm by 4 cm by 3 cm.**

1. $V = 5 \cdot 4 \cdot 3 = 60$ cm³.
2. The three rectangles: $20, 12, 15$.
3. $S = 2 \cdot 47 = 94$ cm².

   Volume in cm^3, area in cm^2 ✓

ANSWER

60 cm³ and 94 cm²

EXAMPLE 2 **A cube has edge 6 cm. Find its volume and surface area.**

1. $V = 6^3 = 216$ cm³.
2. $S = 6 \cdot 6^2 = 6 \cdot 36$.
3. $S = 216$ cm².
   The numbers agree only for edge 6 — a coincidence, not a rule.

ANSWER

216 cm³ and 216 cm²

EXAMPLE 3 **A tank measures 50 cm by 40 cm by 30 cm. How many litres does it hold?**

1. $V = 50 \cdot 40 \cdot 30 = 60\,000$ cm³.
2. 1000 cm³ is 1 litre.
3. $60\,000 \div 1000 = 60$ litres.

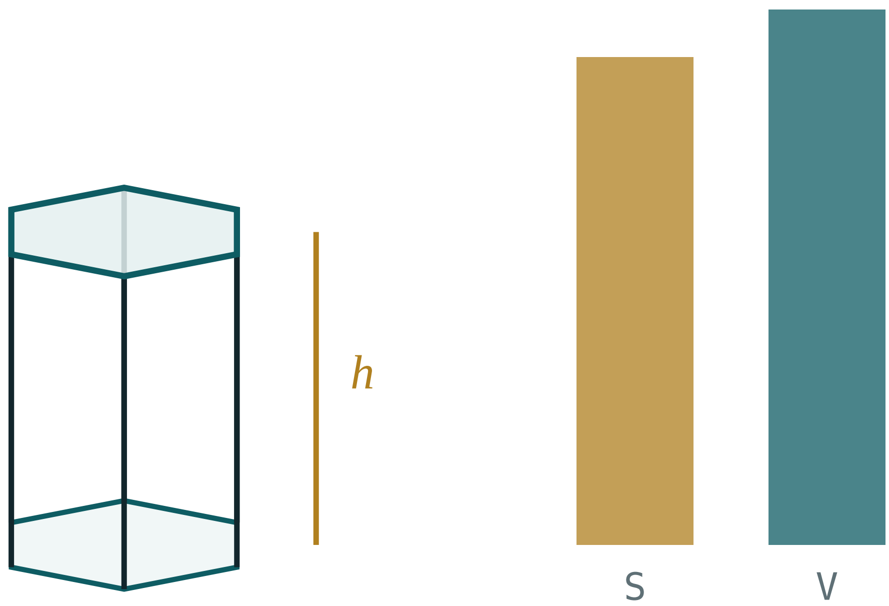
ANSWER

60 litres

03 Interactive model

Hold up a chalk box and ask for the volume and the surface area of the same object; separating “how much fits in” from “how much wraps round” is the whole lesson.

Base area times height



base area B **41.6** | perimeter P **24** | total S **227.1** | volume V **249.4**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Cuboid	Parallelepiped	Прямоугольный параллелепипед
Cube	Kub	Куб
Face	Yoq	Грань
Edge	Qirra	Ребро
Vertex	Uchi	Вершина
Volume	Hajm	Объём
Surface area	Sirt yuzasi	Площадь поверхности
Litre	Litr	Литр

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A cuboid has:

- *4* faces
- *6* faces
- *8* faces
- *12* faces

The volume of a $5 \times 4 \times 3$ cuboid is:

- *12*
- *60*
- *94*
- *47*

The surface area of that cuboid is:

- *47*
- *60*
- *94*
- *188*

A cube of edge 4 cm has volume:

- 12
- 16
- 64
- 96

1 litre is:

- 100 cm³
- 1000 cm³
- 1 m³
- 10 cm³

1 m³ in cm³ is:

- 100
- $10\ 000$
- $1\ 000\ 000$
- 1000

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. The number of faces of a cuboid

ANSWER 6

2. The number of edges of a cuboid

ANSWER 12

3. The number of vertices of a cuboid

ANSWER 8

4. The volume of a cube of edge 4 cm

ANSWER 64 cm³

5. The volume of a $5 \times 4 \times 3$ cm cuboid

ANSWER 60 cm³

6. The surface area of a cube of edge 4 cm

ANSWER 96 cm²

7. 1 litre in cm³

ANSWER 1000 cm³

Medium · the standard question

7 PROBLEMS

1. The volume of an $8 \times 5 \times 2$ cm cuboid

   ANSWER 80 cm³

2. The surface area of a $5 \times 4 \times 3$ cm cuboid

   ANSWER 94 cm²

3. The surface area of an $8 \times 5 \times 2$ cm cuboid

   ANSWER 132 cm²

4. The volume of a cube of edge 10 cm, in litres

   ANSWER 1 litre

5. The volume of a $10 \times 6 \times 4$ cm cuboid

   ANSWER 240 cm³

6. The surface area of a cube of edge 5 cm

   ANSWER 150 cm²

7. 1 m³ in litres

   ANSWER 1000 litres

Hard · stretch and reason

7 PROBLEMS

1. A cube has volume 216 cm³: its edge

   ANSWER 6 cm

2. …and its surface area

   ANSWER 216 cm²

3. A $12 \times 5 \times h$ cm cuboid has volume 300 cm³: find h

   ANSWER 5 cm

4. 2 m³ in cm³

   ANSWER $2\ 000\ 000$ cm³

5. A tank $50 \times 40 \times 30$ cm, in litres

   ANSWER 60 litres

6. How many cubes of edge 3 cm fill a cube of edge 6 cm?

   ANSWER 8

7. Which cube has surface area equal in number to its volume?

   ANSWER Edge 6, both 216

07 Homework

Homework — 5 tasks

Write the unit on every answer; it is the quickest check that the right quantity was found.

1. Find the volume and the surface area of a cuboid 7 cm by 4 cm by 2 cm.
2. Find the volume and the surface area of a cube of edge 5 cm.
3. A box is 1 m by 60 cm by 50 cm. Find its volume in cm³.
4. How many litres does a tank 80 cm by 50 cm by 25 cm hold?
5. Explain in one sentence why 1 m³ is $1\ 000\ 000$ cm³ and not 100 cm³.

Finding an edge of a cuboid

The volume formula run backwards — and the surface-area formula run backwards too.

LESSON 140–142 | 3 × 40 MIN | MATEMATIKA 6, §29 | S7 15 DISTANCE, AREA AND VOLUME

LESSON CLOCK

25' | 30' | 35' | 25' | 5

Reversing the volume formula	25 min
The edge of a cube	30 min
From the surface area	35 min
Depth and height in context	25 min
Homework	5 min

LEARNING OBJECTIVES

- Find the third edge of a cuboid from its volume and two edges.
- Find the edge of a cube from its volume or its surface area.
- Find a missing edge from the total surface area of a cuboid.
- Solve depth-of-water and height-of-box problems in the right units.

TEXTBOOK

National	pp. 404–413
Cambridge	Stage 7, pp. 156–162
Workbook	Exercise 15.2

01 Explanation

Reversing the volume formula

$V = abc$ is a multiplication of three numbers. If two of them and the product are known, the third is a division.

$$c = \frac{V}{ab}$$

VOLUME	BASE	BASE AREA	MISSING EDGE
$120\ cm^3$	5×4	$20\ cm^2$	6 cm
$240\ cm^3$	10×6	$60\ cm^2$	4 cm
$84\ cm^3$	7×3	$21\ cm^2$	4 cm
$150\ cm^3$	5×5	$25\ cm^2$	6 cm

MULTIPLY THE TWO YOU HAVE, THEN DIVIDE ONCE

Finding the base area first turns a three-number problem into a two-number one. It also makes the units obvious: $cm^3 \div cm^2 = cm$.

The edge of a cube

For a cube all three edges are the same, so $V = a^3$ and the edge is the number whose cube is the volume.

VOLUME	QUESTION ASKED	EDGE
$27\ cm^3$	$a^3 = 27$	$3\ cm$
$64\ cm^3$	$a^3 = 64$	$4\ cm$
$343\ cm^3$	$a^3 = 343$	$7\ cm$
$512\ cm^3$	$a^3 = 512$	$8\ cm$
$1000\ cm^3$	$a^3 = 1000$	$10\ cm$

DO NOT DIVIDE THE VOLUME BY THREE

a^3 means $a \cdot a \cdot a$, not $3a$. A cube of volume 27 has edge 3, not 9 — and $3 \cdot 3 \cdot 3$ checks it in one line.

Keeping the first ten cubes in mind — $1, 8, 27, 64, 125, 216, 343, 512, 729, 1000$ — makes almost every question of this kind immediate.

From the surface area

For a cube, $S = 6a^2$: divide by six, then take the square root.

SURFACE AREA	A^2	EDGE
$54\ cm^2$	9	$3\ cm$
$96\ cm^2$	16	$4\ cm$
$150\ cm^2$	25	$5\ cm$
$216\ cm^2$	36	$6\ cm$

For a cuboid with two edges known the same idea gives an equation instead of a division. With $a = 5$, $b = 4$ and $S = 94$:

> $2(20 + 4c + 5c) = 94$, so $40 + 18c = 94$, $18c = 54$, $c = 3$

TWO EDGES	SURFACE AREA	EQUATION	THIRD EDGE
5 and 4	$94\ cm^2$	$40 + 18c = 94$	$3\ cm$
8 and 5	$132\ cm^2$	$80 + 26c = 132$	$2\ cm$
10 and 6	$248\ cm^2$	$120 + 32c = 248$	$4\ cm$
6 and 4	$108\ cm^2$	$48 + 20c = 108$	$3\ cm$

THE FACE YOU KNOW IS A CONSTANT, THE OTHER TWO ARE THE UNKNOWN

Only the pair of faces built from the two known edges contributes a fixed number. The other four faces all carry c, which is why the equation is always linear.

Depth and height in context

Almost every practical question is the volume formula reversed, with a unit conversion in front of it.

QUESTION	VOLUME IN ONE UNIT	BASE AREA	ANSWER
a tank 80×50 cm holding 120 litres: its depth	$120\,000\ cm^3$	$4000\ cm^2$	$30\ cm$
a pool 25×10 m holding 500 m³: its depth	$500\ m^3$	$250\ m^2$	$2\ m$
a box of 360 cm³ on a 12×6 base: its height	$360\ cm^3$	$72\ cm^2$	$5\ cm$
a brick of 1920 cm³, 20 long and 12 wide	$1920\ cm^3$	$240\ cm^2$	$8\ cm$

LITRES ARE NOT LENGTHS

A capacity in litres must become cm^3 — multiply by 1000 — before it can be divided by an area in cm^2. Dividing 120 by 4000 gives 0.03, which is the right arithmetic on the wrong numbers.

02 Worked examples

EXAMPLE 1 **A cuboid has volume 240 cm³ and a base 10 cm by 6 cm. Find its height.**

1. Base area $10 \cdot 6 = 60$ cm².
2. $240 \div 60$.
3. $= 4$ cm.
   Check: $10 \cdot 6 \cdot 4 = 240$ ✓

ANSWER

4 cm

EXAMPLE 2 **A cube has volume 512 cm³. Find its edge.**

1. We need a with $a^3 = 512$.
2. $8 \cdot 8 = 64$ and $64 \cdot 8 = 512$.
3. $a = 8$ cm.
   Not $512 \div 3$ — the edges multiply, they do not add.

ANSWER

8 cm

EXAMPLE 3 **A cuboid is 5 cm by 4 cm and has surface area 94 cm². Find the third edge.**

1. $S = 2(5 \cdot 4 + 4c + 5c) = 94$.
2. $40 + 18c = 94$, so $18c = 54$.
3. $c = 3$ cm.

   Check: $2(20 + 12 + 15) = 94$ ✓

ANSWER

3 cm

03 Interactive model

Give the volume and one pair of edges and let the class find the third by trial before the division is written down; the division then looks like a shortcut, not a rule.

Working the formula backwards

A cuboid of volume 120 cm³ on a 5×4 base has height:

- 6 cm
- 4 cm
- 24 cm
- 3 cm

A cuboid of volume 84 cm³ on a 7×3 base has height:

- 3
- 4
- 12
- 21

A cube of volume 343 cm³ has edge:

- 6
- 7
- 8
- 49

A cube of volume 27 cm³ has edge:

- 9
- 3
- 27
- 13.5

A cube of surface area 150 cm² has edge:

- 5
- 25
- 6
- 12.5

A cuboid 8 by 5 with $S = 132$ cm² has third edge:

- 2
- 3
- 4
- 5

A tank 80×50 cm holding 120 litres is deep:

- 0.03 cm
- 30 cm
- 3 cm
- 300 cm

To find a missing edge from the volume you:

- divide by 3
- divide by the base area
- take the square root
- multiply

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Missing edge	Noma’lum qirra	Неизвестное ребро
Base	Asos	Основание
Height	Balandlik	Высота
Depth	Chuqurlik	Глубина
Base area	Asos yuzasi	Площадь основания
Cube root	Kub ildiz	Кубический корень
Capacity	Sig‘im	Вместимость
To rearrange	O‘zgartirib yozish	Выразить

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

From $V = abc$, the edge c equals:

- $V - ab$
- $V \div (ab)$
- $V \div 3$
- Vab

A cuboid of volume 240 cm³ on a 10×6 base is:

- 4 cm high
- 24 cm high
- 40 cm high
- 60 cm high

A cube of volume 1000 cm³ has edge:

- 10 cm
- 100 cm
- 333 cm
- 31.6 cm

A cube of surface area 96 cm² has edge:

- 16
- 4
- 6
- 8

A capacity in litres must first be turned into:

- cm^2
- cm^3
- metres
- nothing

The units of volume divided by area are:

- cm^3
- cm^2
- cm
- litres

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $V = 120$ cm³ on a 5×4 base: the height

   ANSWER 6 cm

2. $V = 240$ cm³ on a 10×6 base: the height

   ANSWER 4 cm

3. $V = 84$ cm³ on a 7×3 base: the height

   ANSWER 4 cm

4. A cube of volume 27 cm³: the edge

   ANSWER 3 cm

5. A cube of volume 343 cm³: the edge

   ANSWER 7 cm

6. A cube of surface area 96 cm²: the edge

   ANSWER 4 cm

7. From $V = abc$ the third edge is

   ANSWER $V \div (ab)$

Medium · the standard question

7 PROBLEMS

1. A cube of volume 512 cm³: the edge

ANSWER 8 cm

2. A cube of surface area 150 cm²: the edge

ANSWER 5 cm

3. $V = 150$ cm³ on a 5×5 base: the height

ANSWER 6 cm

4. A box of 360 cm³ on a 12×6 base: the height

ANSWER 5 cm

5. A brick of 1920 cm³, 20 cm by 12 cm: the height

ANSWER 8 cm

6. A cube of surface area 54 cm²: the edge

ANSWER 3 cm

7. A cube of volume 1000 cm³: the edge

ANSWER 10 cm

Hard · stretch and reason

7 PROBLEMS

1. A cuboid 5 by 4 with $S = 94$ cm²: the third edge

   ANSWER 3 cm

2. A cuboid 8 by 5 with $S = 132$ cm²: the third edge

   ANSWER 2 cm

3. A cuboid 10 by 6 with $S = 248$ cm²: the third edge

   ANSWER 4 cm

4. A tank 80×50 cm holding 120 litres: the depth

   ANSWER 30 cm

5. A pool 25 m by 10 m holding 500 m³: the depth

   ANSWER 2 m

6. A cuboid 12 by 3 has the volume of a cube of edge 6: its third edge

   ANSWER 6 cm

7. A cuboid 6 by 4 with $S = 108$ cm²: its volume

   ANSWER 72 cm³

07 Homework

Homework — 5 tasks

Check every answer by putting the edge back into the formula it came from.

1. A cuboid has volume 180 cm³ and a base 9 cm by 5 cm. Find its height.
2. A cube has volume 729 cm³. Find its edge.
3. A cube has surface area 294 cm². Find its edge.
4. A tank with base 60 cm by 40 cm holds 96 litres. Find its depth.
5. A cuboid is 7 cm by 3 cm and has surface area 122 cm². Find the third edge.

Finding the area of one face of a cube and of a cuboid

Six faces in three equal pairs — and what a single one of them is worth.

LESSON 143–145 | 3 × 40 MIN | MATEMATIKA 6, §30 | S7 15 DISTANCE, AREA AND VOLUME

LESSON CLOCK

The three pairs of faces	25 min
One face of a cube	30 min
A face from the volume	35 min
Which faces the question needs	25 min
Homework	5 min

LEARNING OBJECTIVES

- Name the three different faces of a cuboid and find each area.
- Find one face of a cube from its total surface area.
- Find a face area from the volume and the perpendicular edge.
- Choose which faces a practical question actually needs.

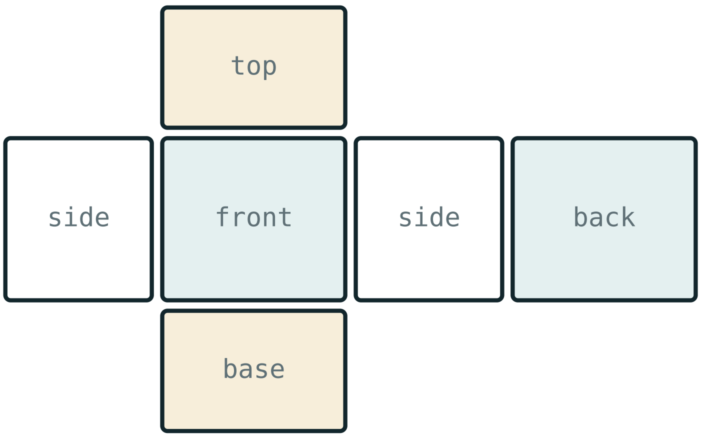
TEXTBOOK

National	pp. 414–423
Cambridge	Stage 7, pp. 160–166
Workbook	Exercise 15.3

01 Explanation

The three pairs of faces

A cuboid with edges a, b, c has only three different faces — ab, bc and ac — and each of them appears twice, on opposite sides.



surface area = the area of the whole net

The net of a cuboid: three pairs of equal rectangles

CUBOID	AB	BC	AC	TOTAL
$5 \times 4 \times 3$	$20\ cm^2$	$12\ cm^2$	$15\ cm^2$	$94\ cm^2$
$8 \times 5 \times 2$	$40\ cm^2$	$10\ cm^2$	$16\ cm^2$	$132\ cm^2$
$10 \times 6 \times 4$	$60\ cm^2$	$24\ cm^2$	$40\ cm^2$	$248\ cm^2$
$6 \times 4 \times 3$	$24\ cm^2$	$12\ cm^2$	$18\ cm^2$	$108\ cm^2$

THE LARGEST FACE USES THE TWO LARGEST EDGES

In the $5 \times 4 \times 3$ cuboid the largest face is $5 \cdot 4 = 20$ and the smallest is $4 \cdot 3 = 12$. Pairing the edges off in the three possible ways is the whole exercise.

One face of a cube

All six faces of a cube are the same square, so the total is six of them and one of them is the total divided by six.

$$one\ face = \frac{S}{6} \text{ and } a = \text{the square root of that}$$

TOTAL SURFACE	ONE FACE	EDGE
$54\ cm^2$	$9\ cm^2$	$3\ cm$
$96\ cm^2$	$16\ cm^2$	$4\ cm$
$150\ cm^2$	$25\ cm^2$	$5\ cm$
$216\ cm^2$	$36\ cm^2$	$6\ cm$
$294\ cm^2$	$49\ cm^2$	$7\ cm$

DIVIDE BY SIX BEFORE TAKING ANY ROOT

The square root of 96 is not the edge of anything here. $96 \div 6 = 16$ is a face, and 4 is the edge. Two steps, always in that order.

A face from the volume

Volume is a face multiplied by the edge standing perpendicular to it. Reverse that and a face area falls out of a division.

$$face = \frac{\text{V}}{\text{perpendicular edge}}$$

VOLUME	EDGE GIVEN	FACE PERPENDICULAR TO IT
120 cm³	*height 6 cm*	*20 cm²*
240 cm³	*height 4 cm*	*60 cm²*
84 cm³	*height 4 cm*	*21 cm²*
360 cm³	*height 5 cm*	*72 cm²*

THE UNITS CONFIRM THE CHOICE

$cm^3 \div cm = cm^2$ — a face. Dividing by an area would have given a length instead, which is the previous topic, not this one.

Which faces the question needs

Practical questions rarely want all six faces. Read what is being covered before adding anything up.

TASK	FACES COUNTED	WORKING	ANSWER
wrapping a $20 \times 15 \times 10$ cm box	all six	$2(300 + 150 + 200)$	$1300\ cm^2$
card for an open box $8 \times 5 \times 4$ cm	five — no lid	$40 + 64 + 40$	$144\ cm^2$
painting the walls of a $5 \times 4 \times 3$ m room	four walls	$2(15) + 2(12)$	$54\ m^2$
the same room with the ceiling	four walls and the top	$54 + 20$	$74\ m^2$

A FLOOR IS NOT PAINTED, A LID MAY NOT EXIST

Each of these questions has the same four numbers in it and four different answers. The reading, not the arithmetic, is what is being tested.

02 Worked examples

EXAMPLE 1 **Find the three different face areas of a $5 \times 4 \times 3$ cm cuboid, and the total surface area.**

1. Pair the edges: $5 \cdot 4 = 20$, $4 \cdot 3 = 12$, $5 \cdot 3 = 15$.
2. Each appears twice.
3. $2(20 + 12 + 15) = 94$ cm².

   The largest face uses the two largest edges ✓

ANSWER

20, 12 and 15 cm²; total 94 cm²

EXAMPLE 2 **A cube has surface area 294 cm². Find the area of one face and the edge.**

1. $294 \div 6 = 49$ cm² for one face.
2. That face is a square, so the edge is its square root.
3. 7 cm.

   Check: $6 \cdot 49 = 294$ ✓

ANSWER

49 cm² and 7 cm

EXAMPLE 3 **How much card is needed for an open box (no lid) 8 cm by 5 cm by 4 cm deep?**

1. Base: $8 \cdot 5 = 40$ cm².
2. Two long walls: $2 \cdot 8 \cdot 4 = 64$. Two short walls: $2 \cdot 5 \cdot 4 = 40$.
3. $40 + 64 + 40 = 144$ cm².

   The closed box would need 184 — one lid more.

ANSWER

144 cm²

03 Interactive model

Open out a cardboard box along its edges before the lesson; laying the net flat makes the three pairs visible in a way no drawing does.

Which face, and how big

The largest face of a $5 \times 4 \times 3$ cuboid is:

- 12
- 15
- 20
- 60

The smallest face of that cuboid is:

- 12
- 15
- 20
- 9

A cube of surface area 150 cm² has one face of:

- 25
- 50
- 30
- 12.5

That cube has edge:

- 25
- 5
- 6
- 12.5

A cuboid of volume 240 cm³ and height 4 cm has base:

- 60 cm²
- 960 cm²
- 60 cm
- 24 cm²

Wrapping a $20 \times 15 \times 10$ cm box needs:

- 650
- 1300
- 3000
- 900

An open box $8 \times 5 \times 4$ cm needs:

- 184
- 144
- 104
- 40

The four walls of a $5 \times 4 \times 3$ m room measure:

- 54 m²
- 74 m²
- 94 m²
- 60 m²

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Face	Yoq	Грань
Opposite faces	Qarama-qarshi yoqlar	Противоположные грани
Base face	Asos yog‘i	Основание
Side face	Yon yoq	Боковая грань
Lateral surface	Yon sirt	Боковая поверхность
Total surface area	To‘la sirt yuzasi	Полная площадь поверхности
Net	Yoyilma	Развёртка
Open box	Qopqog‘i yo‘q quti	Открытая коробка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A cuboid has how many *different* face areas?

- 2
- 3
- 6
- 12

One face of a cube is the total surface:

- times six
- divided by six
- squared
- divided by four

The area of a face of an $8 \times 5 \times 2$ cuboid cannot be:

- 40
- 16
- 10
- 80

Volume divided by an edge gives:

- a length
- an area
- a volume
- nothing

An open box needs how many faces of card?

- 4
- 5
- 6
- 3

Painting only the walls of a room leaves out:

- the walls
- the floor and ceiling
- one wall
- nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. The largest face of a $5 \times 4 \times 3$ cm cuboid

ANSWER 20 cm²

2. The smallest face of that cuboid

ANSWER 12 cm²

3. The three faces of an $8 \times 5 \times 2$ cm cuboid

ANSWER 40, 16 and 10 cm²

4. One face of a cube of edge 4 cm

ANSWER 16 cm²

5. One face of a cube of surface area 96 cm²

ANSWER 16 cm²

6. The number of faces of a cuboid

ANSWER 6

7. The faces of a cube are

ANSWER six equal squares

Medium · the standard question

7 PROBLEMS

1. One face of a cube of surface area 150 cm²

ANSWER 25 cm²

2. One face of a cube of surface area 294 cm²

ANSWER 49 cm²

3. The base of a cuboid of volume 120 cm³ and height 6 cm

ANSWER 20 cm²

4. The base of a cuboid of volume 240 cm³ and height 4 cm

ANSWER 60 cm²

5. The three faces of a $10 \times 6 \times 4$ cm cuboid

ANSWER 60, 40 and 24 cm²

6. The total surface of a $6 \times 4 \times 3$ cm cuboid

ANSWER 108 cm²

7. The base of a cuboid of volume 360 cm³ and height 5 cm

ANSWER 72 cm²

Hard · stretch and reason

7 PROBLEMS

1. A cube of surface area 216 cm²: one face and the edge

   ANSWER 36 cm² and 6 cm

2. The four walls of a 5 m by 4 m by 3 m room

   ANSWER 54 m²

3. The same room including the ceiling

   ANSWER 74 m²

4. Card for an open box $8 \times 5 \times 4$ cm

   ANSWER 144 cm²

5. Paper to wrap a $20 \times 15 \times 10$ cm box

   ANSWER 1300 cm²

6. A cuboid has faces 12, 15 and 20 cm²: its edges

   ANSWER 3, 4 and 5 cm

7. …and its volume

   ANSWER 60 cm³

07 Homework

Homework — 5 tasks

For every task, say first which faces the question is asking about.

1. List the three different face areas of a $9 \times 4 \times 2$ cm cuboid and find the total surface area.
2. A cube has surface area 384 cm². Find one face and the edge.
3. A cuboid has volume 210 cm³ and height 7 cm. Find the area of its base.
4. How much card is needed for an open box 10 cm by 6 cm by 4 cm deep?
5. A room is 6 m by 5 m by 3 m. Find the area of its four walls.

Word problems on volume

Tanks that fill, crates that pack, water that rises, and concrete that has to be paid for.

LESSON 146–149 | 4 × 40 MIN | MATEMATIKA 6, §31 | S7 15 DISTANCE, AREA AND VOLUME

LESSON CLOCK

30' | 35' | 40' | 45' | 10

Filling and emptying	30 min
How many fit	35 min
Rising water and displacement	40 min
Volume, mass and cost	45 min
Homework	10 min

LEARNING OBJECTIVES

- Find a filling or emptying time from a capacity and a rate.
- Decide how many small solids pack into a larger one.
- Use the rise in a water level to find a volume, and the reverse.
- Turn a volume into a mass or a cost with a rate per unit volume.

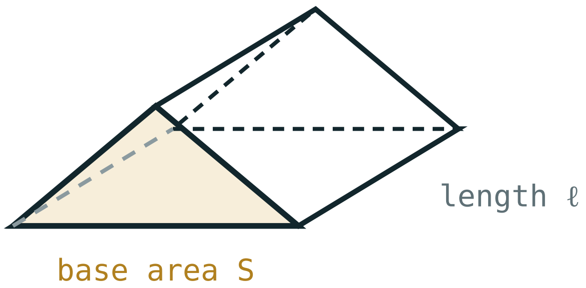
TEXTBOOK

National	pp. 424–435
Cambridge	Stage 7, pp. 162–170
Workbook	Exercise 15.4–15.5

01 Explanation

Filling and emptying

Every one of these problems has the same two steps: find the volume in litres, then divide by the rate. The work is in the units, not the arithmetic.



$$V = S \cdot \ell$$

Base area times height — the volume every tank problem starts from

TANK	CAPACITY	RATE	TIME
$60 \times 40 \times 50$ cm	120 litres	8 l/min	15 min
1 m $\times 80 \times 50$ cm	400 litres	10 l/min	40 min
a *200*-litre drum	200 litres	5 l/min	40 min
a *90*-litre tank draining	90 litres	3 l/min	30 min

THE METRE IN THE SECOND ROW IS THE TRAP

1 m is *100* cm, so the volume is $100 \cdot 80 \cdot 50 = 400\,000$ cm³. Leaving the *1* in place gives *4000* cm³ — four litres, for a tank a person could sit in.

How many fit

When the small solid divides the large one exactly along each edge, count along each edge and multiply the three counts.

CONTAINER	ITEM	ALONG EACH EDGE	HOW MANY
$40 \times 32 \times 30$ cm	$10 \times 8 \times 5$ cm	4, 4, 6	96
cube of edge *12* cm	cube of edge *3* cm	4, 4, 4	64
$10 \times 8 \times 6$ cm	cube of edge *2* cm	5, 4, 3	60
1 m $\times 40 \times 24$ cm	$20 \times 10 \times 8$ cm	5, 4, 3	60

DIVIDING THE VOLUMES IS ONLY AN UPPER LIMIT

$38\,400 \div 400 = 96$ happens to be right in the first row because the boxes fit along every edge. If they did not, the volume division would still give *96* and the true answer would be smaller — so count along the edges, and use the division only as a check.

Rising water and displacement

Water in a tank is a cuboid whose base is the tank's base and whose height is the depth. Two questions follow at once.

$$\textit{volume of water} = \textit{base area} \times \textit{depth} \text{ so } \textit{depth} = \frac{\text{volume}}{\text{base area}}$$

QUESTION	WORKING	ANSWER
water 6 cm deep on a 40×25 cm base	$1000 \cdot 6$	*6 litres*
3 litres poured onto a 30×20 cm base	$3000 \div 600$	*5 cm deep*
a stone raises 6 cm to 7.5 cm on a 40×25 base	$1000 \cdot 1.5$	*1500 cm³ of stone*
a cube of edge 5 cm sunk in a 25×20 tank	$125 \div 500$	*a rise of 0.25 cm*

A SUNKEN OBJECT TAKES EXACTLY ITS OWN VOLUME OF ROOM

That is why the rise in level, multiplied by the base area, gives the volume of an object of any shape at all — the oldest measuring trick in mathematics.

Volume, mass and cost

A rate per unit volume turns the volume into whatever the question is really about.

OBJECT	VOLUME	RATE	ANSWER
an iron block $10 \times 5 \times 2$ cm	$100\ cm^3$	7.8 *g per* cm^3	*780 g*
a full 120-litre tank of water	*120 litres*	*1 kg per litre*	*120 kg*
a concrete slab $4 \times 3 \times 0.1$ m	$1.2\ m^3$	2.4 *t per* m^3	*2.88 t*
sand for a pit $2 \times 1.5 \times 0.4$ m	$1.2\ m^3$	*90 000 so‘m per* m^3	*108 000 so‘m*

MATCH THE UNIT OF THE RATE TO THE UNIT OF THE VOLUME

A rate in g per cm^3 needs the volume in cm^3; a rate per m^3 needs m^3. Converting the volume first is always safer than trying to convert the rate.

Mixed problems

Longer questions simply chain these steps together.

PROBLEM	STEPS	ANSWER
a $50 \times 40 \times 30$ cm tank filled to two-thirds	$60 \cdot \frac{2}{3}$	*40 litres*
two taps, 6 and 4 l/min, on a 120-litre tank	$120 \div 10$	*12 min*
a 300-litre tank filling at 12 and leaking at 2	$300 \div 10$	*30 min*
a 120-litre tank half full, topped up at 4 l/min	$60 \div 4$	*15 min*

TWO TAPS TOGETHER ADD THEIR RATES

Exactly as two people working together add their rates of work. A leak subtracts, which is the same idea with a sign.

02 Worked examples

EXAMPLE 1

A tank measures *60* cm by *40* cm by *50* cm. A tap delivers *8* litres a minute. How long does it take to fill?

1. $60 \cdot 40 \cdot 50 = 120\,000$ cm³.
2. That is 120 litres.
3. $120 \div 8 = 15$ minutes.

   Check: $8 \cdot 15 = 120$ ✓

ANSWER

15 minutes

EXAMPLE 2 **How many boxes 10 cm by 8 cm by 5 cm fit into a crate 40 cm by 32 cm by 30 cm?**

1. Along the edges: $40 \div 10 = 4$, $32 \div 8 = 4$, $30 \div 5 = 6$.
2. $4 \cdot 4 \cdot 6$.
3. $= 96$ boxes.

   Volume check: $38\,400 \div 400 = 96$ ✓

ANSWER

96 boxes

EXAMPLE 3 **A tank with base 40 cm by 25 cm holds water 6 cm deep. A stone is lowered in and the level rises to 7.5 cm. Find the volume of the stone.**

1. Base area $40 \cdot 25 = 1000$ cm².
2. The rise is 1.5 cm.
3. $1000 \cdot 1.5 = 1500$ cm³.

   The stone takes exactly its own volume of room.

ANSWER

1500 cm³

03 Interactive model

Do the displacement question with a real jar and a stone; the number on the board becomes a measurement the class has just watched being made.

Tanks, crates and stones

A $60 \times 40 \times 50$ cm tank holds:

- 12 litres
- 120 litres
- 1200 litres
- $120\,000$ litres

Filling it at 8 l/min takes:

- 12
- 15
- 20
- 960

A 1 m by 80 cm by 50 cm tank holds:

- 4 litres
- 40 litres
- 400 litres
- 4000 litres

Cubes of edge 3 cm in a cube of edge 12 cm:

- 4
- 16
- 64
- 48

3 litres on a 30×20 cm base is deep:

- 5 cm
- 50 cm
- 0.5 cm
- 6 cm

A rise of 1.5 cm on a 1000 cm² base means an object of:

- 150 cm³
- 1500 cm³
- 15 cm³
- 666 cm³

Concrete at 2.4 t per m³ in a 1.2 m³ slab weighs:

- 2 t
- 2.88 t
- 3.6 t
- 0.5 t

Taps of 6 and 4 l/min fill 120 litres in:

- 10
- 12
- 20
- 24

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Capacity	Sig‘im	Вместимость
To fill	To‘ldirmoq	Наполнять
To empty, to drain	Bo‘shatmoq	Опорожнять
Rate of flow	Oqim tezligi	Скорость потока
Water level	Suv sathi	Уровень воды
Displacement	Siqib chiqarish	Вытеснение
Mass	Massa	Масса
Per cubic metre	Bir kub metr uchun	За кубический метр

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A capacity in litres comes from a volume in:

- cm^2, divided by 1000
- cm^3, divided by 1000
- cm, times 1000
- m^3, times 100

To find a filling time you:

- multiply capacity by rate
- divide capacity by rate
- add them
- divide rate by capacity

The number of small boxes that fit is found by:

- dividing the volumes only
- counting along each edge
- adding the edges
- guessing

A sunken object raises the water by:

- its surface area
- its own volume ÷ the base area
- its mass
- its height

A rate of 7.8 g per cm³ needs the volume in:

- m^3
- litres
- cm^3
- cm^2

A tank filling at 12 and leaking at 2 l/min gains:

- 14
- 10
- 6
- 24

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. The capacity of a $60 \times 40 \times 50$ cm tank

   ANSWER 120 litres

2. A 120-litre tank filled at 8 l/min

   ANSWER 15 min

3. A 200-litre drum filled at 5 l/min

   ANSWER 40 min

4. A 90-litre tank draining at 3 l/min

   ANSWER 30 min

5. Cubes of edge 3 cm in a cube of edge 12 cm

   ANSWER 64

6. Water 6 cm deep on a 40×25 cm base

   ANSWER 6 litres

7. The mass of 1 litre of water

   ANSWER 1 kg

Medium · the standard question

7 PROBLEMS

1. The capacity of a 1 m by 80 cm by 50 cm tank

   ANSWER 400 litres

2. …filled at 10 l/min

   ANSWER 40 min

3. Boxes $10 \times 8 \times 5$ cm in a crate $40 \times 32 \times 30$ cm

   ANSWER 96

4. Cubes of edge 2 cm in a $10 \times 8 \times 6$ cm box

   ANSWER 60

5. 3 litres poured onto a 30×20 cm base: the depth

   ANSWER 5 cm

6. An iron block $10 \times 5 \times 2$ cm at 7.8 g per cm³

   ANSWER 780 g

7. The water in a full 120-litre tank

   ANSWER 120 kg

Hard · stretch and reason

7 PROBLEMS

1. A stone raises 6 cm to 7.5 cm on a 40×25 base: its volume

   ANSWER 1500 cm³

2. A cube of edge 5 cm sunk in a 25×20 cm tank: the rise

   ANSWER 0.25 cm

3. A concrete slab 4 m by 3 m by 0.1 m at 2.4 t per m³

   ANSWER 2.88 tonnes

4. Sand at $90\ 000$ so‘m per m³ for a pit $2 \times 1.5 \times 0.4$ m

   ANSWER $108\ 000$ so‘m

5. A $50 \times 40 \times 30$ cm tank filled to two-thirds

   ANSWER 40 litres

6. Taps of 6 and 4 l/min on a 120-litre tank

   ANSWER 12 min

7. A 300-litre tank filling at 12 and leaking at 2 l/min

   ANSWER 30 min

07 Homework

Homework — 5 tasks

Convert every length to one unit before the first multiplication, and write the unit on every line.

1. A tank is 70 cm by 50 cm by 40 cm. Find its capacity in litres.
2. That tank is filled by a tap delivering 7 litres a minute. How long does it take?
3. How many cubes of edge 4 cm fit into a box 20 cm by 16 cm by 12 cm?
4. 4.5 litres of water is poured into a tank with base 45 cm by 20 cm. How deep is the water?
5. Concrete weighs 2.4 tonnes per m³. Find the mass of a slab 5 m by 2 m by 0.2 m.

Think — volume in context

The same volume, six different boxes: which one uses the least card?

LESSON 150 | 1 × 40 MIN | MATEMATIKA 6, O‘YLAB KO‘R | S7 15 IN CONTEXT

LESSON CLOCK

10' | 10' | 12' | 8'

The packaging problem	10 min
Testing the boxes	10 min
What doubling does	12 min
Writing the report	8 min

LEARNING OBJECTIVES

- Find every cuboid with whole-number edges and a given volume.
- Compare their surface areas and identify the most economical.
- State what happens to volume and surface area when every edge is doubled.
- Turn the saving into a quantity of material and a cost.

TEXTBOOK

National	pp. 436–438
Cambridge	Stage 7, pp. 168–170
Workbook	Investigation 15

01 Explanation

The packaging problem

A factory must pack *216* cm³ of tea into a cuboid box with whole-number edges in centimetres. Every box holds the same tea. They do not use the same amount of card.

THE QUESTION

Among all cuboids of volume *216* cm³ with whole-number edges, which uses the least card — and is there a pattern that would let you answer the same question for any volume?

Start by listing boxes. Each one is three whole numbers whose product is *216*.

Testing the boxes

BOX	VOLUME	THE THREE FACES	CARD USED
$6 \times 6 \times 6$	$216\ cm^3$	$36, 36, 36$	$216\ cm^2$
$9 \times 6 \times 4$	$216\ cm^3$	$54, 24, 36$	$228\ cm^2$
$8 \times 9 \times 3$	$216\ cm^3$	$72, 27, 24$	$246\ cm^2$
$12 \times 6 \times 3$	$216\ cm^3$	$72, 18, 36$	$252\ cm^2$
$12 \times 9 \times 2$	$216\ cm^3$	$108, 18, 24$	$300\ cm^2$
$216 \times 1 \times 1$	$216\ cm^3$	$216, 1, 216$	$866\ cm^2$

THE CONJECTURE

The nearer the box is to a cube, the less card it uses; the long thin box is by far the worst. The cube is the cheapest cuboid for any given volume.

What doubling does

Now change the size instead of the shape. Double every edge of a cube and both quantities grow — but not at the same rate.

CUBE	VOLUME	SURFACE AREA	CARD PER CM³
edge 3 cm	$27\ cm^3$	$54\ cm^2$	2
edge 6 cm	$216\ cm^3$	$216\ cm^2$	1
edge 12 cm	$1728\ cm^3$	$864\ cm^2$	0.5

VOLUME GROWS FASTER THAN SURFACE AREA

Doubling the edges multiplies the volume by 8 and the surface area by only 4. That is why a large packet is cheaper per gram than a small one — and why a small animal loses heat faster than a large one.

Writing the report

Turn the finding into money, which is the form a factory understands.

QUANTITY	WORKING	ANSWER
card saved per box	$252 - 216$	36 cm²
cost saved per box at *2* so‘m per cm²	$36 \cdot 2$	*72 so‘m*
card saved on *10 000* boxes	$36 \cdot 10\,000$	36 m²
money saved on *10 000* boxes	$72 \cdot 10\,000$	*720 000 so‘m*

AND WHY IS THE CUBE NOT ALWAYS USED?

Because boxes must stand on shelves, be gripped by a hand, and hold a shape that is not itself a cube. A good report says what the mathematics recommends *and* what overrules it.

02 Worked examples

EXAMPLE 1 **Compare the card used by a $12 \times 6 \times 3$ box and a $6 \times 6 \times 6$ box of the same volume.**

1. Both hold *216* cm³.
2. $2(72 + 18 + 36) = 252$ cm² against $6 \cdot 36 = 216$ cm².
3. The cube saves *36* cm² per box.
   About *14%* less card ✓

ANSWER

252 against *216* cm²

EXAMPLE 2 **Find the card used by a $9 \times 6 \times 4$ box.**

1. The three faces: $54, 24, 36$.
2. $2(54 + 24 + 36)$.
3. $= 228$ cm².
   Between the cube and the $12 \times 6 \times 3$ box ✓

ANSWER

228 cm²

EXAMPLE 3 **A factory makes $10\ 000$ boxes a day. How much card does the cube save, in square metres?**

1. 36 cm² per box.
2. $36 \cdot 10\ 000 = 360\ 000$ cm².
3. $360\ 000 \div 10\ 000 = 36$ m².
   1 m² is $10\ 000$ cm².

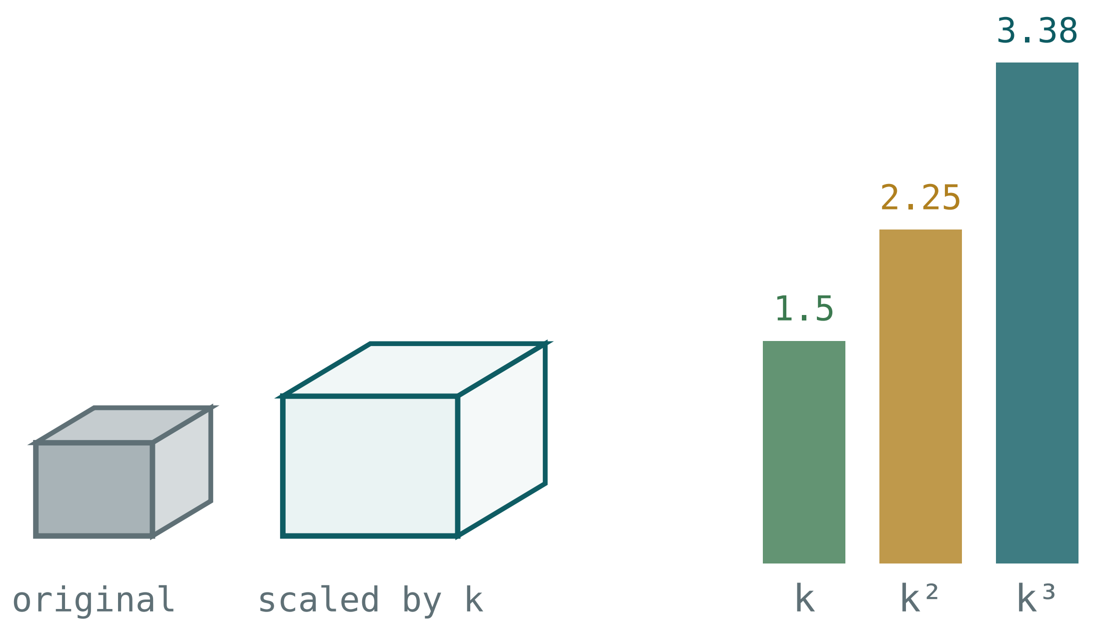
ANSWER

36 m² a day

03 Interactive model

Set k = 2 in the model and read the three multipliers aloud: 2, 4, 8. That single line is the whole of the third section.

Lengths, areas and volumes under a scale factor



k **1.5** | length × **1.5** | area ×k² **2.25** | volume ×k³ **3.38**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Packaging	Qadoqlash	Упаковка
Economical	Tejamkor	Экономичный
Material	Material	Материал
Whole-number edge	Butun sonli qirra	Целое ребро
Scale factor	O‘lchov koeffitsiyenti	Коэффициент подобия
To conjecture	Taxmin qilmoq	Выдвигать гипотезу
Saving	Tejamkorlik	Экономия
Per unit	Bir birlikka	На единицу

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Among cuboids of a fixed volume, the least card is used by:

- the longest
- the cube
- the flattest
- they are all equal

A $216 \times 1 \times 1$ box uses:

- 216 cm²
- 433 cm²
- 866 cm²
- 218 cm²

Doubling every edge multiplies the volume by:

- 2
- 4
- 6
- 8

It multiplies the surface area by:

- 2
- 4
- 6
- 8

Card per cm³ for a large cube compared with a small one is:

- larger
- smaller
- the same
- zero

The cube is not always used because:

- it is hard to make
- shelves and contents have shapes too
- it uses more card
- it holds less

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. The volume of a $6 \times 6 \times 6$ box

   ANSWER 216 cm³

2. The card used by that box

   ANSWER 216 cm²

3. The card used by a $12 \times 6 \times 3$ box

   ANSWER 252 cm²

4. The card used by a $9 \times 6 \times 4$ box

   ANSWER 228 cm²

5. Which of those three is the most economical?

   ANSWER The cube

6. The volume of a $9 \times 6 \times 4$ box

   ANSWER 216 cm³

7. The card used by a $216 \times 1 \times 1$ box

   ANSWER 866 cm²

Medium · the standard question

7 PROBLEMS

1. The card used by an $8 \times 9 \times 3$ box

   ANSWER 246 cm²

2. The card used by a $12 \times 9 \times 2$ box

   ANSWER 300 cm²

3. The most economical shape for a fixed volume

   ANSWER The cube

4. Order $6 \times 6 \times 6$, $9 \times 6 \times 4$, $12 \times 6 \times 3$ by card used

   ANSWER $216, 228, 252$ cm²

5. The volume of a cube of edge 3 cm

   ANSWER 27 cm³

6. The surface area of a cube of edge 3 cm

   ANSWER 54 cm²

7. The surface area of a cube of edge 6 cm

   ANSWER 216 cm²

Hard · stretch and reason

7 PROBLEMS

1. Doubling every edge multiplies the volume by

   ANSWER 8

2. …and the surface area by

   ANSWER 4

3. A cube of edge 10 cm: its volume and surface area

   ANSWER 1000 cm³ and 600 cm²

4. Card per cm³ for cubes of edge 3 and 6

   ANSWER 2 against 1

5. At 2 so‘m per cm², the cube saves against $12 \times 6 \times 3$

   ANSWER 72 so‘m a box

6. On $10\ 000$ boxes, the card saved

   ANSWER 36 m²

7. Why is a cube not always used?

   ANSWER Shelves, handling and the shape of the contents

07 Homework

Homework — the report

One page: the table of boxes, the conjecture, the doubling result, and the cost in so‘m.

1. List every cuboid with whole-number edges and volume 64 cm³, and find the card each uses.
2. State which is the most economical and whether it agrees with the conjecture.
3. Work out what happens to the volume and the surface area of a cube when every edge is trebled.
4. A factory makes 5000 boxes a day. Find the card saved by the best shape, in m².

5. Write two sentences on why a factory might still not choose the best shape.

Revision — mensuration of solids

Every formula of the block, forwards and backwards, with the units that decide the marks.

LESSON 151–154 | 4 × 40 MIN | MATEMATIKA 6, TAKRORLASH | S7 15 CONSOLIDATION

LESSON CLOCK

30' | 30' | 35' | 35' | 25' | 5

The formulae of the block	30 min
Forward questions	30 min
Reverse questions	35 min
Context and units	35 min
The mistakes that cost marks	25 min
Homework	5 min

LEARNING OBJECTIVES

- State and use every formula of the block from memory.
- Answer forward questions — volume and surface area from the edges.
- Answer reverse questions — an edge or a face from the volume or the surface area.
- Name the five mistakes that lose the most marks and avoid each one.

TEXTBOOK

National	pp. 439–448
Cambridge	Stage 7, pp. 156–172
Workbook	Review 15

01 Explanation

The formulae of the block

WANTED	FOR A CUBOID	FOR A CUBE
volume	$V = abc$	$V = a^3$
surface area	$S = 2(ab + bc + ac)$	$S = 6a^2$
one face	ab, bc or ac	$\frac{S}{6}$
a missing edge	$\frac{V}{ab}$	the cube root of V
the edge from S	solve $2(ab + bc + ac) = S$	the square root of $\frac{S}{6}$

TWO CONVERSIONS CARRY MOST OF THE MARKS

1 litre = 1000 cm³, and 1 m³ = $1\,000\,000$ cm³ = 1000 litres. Everything else in the block is a multiplication or its reverse.

Forward questions

SOLID	VOLUME	THE THREE FACES	SURFACE AREA
$5 \times 4 \times 3$ cm	60 cm³	$20, 12, 15$	94 cm²
$8 \times 5 \times 2$ cm	80 cm³	$40, 10, 16$	132 cm²
$10 \times 6 \times 4$ cm	240 cm³	$60, 24, 40$	248 cm²
cube, edge 4 cm	64 cm³	16 each	96 cm²
cube, edge 7 cm	343 cm³	49 each	294 cm²

THREE NUMBERS MULTIPLIED, THREE PRODUCTS DOUBLED

Every forward question in the block is one of those two sentences. Write the three faces on their own line and the surface area cannot go wrong.

Reverse questions

GIVEN	WANTED	WORKING	ANSWER
$V = 120$, base 5×4	the height	$120 \div 20$	6 cm
$V = 240$, base 10×6	the height	$240 \div 60$	4 cm
cube, $V = 512$	the edge	$a^3 = 512$	8 cm
cube, $S = 150$	the edge	$150 \div 6 = 25$	5 cm
$a = 5, b = 4, S = 94$	the third edge	$40 + 18c = 94$	3 cm

THE DIRECTION DECIDES THE OPERATION

Forward is multiplication; reverse is division, a root, or a linear equation. Deciding which of the two a question is before touching the numbers saves more marks than any amount of arithmetic care.

Context and units

PROBLEM	STEPS	ANSWER
a $60 \times 40 \times 50$ cm tank, in litres	$120\ 000 \div 1000$	120 *litres*
…filled at 8 litres a minute	$120 \div 8$	15 *min*
water 6 cm deep on a 40×25 cm base	$1000 \cdot 6$	6 *litres*
a stone raising that level by 1.5 cm	$1000 \cdot 1.5$	$1500\ cm^3$
a $4 \times 3 \times 0.1$ m slab at 2.4 t per m³	$1.2 \cdot 2.4$	$2.88\ t$

CONVERT FIRST, CALCULATE SECOND

Every context question in this block is a formula from the first section with a unit conversion in front of it. Doing the conversion first turns a hard question into an easy one.

The mistakes that cost marks

MISTAKE	LOOKS LIKE	CORRECT
a metre left as 1	$1 \cdot 80 \cdot 50 = 4000\ cm^3$	$100 \cdot 80 \cdot 50 = 400\ 000\ cm^3$
a cube edge by dividing by three	$27 \div 3 = 9$	$a^3 = 27, a = 3$
the root taken before the six	$\sqrt{96}$	$\sqrt{(96 \div 6)} = 4$
litres divided by an area	$120 \div 4000$	$120\ 000 \div 4000 = 30\ cm$
a volume written in cm^2	$60\ cm^2$	$60\ cm^3$

FOUR OF THE FIVE ARE UNITS, NOT ARITHMETIC

Which is the good news: reading the units off the working, line by line, removes almost every lost mark in this block.

02 Worked examples

EXAMPLE 1 **Find the volume and the surface area of a cuboid 10 cm by 6 cm by 4 cm.**

1. $V = 10 \cdot 6 \cdot 4 = 240$ cm³.
2. Faces: $60, 24, 40$.
3. $S = 2 \cdot 124 = 248$ cm².

   Volume cubed, area squared ✓

ANSWER

240 cm³ and 248 cm²

EXAMPLE 2 **A cube has surface area 150 cm². Find its edge and its volume.**

1. $150 \div 6 = 25$ cm² for one face.
2. The edge is 5 cm.
3. $V = 5^3 = 125$ cm³.

   Check: $6 \cdot 25 = 150$ ✓

ANSWER

5 cm and 125 cm³

EXAMPLE 3 **A tank with base *80* cm by *50* cm holds *120* litres when full. Find its depth.**

1. 120 litres is $120\,000$ cm^3.

2. Base area $80 \cdot 50 = 4000$ cm^2.

3. $120\,000 \div 4000 = 30$ cm.

   Convert first, divide second.

ANSWER

30 cm

03 Interactive model

Give the class the five-row mistake table before the revision problems, not after; most of them recognise at least one of their own habits in it.

Revision of the solids block

The volume of a $10 \times 6 \times 4$ cm cuboid:

- 124
- 240
- 248
- 20

Its surface area:

- 124
- 240
- 248
- 480

A cube of surface area 150 cm² has edge:

- 5
- 25
- 6
- 12.5

A cuboid of volume 120 cm³ on a 5×4 base is high:

- 6 cm
- 24 cm
- 20 cm
- 4 cm

A cube of volume 512 cm³ has edge:

- 7
- 8
- 9
- 171

A $60 \times 40 \times 50$ cm tank holds:

- 12
- 120
- 1200
- $120\,000$

A stone raising the level 1.5 cm on a 1000 cm² base has volume:

- 1500 cm³
- 150 cm³
- 666 cm³
- 1.5 cm³

A volume must be written in:

- *cm*
- cm^2
- cm^3
- any unit

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Volume	Hajm	Объём
Surface area	Sirt yuzasi	Площадь поверхности
Cuboid	Parallelepiped	Прямоугольный параллелепипед
Cube	Kub	Куб
Base area	Asos yuzasi	Площадь основания
Capacity	Sig‘im	Вместимость
Displacement	Siqib chiqarish	Вытеснение
Unit	Birlik	Единица измерения

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The surface area of a cuboid is:

- abc
- $2(ab + bc + ac)$
- $6a^2$
- ab

The edge of a cube from its volume is:

- $V \div 3$
- the cube root of V
- the square root of V
- $V \div 6$

The edge of a cube from S is the square root of:

- S
- $S \div 6$
- $6S$
- $S \div 2$

1 m³ equals:

- 1000 cm³
- $10\ 000$ cm³
- $1\ 000\ 000$ cm³
- 100 cm³

A missing edge from the volume is found by dividing by:

- three
- the base area
- the surface area
- six

The most common lost mark in this block is:

- arithmetic
- the units
- the diagram
- the formula

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. The volume of a $5 \times 4 \times 3$ cm cuboid

   ANSWER 60 cm³

2. Its surface area

   ANSWER 94 cm²

3. The volume of a cube of edge 4 cm

   ANSWER 64 cm³

4. Its surface area

   ANSWER 96 cm²

5. 1 litre in cm³

   ANSWER 1000 cm³

6. The volume of an $8 \times 5 \times 2$ cm cuboid

   ANSWER 80 cm³

7. The number of faces of a cuboid

   ANSWER 6

Medium · the standard question

7 PROBLEMS

1. The surface area of an $8 \times 5 \times 2$ cm cuboid

   ANSWER 132 cm²

2. The volume and surface area of a cube of edge 7 cm

   ANSWER 343 cm³ and 294 cm²

3. $V = 120$ cm³ on a 5×4 base: the height

   ANSWER 6 cm

4. A cube of volume 512 cm³: the edge

   ANSWER 8 cm

5. A cube of surface area 150 cm²: the edge

   ANSWER 5 cm

6. The capacity of a $60 \times 40 \times 50$ cm tank

   ANSWER 120 litres

7. One face of a cube of surface area 294 cm²

   ANSWER 49 cm²

Hard · stretch and reason

7 PROBLEMS

1. A cuboid 5 by 4 with $S = 94$ cm²: the third edge

   ANSWER 3 cm

2. A tank with base 80×50 cm holding 120 litres: the depth

   ANSWER 30 cm

3. A stone raising the level 1.5 cm on a 1000 cm² base

   ANSWER 1500 cm³

4. A 4 m by 3 m by 0.1 m slab at 2.4 t per m³

   ANSWER 2.88 tonnes

5. Card for an open box $8 \times 5 \times 4$ cm

   ANSWER 144 cm²

6. Doubling every edge multiplies V and S by

   ANSWER 8 and 4

7. Boxes $10 \times 8 \times 5$ cm in a crate $40 \times 32 \times 30$ cm

   ANSWER 96

07 Homework

Homework — 5 tasks

Write the unit on every line of working, not only on the answer.

1. Find the volume and the surface area of a cuboid 9 cm by 5 cm by 4 cm.
2. A cube has surface area 216 cm². Find its edge and its volume.
3. A cuboid has volume 252 cm³ and a base 9 cm by 7 cm. Find its height.
4. A tank 90 cm by 50 cm by 40 cm is filled at 12 litres a minute. How long does it take?
5. List the five mistakes of the last section and write one line on how you will avoid each.

Control work 8 — volume and surface, and work on the mistakes

Twenty-five marks on the solids block, then the diagnosis and the rewriting.

LESSON 155–156 | 2 × 40 MIN | MATEMATIKA 6, NAZORAT ISHI 8 | S7 15 REVIEW

LESSON CLOCK

3 | 40' | 27' | 10'

Instructions	3 min
The paper	40 min
Answers and diagnosis	27 min
What comes next	10 min

LEARNING OBJECTIVES

- Find the volume and the surface area of a cuboid and of a cube.
- Run both formulae backwards to a missing edge.
- Solve a capacity, a filling-time and a displacement problem.
- Name each lost mark by its type and rewrite the whole solution.

TEXTBOOK

National	pp. 400–448
Cambridge	Stage 7, pp. 156–172
Workbook	Control paper 8

01 Explanation

The paper — 25 marks, 40 minutes

Q	TASK	MARKS	FROM
1	The volume and surface area of a $6 \times 5 \times 4$ cm cuboid	3	L139, L143–145
2	The volume and surface area of a cube of edge 5 cm	3	L139
3	A cuboid of volume 168 cm³ on a 7×6 base: the height	3	L140–142
4	A cube of surface area 384 cm²: the edge	3	L140–142
5	A cuboid 8 by 3 cm with $S = 158$ cm²: the third edge	4	L140–142
6	A tank $70 \times 40 \times 50$ cm: its capacity, and the time to fill it at 14 l/min	4	L146–149
7	A stone raises the water from 8 cm to 9.2 cm on a 50×30 cm base: its volume	3	L146–149
8	The card for an open box $12 \times 8 \times 5$ cm	2	L143–145

THE ANSWERS

120 cm³ and 148 cm²; 125 cm³ and 150 cm²; 4 cm; 8 cm; 5 cm; 140 litres and 10 min; 1800 cm³; 296 cm².

Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
a metre left as 1	$1 \cdot 80 \cdot 50$	$100 \cdot 80 \cdot 50$
a cube edge by dividing by three	$125 \div 3$	$a^3 = 125, a = 5$
the root taken before the six	$\sqrt{384}$	$\sqrt{(384 \div 6)} = 8$
litres divided by an area	$140 \div 2800$	$140\,000 \div 2800$
a volume written in cm^2	$120\ cm^2$	$120\ cm^3$
the lid counted on an open box	$392\ cm^2$	$296\ cm^2$
the new level used instead of the rise	$1500 \cdot 9.2$	$1500 \cdot 1.2$

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

What comes next

IF YOU LOST MARKS ON	REVISE
Q1–Q2	the two formulae, L139 and L143–145
Q3–Q5	the formulae run backwards, L140–142
Q6–Q7	capacity, filling and displacement, L146–149
Q8	choosing which faces to count, L143–145

LOOKING FORWARD

Quarter IV opens with data handling — collecting it, drawing pie charts and reading them — then probability, the three-dimensional shapes and their nets, and the revision of the year.

02 Worked examples

EXAMPLE 1 **Model answer, Q5: a cuboid 8 cm by 3 cm with surface area 158 cm².**

1. $S = 2(8 \cdot 3 + 3c + 8c) = 158$.
2. $48 + 22c = 158$, so $22c = 110$.
3. $c = 5$ cm.

   Check: $2(24 + 15 + 40) = 158$ ✓

ANSWER

5 cm

EXAMPLE 2 **Model answer, Q6: a tank *70* cm by *40* cm by *50* cm, filled at *14* litres a minute.**

1. $70 \cdot 40 \cdot 50 = 140\ 000$ cm³.

2. That is *140* litres.
   Divide by *1000*.

3. $140 \div 14 = 10$ minutes.

ANSWER

140 litres and *10* minutes

EXAMPLE 3 **Model answer, Q7: the water rises from *8* cm to *9.2* cm on a *50* × *30* cm base.**

1. Base area $50 \cdot 30 = 1500$ cm².

2. The rise is $9.2 - 8 = 1.2$ cm.
   The rise, not the new level.

3. $1500 \cdot 1.2 = 1800$ cm³.

ANSWER

1800 cm³

03 Interactive model

Return Q7 first and ask who multiplied by 9.2; naming the error as “the level instead of the rise” fixes it for the year.

The paper in eight questions

The volume of a $6 \times 5 \times 4$ cm cuboid:

- 74
- 120
- 148
- 30

Its surface area:

- 74
- 120
- 148
- 296

A cube of edge 5 cm has volume:

- 25
- 125
- 150
- 15

A cuboid of volume 168 cm³ on a 7×6 base is high:

- 4 cm
- 42 cm
- 24 cm
- 8 cm

A cube of surface area 384 cm² has edge:

- 64
- 8
- 19.6
- 6

A $70 \times 40 \times 50$ cm tank holds:

- 14 litres
- 140 litres
- 1400 litres
- $140\ 000$ litres

A rise from 8 to 9.2 cm on 1500 cm² means:

- $13\ 800$ cm³
- 1800 cm³
- $12\ 000$ cm³
- 1.2 cm³

An open box $12 \times 8 \times 5$ cm needs:

- 392
- 296
- 200
- 96

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Volume	Hajm	Объём
Surface area	Sirt yuzasi	Площадь поверхности
Capacity	Sig‘im	Вместимость
Displacement	Siqib chiqarish	Вытеснение
Open box	Qopqog‘i yo‘q quti	Открытая коробка
Unit	O‘lchov birligi	Единица измерения
Diagnosis	Tashxis	Диагностика

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Q1 wants the surface area in:

- cm
- cm^2
- cm^3
- litres

Q4 divides by six:

- after the root
- before the root
- never
- twice

Q5 gives an equation that is:

- quadratic
- linear
- impossible
- a fraction

Q6 needs the volume converted to:

- cm^2
- litres
- metres
- nothing

Q7 multiplies the base area by:

- the new level
- the rise
- the old level
- their sum

Q8 leaves out:

- the base
- the lid
- two walls
- nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. The volume of a $6 \times 5 \times 4$ cm cuboid

   ANSWER 120 cm³

2. Its surface area

   ANSWER 148 cm²

3. The volume of a cube of edge 5 cm

   ANSWER 125 cm³

4. Its surface area

   ANSWER 150 cm²

5. $V = 168$ cm³ on a 7×6 base: the height

   ANSWER 4 cm

6. One face of a cube of surface area 384 cm²

   ANSWER 64 cm²

7. The capacity of a $70 \times 40 \times 50$ cm tank

   ANSWER 140 litres

Medium · the standard question

7 PROBLEMS

1. The edge of a cube of surface area 384 cm²

ANSWER 8 cm

2. A cuboid 8 by 3 cm with $S = 158$ cm²: the third edge

ANSWER 5 cm

3. A 140-litre tank filled at 14 l/min

ANSWER 10 min

4. A stone raising 8 cm to 9.2 cm on a 50×30 base

ANSWER 1800 cm³

5. Card for an open box $12 \times 8 \times 5$ cm

ANSWER 296 cm²

6. The closed box of the same size

ANSWER 392 cm²

7. The volume of that box

ANSWER 480 cm³

Hard · stretch and reason

7 PROBLEMS

1. A 1 m by 80 cm by 50 cm tank, in litres

   ANSWER 400 litres

2. The commonest lost mark in this paper

   ANSWER A unit not converted

3. A cube of volume 1728 cm³: its edge and surface area

   ANSWER 12 cm and 864 cm²

4. The mass of the water in a full 140-litre tank

   ANSWER 140 kg

5. Doubling every edge of the Q1 cuboid: the new volume

   ANSWER 960 cm³

6. …and the new surface area

   ANSWER 592 cm²

7. The depth of 28 litres in a tank of base 70×40 cm

   ANSWER 10 cm

07 Homework

Homework — 5 tasks

Rewrite every question you lost a mark on in full, with the units on every line.

1. Rewrite in full every question on which you lost a mark, naming the slip in the margin.
2. Find the volume and the surface area of a cuboid 7 cm by 6 cm by 3 cm.
3. A cube has volume 1331 cm³. Find its edge and its surface area.
4. A tank 100 cm by 60 cm by 45 cm is filled at 15 litres a minute. How long does it take?
5. How much card is needed for an open box 15 cm by 10 cm by 6 cm deep?